

Rough Polynomial Fluctuation Theory: Coefficient-Uniform Jump, Variation, and Endpoint Bounds

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Abstract

We prove coefficient-uniform critical 2-jump inequalities for the full sharp-truncation trajectory

$$T_{\Omega, \varepsilon}^{P, b} f(x) = \int_{|x-y| > \varepsilon} e^{iP(x, y)} b(|x-y|) \frac{\Omega((x-y)/|x-y|)}{|x-y|^n} f(y) dy,$$

where P is an arbitrary real polynomial on $\mathbb{R}^n \times \mathbb{R}^n$ of bounded degree and $\Omega \in L \log L(\mathbb{S}^{n-1})$ has mean zero. When $b \equiv 1$, the estimate holds on $L^p(\mathbb{R}^n)$ for every $1 < p < \infty$ and at spatial weak type $(1, 1)$. The constants depend only on the dimension, the degree bound, and the relevant Lebesgue exponent, and are independent of every coefficient of P ; no nondegeneracy, smallness, or genericity assumption is imposed on the phase.

At $P = 0$, the weak endpoint independently resolves the critical-jump problem posed by Jones–Seeger–Wright [16]; for general P , it extends that endpoint to all bounded-degree two-variable polynomial phases. The jump theorem implies strong and weak r -variation inequalities for every $r > 2$, maximal-truncation bounds, almost-everywhere principal-value convergence, and a canonical trajectory determined by its finite-annulus increments and zero-at-infinity anchor. For bounded radial b of finite total variation, an anchored Stieltjes representation transfers the entire theory with linear dependence on $\|b\|_\infty + \text{Var}_{(0, \infty)}(b)$.

The proof combines a finite global algebraic flag on polynomial phases modulo pure input and output phases, a phase-adapted Calderón–Zygmund decomposition with anchored phase–packet induction, and a selector-free fixed-rank Rademacher–Menshov reduction supported by concentration-stable equal- and unequal-shell correlation estimates. The unphased endpoint is proved independently within the same framework.

AI-use disclosure. The research direction and target theorem of this paper were formulated through iterative interaction between the author and an artificial-intelligence system. The author directed and coordinated the mathematical development, including the selection and refinement of the problems and intermediate objectives pursued during that process. The AI system generated the mathematical proof arguments and derivations establishing the results of the paper; these contributions occur throughout Sections 2–12 rather than being confined to isolated lemmas or theorems. The author examined, selected, revised, and organized the resulting arguments.

1 Introduction and main results

A maximal-truncation estimate controls only the largest excursion of an operator trajectory. An r -variation estimate with $r > 2$ also quantifies convergence, but the endpoint $r = 2$ is generally inaccessible. The critical replacement is the 2-jump functional

$$\sup_{\lambda > 0} \lambda N_\lambda^{1/2},$$

which measures, uniformly over all amplitudes λ , the number of disjoint oscillations of that size. Its boundedness implies every r -variation estimate with $r > 2$ through an abstract endpoint principle. The

purpose of this paper is to prove this critical estimate, including spatial weak type $(1, 1)$, for the complete sharp-truncation trajectory of rough polynomial singular integrals.

The weak endpoint at $P = 0$ is exactly the problem posed by Jones–Seeger–Wright [16]. We resolve it independently and prove, in the same framework, the coefficient-uniform extension to arbitrary real two-variable polynomial phases of bounded degree. The angular kernel is assumed only to belong to the endpoint class $L \log L$; the phase is subject to no nondegeneracy, smallness, or genericity hypothesis. Thus the theorem simultaneously treats rough angular data, genuinely nonconvolution phases, critical fluctuations, and the spatial endpoint.

Let $d\sigma$ be the usual surface measure on $\mathbb{S}^{n-1}$. Put

$$K_\Omega(z) = \frac{\Omega(z/|z|)}{|z|^n}, \quad z \neq 0.$$

For $A = \|\Omega\|_1 > 0$ define the homogeneous gauge

$$\mathfrak{L}(\Omega) = A + \int_{\mathbb{S}^{n-1}} |\Omega(\theta)| \log^+ \left(\frac{|\Omega(\theta)|}{A} \right) d\sigma(\theta), \quad (1)$$

and set $\mathfrak{L}(0) = 0$. This is equivalent, with dimensional constants, to the homogeneous Luxemburg $L \log L$ norm. Its homogeneity and the boundedness of the odd and even projections give

$$\mathfrak{L}(\Omega_o) + \mathfrak{L}(\Omega_e) \leq C \mathfrak{L}(\Omega). \quad (2)$$

The elementary normalization details are recorded in Lemma 2.1.

On bounded compactly supported inputs define

$$T_{\Omega, \varepsilon}^P f(x) = \int_{|x-y| > \varepsilon} e^{iP(x,y)} K_\Omega(x-y) f(y) dy.$$

The final construction for general data is given in Section 12; it is characterized by the finite-annulus identity and by convergence to zero as $\varepsilon \rightarrow \infty$.

For a scalar path $F = (F_t)_{t>0}$, the paired strict jump number is

$$N_\lambda(F) = \sup \{N : \exists 0 < s_1 < t_1 \leq s_2 < t_2 \leq \dots \leq s_N < t_N, |F_{t_i} - F_{s_i}| > \lambda\}.$$

The homogeneous r -variation is

$$V^r(F_t : t > 0) = \sup_{t_0 < \dots < t_N} \left(\sum_{i=1}^N |F_{t_i} - F_{t_{i-1}}|^r \right)^{1/r}.$$

For a measurable function H , set $\|H\|_{L^{1,\infty}} = \sup_{\tau>0} \tau |\{x : |H(x)| > \tau\}|$. For $1 < p < \infty$ set

$$\mathcal{J}_{2,p}(F) = \sup_{\lambda>0} \|\lambda N_\lambda(F)^{1/2}\|_p, \quad \mathcal{J}_{2,1,\infty}(F) = \sup_{\lambda>0} \|\lambda N_\lambda(F)^{1/2}\|_{L^{1,\infty}}.$$

Throughout, C and c may change from line to line. Unless an additional dependence is displayed, they depend only on the ambient dimension, the fixed degree bound, the indicated Lebesgue or variation exponents, and fixed compact support or chart data. They never depend on the coefficients of P , on truncation or dyadic scales, or on auxiliary finite cutoffs. Every coefficient norm is a fixed Euclidean norm on the relevant finite-dimensional polynomial space; changing that norm changes only the permitted structural constants.

Theorem 1.1 (Coefficient-uniform critical jump inequalities). *Let $n \in \mathbb{N}$ and $d \in \mathbb{N}_0$. Let P be a real polynomial on $\mathbb{R}^n \times \mathbb{R}^n$ of total degree at most d , and let $\Omega \in L \log L(\mathbb{S}^{n-1})$ satisfy $\int \Omega d\sigma = 0$. Then the sharp-truncation trajectory satisfies the coefficient-uniform estimates*

$$\mathcal{J}_{2,p}(T_{\Omega, \varepsilon}^P f : \varepsilon > 0) \leq C_{n,d,p} \mathfrak{L}(\Omega) \|f\|_p, \quad 1 < p < \infty, \quad (3)$$

$$\mathcal{J}_{2,1,\infty}(T_{\Omega, \varepsilon}^P f : \varepsilon > 0) \leq C_{n,d} \mathfrak{L}(\Omega) \|f\|_1. \quad (4)$$

Estimate (3) holds for every $1 < p < \infty$ and $f \in L^p(\mathbb{R}^n)$; estimate (4) holds for every $f \in L^1(\mathbb{R}^n)$. The constants are independent of every coefficient of P .

For each admissible pair (Ω, f) , these estimates determine a unique trajectory with the prescribed finite-annulus increments and satisfying

$$T_{\Omega, \varepsilon}^P f(x) \longrightarrow 0 \quad \text{as } \varepsilon \rightarrow \infty \quad (5)$$

almost everywhere. When $1 < p < \infty$, the convergence also holds in $L^p(\mathbb{R}^n)$.

Corollary 1.2 (Variation and convergence consequences). *Under the hypotheses of Theorem 1.1, for every $r > 2$,*

$$\|V^r(T_{\Omega, \varepsilon}^P f : \varepsilon > 0)\|_p \leq C_{n,d,p,r} \mathfrak{V}(\Omega) \|f\|_p, \quad 1 < p < \infty, \quad (6)$$

$$\|V^r(T_{\Omega, \varepsilon}^P f : \varepsilon > 0)\|_{L^{1,\infty}} \leq C_{n,d} \frac{r}{r-2} \mathfrak{V}(\Omega) \|f\|_1. \quad (7)$$

where (6) holds for every $1 < p < \infty$ and $f \in L^p(\mathbb{R}^n)$, and (7) holds for every $f \in L^1(\mathbb{R}^n)$. The maximal truncation satisfies the corresponding strong or weak estimate, and $\lim_{\varepsilon \downarrow 0} T_{\Omega, \varepsilon}^P f$ exists almost everywhere in the respective ranges.

For a bounded function $b : (0, \infty) \rightarrow \mathbb{C}$ of finite total variation, fix its right-continuous representative, put

$$\|b\|_{\text{BV}} = \|b\|_{\infty} + \text{Var}_{(0,\infty)}(b),$$

and, initially on bounded compactly supported inputs, define

$$T_{\Omega, \varepsilon}^{P,b} f(x) = \int_{|x-y|>\varepsilon} e^{iP(x,y)} b(|x-y|) K_{\Omega}(x-y) f(y) dy.$$

Corollary 1.3 (Radial BV transfer). *Under the hypotheses of Theorem 1.1, let $b : (0, \infty) \rightarrow \mathbb{C}$ be bounded and of finite total variation. All conclusions of Theorem 1.1 and Corollary 1.2 remain valid with $T_{\Omega, \varepsilon}^P$ replaced by $T_{\Omega, \varepsilon}^{P,b}$ and with each right-hand side multiplied by*

$$\|b\|_{\text{BV}} = \|b\|_{\infty} + \text{Var}_{(0,\infty)}(b).$$

The constants are independent of where the variation of b is concentrated.

The primary result is the weak critical-jump estimate (4): it independently resolves the Jones–Seeger–Wright problem when $P = 0$ and yields its coefficient-uniform nonconvolution extension for general P . The strong critical-jump estimate is proved within the same architecture. The variation, maximal, and convergence assertions then follow from the abstract jump calculus and the zero anchor, while Corollary 1.3 follows from the anchored Stieltjes transfer in Section 12.

Position in the literature

Static rough polynomial theory. For $P = 0$, the strong and weak theory of rough homogeneous singular integrals at the $L \log L$ angular scale was developed in [6, 13, 11, 25]. For arbitrary $P(x, y)$, Jiang–Lu [15] studied the static principal-value operator with $\Omega \in L \log L$ and a radial factor in BV, including an L^p boundedness criterion and weighted estimates. Ojanen [23] proved weighted strong estimates for the same polynomial-phase, angular- $L \log L$, radial-BV class under the geometric hypotheses formulated there. Al-Salman and Grafakos [3] more recently obtained static L^p bounds for arbitrary $P(x, y)$, angular $L \log L$ data, and a rough radial factor satisfying an L^γ -type condition. These results concern the scalar principal-value or maximal operator. The object here is the complete sharp-truncation trajectory at the critical fluctuation and spatial endpoints.

Smooth coefficient-uniform weak theory. Ricci and Stein [24] established coefficient-uniform strong L^p estimates for smooth kernels. Chanillo and Christ [9] proved coefficient-uniform weak type $(1, 1)$ for arbitrary polynomial phases and smooth Calderón–Zygmund convolution kernels. Their argument removes pure phases, normalizes a leading coefficient by dilation, and uses a phase-adapted double induction; their final remarks also yield L^p bounds, $1 < p < \infty$, for the maximal truncation. Rough $L \log L$ kernels fall outside that theory. The global phase flag introduced below is designed to retain coefficient uniformity while accommodating rough annular decay and phases that degenerate on the diagonal.

Maximal rough polynomial theory. Representative rough polynomial estimates include [21, 2]; polynomial maximal-truncation estimates for smooth Calderón–Zygmund kernels include [17]. Choudhary, Shrivastava, and Shuin [10] state coefficient-uniform sparse and strong maximal estimates for arbitrary P when $\Omega \in L^\infty$. For $P = 0$, Lai [18, Theorem 1.1] proves the weak $(1, 1)$ maximal estimate at the $L \log L$ angular endpoint. In the present paper, maximal truncation is obtained as a consequence of the stronger critical-jump theorem.

Fluctuation theory. For sharply truncated Hilbert transforms, Campbell, Jones, Reinhold, and Wierdl [7, Theorems 1.2–1.4 and Conjecture 7.2] established strong and weak r -variation estimates for $r > 2$, the associated noncritical jump bounds, and short 2-variation estimates, and isolated the critical jump exponent. Their higher-dimensional work [8] developed singular-integral variation theory at the rough $L \log L$ scale. Jones–Seeger–Wright [16, Theorem 1.6] proved the strong critical jump inequality for the unphased rough trajectory under $\Omega \in L^a$, $a > 1$, and posed the corresponding weak-type $(1, 1)$ estimate for rough homogeneous kernels [16]. In the notation of this paper, that problem is exactly the $P = 0$ case of (4). Lemma 2.4 derives the one-dimensional terminal estimate needed below from [7, Theorem 1.4] and [16, Theorem 1.2(ii) and Lemma 1.3].

Ding, Hong, and Liu [12, Theorem 1.2(ii)] state, with the angular size tacitly normalized, the unphased strong critical-jump theorem at the $L \log L$ endpoint. Bhojak and Shrivastava [5, Theorem 1.1] subsequently state the unphased weak critical-jump and weak r -variation endpoints. Remark 1.4 explains why neither argument is used here.

Recent polynomial-modulation variation theory. Becker, Jamneshan, and Thiele [4, Theorem 1.3 and Corollary 1.4] prove strong L^p r -variation estimates, $1 < p < \infty$ and $r > 2$, for sharp truncations of Hölder Calderón–Zygmund kernels satisfying exact two-sided annular cancellation. Their estimate takes the supremum over bounded-degree polynomial input modulations pointwise inside the L^p norm and holds more generally on homogeneous Lie groups. In Euclidean space, this formulation already contains arbitrary two-variable phases: for each output point x , set

$$Q_x(y) = P(x, y) - P(x, 0), \quad T_{\Omega, u}^{P, 1} f(x) = e^{iP(x, 0)} S_u(K_\Omega, Q_x, f)(x).$$

Because the output factor is independent of u , their smooth-kernel theorem is not restricted to pure-input phases. Combining their one-dimensional result with the method of rotations also gives the fixed-phase, coefficient-uniform strong $r > 2$ estimates considered here for odd rough angular kernels in L^1 (and hence for the entire mean-zero class when $n = 1$). It does not cover the general even rough component at the $L \log L$ scale, the critical 2-jump estimate, or spatial weak type $(1, 1)$. Conversely, the estimates proved here are uniform in the coefficients of each fixed phase but do not place a pointwise supremum over phases inside the spatial norm.

Contribution of the present paper. The central result is the weak critical-jump estimate (4). At $P = 0$ it independently resolves the Jones–Seeger–Wright problem at the $L \log L$ angular endpoint; for general P it gives a coefficient-uniform extension to all bounded-degree two-variable polynomial phases, with no nondegeneracy assumption. The same theorem covers convolution phases, genuinely nonconvolution

phases, and phase geometries that degenerate on the diagonal. The strong critical 2-jump estimate is proved in parallel. The strong $r > 2$ consequence extends the smooth and odd-kernel subclasses covered by Becker, Jamneshan, and Thiele to the full rough $L \log L$ class, including its general even component. The critical jump, the spatial weak endpoint, and the radial BV transfer are separate conclusions not contained in that result.

Remark 1.4 (Independence from prior proofs). The proof below does not use the coefficient-detection and annular-decay arguments in [2, 10, 15, 19, 20, 21] or the endpoint fluctuation arguments in [12, 5]. The observations in this remark concern the cited proofs as written, not the truth of their theorem statements. The discussion of [5] refers to version 2, accessed on August 27, 2026.

The obstruction in [10] is inherited from the one-ray reduction of Lu–Zhang [21, pp. 210–213], which [10, Lemma 3.2] explicitly adapts. For the top bidegree block

$$P_{m,\ell}(x, y) = \sum_{\substack{|\alpha|=m \\ |\beta|=\ell}} a_{\alpha,\beta} x^\alpha y^\beta,$$

that reduction detects the coefficient tensor only through its diagonal restriction

$$A_{m,\ell}(\theta) = P_{m,\ell}(\theta, \theta) = \sum_{\substack{|\alpha|=m \\ |\beta|=\ell}} a_{\alpha,\beta} \theta^{\alpha+\beta}.$$

The map $P_{m,\ell} \mapsto A_{m,\ell}$ is not injective. For example, the alternating bilinear phase $P(x, y) = x_1 y_2 - x_2 y_1$ has a nonzero top bidegree block, but $A_{1,1} \equiv 0$. Thus the nonzero top-bidegree hypothesis in Lemma 3.2 does not supply the nonvanishing needed to apply Corollary 3.4 in the passage to (4.7). Related coefficient-detection reductions are invoked or reproduced by Jiang–Lu [15, p. 142], Lu–Wu [19, p. 54] and [20, p. 308], and Al-Salman [2, pp. 82–83]. The global phase flag used here avoids this diagonal compression and retains the full mixed coefficient data.

In [12], the geometric tail preceding (2.16) carries a factor comparable to m that is not retained in the displayed estimate. Independently, Lemma 2.11 is false as stated for $0 < \theta < 1$: as $t \downarrow 0$, the ratio of its left-hand side to its right-hand side is comparable to $t^{\theta-1}$. The lemma is used in precisely this range to reduce a loss of order $m^{2-\theta}$ to the m -weighted angular summability available at the $L \log L$ endpoint. Consequently, the published proof does not justify that endpoint summation in its printed form.

In version 2 of [5], the short-jump packets $B_{i,\text{in}}^{k-s,m}$ depend on the output point x , whereas (2.2) and Lemma A.2 control sums whose coefficients are chosen before x is evaluated; the variational Rademacher–Menshov theorem does not by itself supply the missing selector estimate. In the long-jump argument, the displayed fixed-shift families $\mathcal{D}_u$ are not nested: for $n = 1$ and $u = 1/3$, the intervals $[1/3, 4/3)$ and $[2/3, 8/3)$ overlap without either containing the other. These points require additional arguments. For these reasons, the present paper proves the relevant critical-jump endpoints independently.

Endpoints and sharpness

The theorem reaches both critical thresholds: jump exponent 2 and spatial weak type $(1, 1)$. For $n \geq 2$, the angular $L \log L$ assumption is also sharp for coefficient-uniform estimates of this generality in the following inherited sense. Al-Qassem, Cheng, and Pan [1, Theorem 1.3(b)] construct mean-zero angular functions in $L(\log L)^{1-\varepsilon}$ for every $\varepsilon > 0$ such that the associated scalar rough oscillatory integrals fail to be uniformly bounded over polynomial phases. Thus the theorem meets the known coefficient-uniform angular obstruction. It does not address whether finite radial variation is necessary or whether the factor $r/(r-2)$ is optimal for this operator family.

Three obstructions and their resolution

Phase degeneracy. Neither a displayed monomial coefficient nor the diagonal restriction of the phase provides a stable normalization. Simultaneous translation mixes individual coefficients, and the alternating

bilinear phase $x_1 y_2 - x_2 y_1$ vanishes on the diagonal. Section 3 constructs a finite global flag on polynomial phases modulo pure input and output phases. The transverse rows are placed in reduced row-echelon form, and the final nonterminal rows consist of convolution phases and alternating bilinear forms. Dilation normalizes the first nonzero flag coordinate, while simultaneous translation preserves that coordinate and creates only later ones. This triangular structure yields coefficient-uniform annular decay and closes the local analysis by a finite backward induction over the flag.

Modulation-sensitive cancellation. The parent-dependent input modulations generated by a non-convolution phase destroy ordinary mean-zero cancellation. The phase-adapted Calderón–Zygmund decomposition replaces it by

$$\int_Q e^{iP(c_Q, y)} b_Q(y) dy = 0$$

and subtracts the same parent anchor from the operator phase. The anchored packet proposition then performs a simultaneous phase–packet induction. A large first coordinate gives scale-covariant decay before any expansion; a small coordinate is Fourier-expanded with packet coefficients fixed before the output point is evaluated. The all-low branch terminates in the separated-packet theory. Proposition 6.2 isolates this interaction among phase reduction, modulation-sensitive cancellation, and separated-packet estimates.

Output-dependent selectors. A jump linearization chooses parameter intervals as functions of the output point. The endpoint argument first imposes finite cube, shell, and parameter cutoffs and assigns every relevant increment a fixed rank. A variational Rademacher–Menshov estimate then controls every interval sum through that ranking, without introducing an output-dependent sign choice. The phase-adapted bad functions enter directly through cancellation and the average density bound $\|h_Q\|_1 \lesssim \alpha|Q|$. On each shell, a descending homogeneous-degree packet induction combines Fourier expansion of low blocks with a concentration-stable high-block estimate. Across high shells, an exact oriented two-ray calculation retains a coefficient of size R^m after all projective cancellations and lower-degree perturbations. A one-sided polynomial incidence estimate then sums the small-cube mesh errors geometrically while the larger atom remains fixed. The resulting fixed-sign all-shell estimate is precisely the input needed by the rank argument, and it requires no L^∞ decomposition of the bad data.

The proof uses four specialized fluctuation-theoretic or microlocal inputs: the normability of jump spaces and the weak jump-to-variation principle of Mirek, Stein, and Zorin-Kranich; the one-dimensional jump inequalities and strong jump-to-variation lemma of Jones, Seeger, and Wright; the Hilbert-transform short-variation theorem of Campbell, Jones, Reinhold, and Wierdl; and Seeger’s two separated-packet estimates. We also use the classical M. Riesz theorem and the Fefferman–Stein H^1 –BMO theory. Each input is stated in the precise form required at its point of application. All polynomial, oscillatory, incidence, packet, angular-summation, and limiting arguments specific to the present theorem are proved here.

The proof is organized as follows. Section 2 develops the abstract jump calculus and establishes the unphased strong endpoint. Sections 3 and 4 construct the global phase flag, prove coefficient-uniform annular decay, and close the strong polynomial jump theorem by angular extrapolation and triangular induction. Sections 5 and 6 establish the terminal separated-packet estimates and the anchored phase–packet induction. Section 7 provides the concentration-stable equal- and unequal-shell estimates, while Section 8 proves the unphased weak endpoint. Sections 9–11 combine the phase-adapted Calderón–Zygmund decomposition with far-field and local homogeneous-degree induction to obtain the polynomial weak endpoint. Finally, Section 12 constructs the trajectory for general data, derives the variation and convergence consequences, and proves the radial Stieltjes transfer.

2 Abstract jump inequalities and the unphased strong endpoint

This section assembles the jump calculus and the one-dimensional estimates used throughout the paper, then combines them with a Riesz-transform factorization to prove the unphased strong critical-jump

theorem.

Lemma 2.1 (Gauge normalization). *For every $\Omega \in L \log L(\mathbb{S}^{n-1})$, the quantity $\mathfrak{L}$ in (1) is equivalent, with dimensional constants, to the homogeneous Luxemburg norm associated with $\Phi_0(t) = (1+t) \log(1+t) - t$. It also satisfies (2).*

Proof. The assertion is immediate for $\Omega = 0$, so assume $\mathfrak{L}(\Omega) > 0$. Convexity shows that $\int \Phi_0(|\Omega|/(C\mathfrak{L}(\Omega))) \leq 1$ for a sufficiently large absolute C . Conversely, if $\int \Phi_0(|\Omega|/L) \leq 1$, then $\|\Omega\|_1 \leq CL$ and

$$\int |\Omega| \log^+(|\Omega|/\|\Omega\|_1) \leq CL + \|\Omega\|_1 \log^+(L/\|\Omega\|_1) \leq C'L.$$

Thus the two quantities are equivalent. Reflection is an isometry of the Luxemburg space, and convexity makes the odd and even projections bounded; homogeneity and the equivalence just proved give (2). $\square$

2.1 Jump calculus and one-dimensional inputs

For a scalar path indexed by positive radii, write

$$N_\lambda^{\text{dyad}}(F) := N_\lambda(F_{2^j} : j \in \mathbb{Z}). \quad (8)$$

Lemma 2.2 (Jump and variation calculus). *For scalar paths F, G and $\lambda > 0$,*

$$N_\lambda(F + G) \leq N_{\lambda/2}(F) + N_{\lambda/2}(G), \quad (9)$$

$$\lambda N_\lambda(F)^{1/2} \leq V^2(F), \quad (10)$$

$$V^2(F + G) \leq V^2(F) + V^2(G). \quad (11)$$

If F is decomposed into dyadic parameter blocks, then

$$\lambda N_\lambda(F)^{1/2} \leq C \left[\left(\sum_{j \in \mathbb{Z}} V^2(F_\varepsilon : 2^j \leq \varepsilon \leq 2^{j+1})^2 \right)^{1/2} + \lambda N_{\lambda/3}^{\text{dyad}}(F)^{1/2} \right]. \quad (12)$$

Moreover $\lambda N_\lambda^{\text{dyad}}(F)^{1/2} \leq V^2(F_{2^j} : j \in \mathbb{Z})$.

Proof. For each strict λ -jump of $F + G$, one of the two corresponding increments has modulus $> \lambda/2$. Assign the pair to such a summand; each assigned subcollection remains ordered and paired, proving (9). Inserting the endpoints of a paired collection into a variation partition gives (10); Minkowski in the finite ℓ^2 sum gives (11).

For (12), let $j(t)$ be the integer determined by $2^{j(t)} \leq t < 2^{j(t)+1}$. Fix an ordered paired family (s_i, t_i) of strict λ -jumps. If $j(s_i) = j(t_i)$, assign the pair to that dyadic block. Otherwise put

$$\widehat{s}_i = 2^{j(s_i)+1}, \quad \widehat{t}_i = 2^{j(t_i)}.$$

Then $s_i \leq \widehat{s}_i \leq \widehat{t}_i \leq t_i$, and

$$F_{t_i} - F_{s_i} = (F_{t_i} - F_{\widehat{t}_i}) + (F_{\widehat{t}_i} - F_{\widehat{s}_i}) + (F_{\widehat{s}_i} - F_{s_i}).$$

One of the three terms has modulus $> \lambda/3$. The middle terms form an ordered paired family on the dyadic sequence. The initial and terminal terms assigned to any fixed dyadic block have pairwise disjoint parameter intervals, because the original pairs are ordered. Consequently, if N_{sh} and N_{long} denote the numbers assigned to within-block and middle terms, respectively, then

$$\frac{\lambda^2}{9} N_{\text{sh}} \leq \sum_j V^2(F_\varepsilon : 2^j \leq \varepsilon \leq 2^{j+1})^2, \quad N_{\text{long}} \leq N_{\lambda/3}^{\text{dyad}}(F).$$

Since $N_\lambda(F) \leq N_{\text{sh}} + N_{\text{long}}$, taking square roots proves (12). The last assertion follows from (10) on the dyadic sequence. $\square$

Normability at the critical jump exponent is essential for the angular factorizations below. We therefore isolate the exact consequence of [22, Corollary 2.11 and Lemma 2.12] that is used.

Let $N_\lambda^{\text{ch}}(F)$ be the largest J for which there are $t_0 < \dots < t_J$ with $|F_{t_j} - F_{t_{j-1}}| > \lambda$. Then

$$N_\lambda^{\text{ch}}(F) \leq 2N_\lambda(F), \quad N_\lambda(F) \leq N_{\lambda/2}^{\text{ch}}(F). \quad (13)$$

The first inequality follows by taking alternating edges of a chain. For the second, after an endpoint has been chosen from one admissible pair, at least one endpoint of the next pair is more than $\lambda/2$ away; otherwise the endpoints of that pair would be at distance at most λ . For a finite ordered index set I , we use the notation

$$\mathcal{J}_{2,p}^{\text{ch}}(F_t : t \in I) := \sup_{\lambda > 0} \|\lambda N_\lambda^{\text{ch}}(F_t : t \in I)^{1/2}\|_p. \quad (14)$$

Lemma 2.3 (Critical-jump normability and integration). *Let $1 < p < \infty$ and let I be a finite ordered set. The quantity*

$$\sup_{\lambda > 0} \|\lambda N_\lambda^{\text{ch}}(F_t : t \in I)^{1/2}\|_p$$

is equivalent, with constants depending only on p , to a subadditive seminorm, uniformly in I . Consequently, suppose that for every $t \in I$ the map $z \mapsto F_t^z$ is strongly measurable on a measure space Z and Bochner integrable in L^p with respect to a complex measure μ on Z , and that the right side below is finite. Then

$$\mathcal{J}_{2,p}^{\text{ch}}\left(\int F^z d\mu(z)\right) \leq C_p \int \mathcal{J}_{2,p}^{\text{ch}}(F^z) d|\mu|(z). \quad (15)$$

Suppose next that $J_0 \subseteq (1, \infty)$ is a compact interval whose interior contains both p and 2 , and that a linear family $\{S_t\}$ satisfies

$$\mathcal{J}_{2,q}^{\text{ch}}(S_t f : t \in I) \leq A_q \|f\|_q, \quad q \in J_0, \quad (16)$$

uniformly in finite ordered sets I , where $A_{J_0} := \sup_{q \in J_0} A_q < \infty$. Then, for every $r > 2$,

$$\|V^r(S_t f : t \in I)\|_p \leq C_{p,r,J_0} A_{J_0} \|f\|_p, \quad (17)$$

uniformly in finite I . At the endpoint,

$$\|V^r(F_t : t \in I)\|_{L^{1,\infty}} \leq C \frac{r}{r-2} \sup_{\lambda > 0} \|\lambda N_\lambda^{\text{ch}}(F)^{1/2}\|_{L^{1,\infty}}. \quad (18)$$

Proof. The normability and hence (15) are [22, Corollary 2.11], applied with their Lorentz indices $q = p$ and jump exponent $\rho = 2$; the equivalence constants are independent of I and of the target Banach space. That source counts increments of size $\geq \lambda$, whereas we use strict jumps. If $N_\lambda^{\geq}$ and $N_\lambda^>$ denote the two chain counts, then $N_\lambda^> \leq N_\lambda^{\geq} \leq N_{\lambda/2}^>$; hence the associated suprema are equivalent with an absolute constant. The strong implication is [16, Lemma 2.1]; its exponent interval must contain the target exponent and the jump exponent 2. Only finitely many auxiliary exponents in J_0 enter that argument, and homogeneity gives the displayed dependence on A_{J_0} . Formula (18) is [22, Lemma 2.12] with $p = 1$ and $\rho = 2$. Passing between the chain and paired conventions uses (13) and changes only absolute constants. It remains to justify (15) for a complex measure. Write $d\mu = \omega d|\mu|$, where $|\omega| = 1$ $|\mu|$ -almost everywhere, and fix $t_* \in I$. Choose a subadditive seminorm $\|\cdot\|_{\mathfrak{J}}$ equivalent to $\mathcal{J}_{2,p}^{\text{ch}}$. Since both seminorms are invariant under addition of a constant trajectory, replace F^z by $F^z - F_{t_*}^z$. For a simple function of z , subadditivity and homogeneity give

$$\left\| \int (F^z - F_{t_*}^z) d\mu(z) \right\|_{\mathfrak{J}} \leq C_p \int \|F^z - F_{t_*}^z\|_{\mathfrak{J}} d|\mu|(z).$$

If $I = \{t_0 < \dots < t_M\}$, then

$$\mathcal{J}_{2,p}^{\text{ch}}(F) \leq \|V_I^1(F)\|_p \leq \sum_{j=1}^M \|F_{t_j} - F_{t_{j-1}}\|_p \leq 2 \sum_{t \in I} \|F_t\|_p.$$

Coordinatewise Bochner-simple approximation therefore converges in the normed jump quotient, and the simple-function inequality passes to the integral. Equivalence of the two seminorms proves (15). $\square$

We next record the one-dimensional strong estimates used in the factorization. If $u \in \mathbb{S}^{n-1}$, put

$$H_{\varepsilon,u}f(x) = \int_{|s|>\varepsilon} f(x-su) \frac{ds}{s}, \quad A_{t,u}^+f(x) = \frac{1}{t} \int_0^t f(x-su) ds.$$

[16, Theorem 1.6] gives

$$\mathcal{J}_{2,p}^{\text{ch}}(H_{\varepsilon,u}f : \varepsilon > 0) \leq C_p \|f\|_p, \quad 1 < p < \infty, \quad (19)$$

uniformly in u . The source theorem is stated for the paired jump count; (13) gives the displayed chain form. The combination of [16, Theorem 1.1(i) and Lemmas 6.1 and 1.3] gives

$$\mathcal{J}_{2,p}^{\text{ch}}(A_{t,u}^+f : t > 0) \leq C_p \|f\|_p. \quad (20)$$

Indeed, for the density $\mathbf{1}_{[0,1]}$ the Fourier decay exponent is $1 = (1+1)/2$, because $\widehat{\mathbf{1}_{[0,1]}}(\xi) = (1 - e^{-i\xi})/(i\xi)$ (with the value at zero understood by continuity). Theorem 1.1(i) controls the dyadic jump family, while Lemma 6.1 controls its short 2-variation in the full range $1 < p < \infty$; Lemma 1.3 recombines them. The results in [16] use the paired convention, and (13) transfers the estimate to the chain seminorm displayed here. Directional forms follow by slicing along lines parallel to u .

We also require the weak endpoint for the one-dimensional Hilbert truncation path. The next lemma derives it from the available dyadic-jump and short-variation estimates.

Lemma 2.4 (Weak critical jumps for the Hilbert path). *For $f \in L^1(\mathbb{R})$,*

$$\sup_{\lambda>0} \left\| \lambda N_{\lambda}(H_{\varepsilon}f : \varepsilon > 0)^{1/2} \right\|_{L^{1,\infty}(\mathbb{R})} \leq C \|f\|_1. \quad (21)$$

The same estimate holds on lines parallel to a fixed direction in $\mathbb{R}^n$, with the other coordinates treated as parameters.

Proof. Let

$$\phi(s) = \frac{\mathbf{1}_{\{1 \leq |s| < 2\}}(s)}{s}, \quad d\nu(s) = \phi(s) ds, \quad \phi_j(s) = 2^{-j} \phi(2^{-j}s).$$

The measure ν is compactly supported and has integral zero. Its density has bounded variation, including its four jump discontinuities; hence

$$\int_{\mathbb{R}} |\phi(s+h) - \phi(s)| ds \leq \text{Var}(\phi) |h| \quad (|h| \leq 1), \quad (22)$$

while for $|h| > 1$ the left side is at most $2\|\phi\|_1 \leq 2\|\phi\|_1 |h|$. Thus the hypothesis in [16, condition (11)] holds with exponent $b = 1$. Moreover, first for compactly supported bounded f and then by L^1 approximation,

$$H_{2^k}f = \sum_{j \geq k} \phi_j * f. \quad (23)$$

The endpoint assertion in [16, Theorem 1.2(ii)], applied to ν and the ordinary isotropic dilations, therefore bounds the dyadic critical jumps of $H_{2^k}f$ weakly in L^1 .

Let $S_2(Hf)$ denote the square sum, over $k \in \mathbb{Z}$, of the 2-variations of $H_{\varepsilon}f$ on $[2^k, 2^{k+1}]$. The weak $(1, 1)$ estimate for this precise short variation is [7, Theorem 1.4]; their Hilbert transform uses the normalization factor $1/\pi$. [16, Lemma 1.3], with $\rho = 2$, gives pointwise

$$\lambda N_{\lambda}(H_{\varepsilon}f : \varepsilon > 0)^{1/2} \leq C(S_2(Hf) + \lambda N_{\lambda/3}^{\text{dyad}}(Hf)^{1/2}).$$

The quasi-triangle inequality in $L^{1,\infty}$ proves (21). Reversing the order used in [7], or changing half-open shell endpoints, does not alter a variation or a strict jump count. Finally, Fubini in coordinates $x = x_{\perp} + su$ applies the one-dimensional estimate on almost every line and integrates the resulting distribution bound, proving the directional assertion. $\square$

Lemma 2.5 (Radial profiles). *Let $1 < p < \infty$, let $f \in L^p(\mathbb{R}^n)$, let $u \in \mathbb{S}^{n-1}$, and let φ be represented by a right-continuous function that is locally of bounded variation on $(0, \infty)$, $\varphi(t) \rightarrow 0$ as $t \rightarrow \infty$, and*

$$B(\varphi) = \int_{(0, \infty)} t d|D\varphi|(t) < \infty.$$

For $P_{\varepsilon, u}^\varphi f(x) = \int_0^\infty \varphi(s) f(x - \varepsilon su) ds$,

$$\mathcal{J}_{2, p}^{\text{ch}}(P_{\varepsilon, u}^\varphi f : \varepsilon > 0) \leq C_p B(\varphi) \|f\|_p. \quad (24)$$

Suppose, more generally, that $(\varphi_u)_{u \in \mathbb{S}^{n-1}}$ is a family satisfying the preceding hypotheses, with right-continuous representatives chosen so that $(u, s) \mapsto \varphi_u(s)$ is jointly measurable on $\mathbb{S}^{n-1} \times (0, \infty)$, and suppose that $\int_{\mathbb{S}^{n-1}} B(\varphi_u) d\sigma(u) < \infty$. Then, for every $\varepsilon > 0$, the map $u \mapsto P_{\varepsilon, u}^{\varphi_u} f$ is strongly measurable in L^p , its Bochner integral is well defined, and

$$\mathcal{J}_{2, p}^{\text{ch}}\left(\int_{\mathbb{S}^{n-1}} P_{\varepsilon, u}^{\varphi_u} f d\sigma(u) : \varepsilon > 0\right) \leq C_p \left(\int_{\mathbb{S}^{n-1}} B(\varphi_u) d\sigma(u)\right) \|f\|_p. \quad (25)$$

Proof. Stieltjes integration and the zero limit give

$$\varphi(s) = - \int_{(s, \infty)} d\varphi(t), \quad P_{\varepsilon, u}^\varphi f = - \int_{(0, \infty)} t A_{\varepsilon t, u}^+ f d\varphi(t).$$

Tonelli gives the quantitative integrability estimate

$$\int_0^\infty |\varphi(s)| ds \leq \int_0^\infty \int_{(s, \infty)} d|D\varphi|(t) ds = \int_{(0, \infty)} t d|D\varphi|(t) = B(\varphi).$$

Thus Fubini is legitimate. Apply (15) and (20). In the parameterized case, $(u, s) \mapsto f(\cdot - \varepsilon su)$ is strongly measurable in L^p , because translations act strongly continuously on L^p . The preceding estimate and $\int B(\varphi_u) d\sigma(u) < \infty$ justify both Bochner integrals and the second application of (15), proving (25). $\square$

We repeatedly use the following elementary compactness principle.

Lemma 2.6 (Finite witnesses and dense parameter sets). *Let I be a finite ordered set, and suppose that $F_t^{(m)} \rightarrow F_t$ for every $t \in I$. Then, for every $\lambda > 0$ and $r \geq 1$,*

$$N_\lambda(F_t : t \in I) \leq \liminf_{m \rightarrow \infty} N_\lambda(F_t^{(m)} : t \in I), \quad V^r(F_t : t \in I) \leq \liminf_{m \rightarrow \infty} V^r(F_t^{(m)} : t \in I).$$

If a path F is continuous on compact subintervals of an interval J and $D \subset J$ is dense, then its paired strict jump counts and its r -variation on D agree with those on J .

Proof. Any finite family of strict λ -jumps of F has a positive minimum excess over λ ; all of its witnessing increments therefore persist for sufficiently large m . This proves the first inequality. Applying the same argument to a fixed finite partition and then taking the supremum proves the second. Finally, continuity permits the finitely many points in any strict-jump witness, or in any variation partition, to be perturbed into D while preserving their order and changing every increment by an arbitrarily small amount. This proves the dense-set assertions. $\square$

Lemma 2.7 (Homogeneous tail counting). *Let $\Omega \in L \log L(\mathbb{S}^{n-1})$ and put $A = \|\Omega\|_1$. For every $\delta > 0$,*

$$\sum_{s \geq 0} \|\Omega \mathbf{1}_{\{|\Omega| > 2^{\delta s} A\}}\|_1 \leq C_\delta \mathfrak{L}(\Omega). \quad (26)$$

If $\int \Omega = 0$, put $m_n = \sigma(\mathbb{S}^{n-1})$ and $E_s = \{|\Omega| > 2^{\delta s} A\}$, $c_s = m_n^{-1} \int_{E_s} \Omega d\sigma$. Then

$$\Omega_s^\# = \Omega \mathbf{1}_{E_s} - c_s, \quad \omega_s = \Omega \mathbf{1}_{E_s^c} + c_s \quad (27)$$

have mean zero and satisfy

$$\|\omega_s\|_\infty \leq C_n 2^{\delta s} A, \quad \|\omega_s\|_1 \leq 2A, \quad \|\Omega_s^\#\|_1 \leq 2\|\Omega \mathbf{1}_{E_s}\|_1. \quad (28)$$

Proof. If $A = 0$, all assertions are immediate. Hence assume $A > 0$. Tonelli gives

$$\sum_s \|\Omega \mathbf{1}_{E_s}\|_1 = \int |\Omega(\theta)| \#\{s : 2^{\delta s} A < |\Omega(\theta)|\} d\sigma(\theta),$$

and the counting function is bounded by $C_\delta(1 + \log^+(\|\Omega\|/A))$. This proves (26). The identities and bounds in (27)–(28) follow from $\int_{E_s^c} \Omega = -m_n c_s$ and $m_n |c_s| \leq \|\Omega \mathbf{1}_{E_s}\|_1 \leq A$. $\square$

Lemma 2.8 (Compact $L \log L$ data in H^1). *Suppose that F is supported in a fixed ball and $\int F = 0$. Define $\mathcal{E}(0) = 0$, while for $F \neq 0$ put*

$$\mathcal{E}(F) = \|F\|_1 + \int |F(x)| \log\left(e + \frac{|F(x)|}{\|F\|_1}\right) dx.$$

If $\mathcal{E}(F) < \infty$, then $\|F\|_{H^1(\mathbb{R}^n)} \leq C\mathcal{E}(F)$, with a constant depending only on the dimension and the supporting ball.

Proof. If $F = 0$ there is nothing to prove. Put $A = \|F\|_1$. Let $\phi \in C_c^\infty(B(0, 1))$ have integral one and let $\mathcal{M}_\phi F = \sup_{t>0} |F * \phi_t|$. On a fixed enlargement of the support, $\mathcal{M}_\phi F \leq CMF$. The refined weak estimate

$$|\{MF > 2s\}| \leq \frac{C}{s} \int_{\{|F|>s\}} |F|$$

follows by decomposing $F = F\mathbf{1}_{\{|F|\leq s\}} + F\mathbf{1}_{\{|F|>s\}}$: the maximal function of the first summand is at most s , and the weak $(1, 1)$ maximal theorem applies to the second. Layer cake, with threshold $s_0 = A$, gives

$$\begin{aligned} \int_{B(0,C)} MF &\leq CA + C \int_A^\infty \frac{1}{s} \int_{\{|F|>s/2\}} |F(x)| dx ds \\ &\leq CA + C \int |F(x)| \log^+\left(\frac{2|F(x)|}{A}\right) dx \leq C\mathcal{E}(F), \end{aligned}$$

where Tonelli justifies the interchange. Outside that ball, cancellation gives

$$|F * \phi_t(x)| \leq \int |F(y)| |\phi_t(x-y) - \phi_t(x)| dy \leq C|x|^{-n-1} \|F\|_1;$$

whenever the integrand is nonzero, t is comparable to or larger than $|x|$. Thus $\|\mathcal{M}_\phi F\|_1 \leq C\mathcal{E}(F)$. The radial maximal characterization of real H^1 is [14, Theorem 11 and the following Remark 2]; this proves the assertion. The Riesz-transform characterization $R_j : H^1 \rightarrow L^1$ is recorded in [14, Section II.2, p. 144]. $\square$

Proposition 2.9 (Unphased strong critical-jump inequality). *If $\int \Omega = 0$ and $\Omega \in L \log L(\mathbb{S}^{n-1})$, then, for every $1 < p < \infty$ and every $f \in L^p(\mathbb{R}^n)$,*

$$\mathcal{J}_{2,p}(T_{\Omega,\varepsilon}^0 f : \varepsilon > 0) \leq C_{n,p} \mathfrak{L}(\Omega) \|f\|_p. \quad (29)$$

Proof. We first work on a finite parameter set and use the chain convention; all constants are independent of that set. Decompose $\Omega = \Omega_o + \Omega_e$ into its odd and even parts. The gauge is stable under this decomposition: this is (2).

The odd component.

For odd Ω , polar coordinates give the exact rotation identity

$$T_{\Omega,\varepsilon}^0 f = \frac{1}{2} \int_{\mathbb{S}^{n-1}} \Omega(u) H_{\varepsilon,u} f d\sigma(u). \quad (30)$$

Equations (15) and (19) yield

$$\mathcal{J}_{2,p}^{\text{ch}}(T_{\Omega,\varepsilon}^0 f) \leq C_p \|\Omega\|_1 \|f\|_p. \quad (31)$$

The even component and the Riesz-transform factorization.

Assume now that Ω is even; when $n = 1$ its mean-zero condition makes it identically zero. Let K be the homogeneous principal-value distribution $K(x) = \Omega(x/|x|)|x|^{-n}$. Its action on a Schwartz function is obtained by integrating over $\{\delta < |x| < R\}$ and passing to $\delta \downarrow 0, R \uparrow \infty$; angular cancellation permits subtraction of the test function at the origin, so the limit exists and is tempered. Choose a smooth radial function Φ with $\Phi = 0$ on $[0, 1/4]$ and $\Phi = 1$ on $[3/4, \infty)$, and put $\psi = 1 - \Phi$. Normalize the Riesz transforms by $\sum_{j=1}^n R_j^2 = -I$. Define $N_j = R_j K$ by the multiplier $-i\xi_j/|\xi|$ away from the frequency origin and take its homogeneous extension as a tempered distribution.

We repeatedly use the following explicit fixed-annulus consequence of Lemma 2.8. If $\eta \in C_c^\infty((0, \infty))$ is supported in $c \leq r \leq C$, then

$$F_\eta(x) = \eta(|x|)K(x), \quad \int F_\eta = 0, \quad \|F_\eta\|_{H^1} \leq C_{\eta,n} \mathfrak{L}(\Omega). \quad (32)$$

The cancellation follows in polar coordinates. Since r stays in a fixed compact subinterval of $(0, \infty)$,

$$\|F_\eta\|_1 + \int |F_\eta| \log \left(e + \frac{|F_\eta|}{\|F_\eta\|_1} \right) \leq C_{\eta,n} \mathfrak{L}(\Omega),$$

with the zero case read separately; the asserted H^1 bound follows.

We first establish

$$N_j(x) = \frac{\nu_j(x/|x|)}{|x|^n}, \quad \nu_j \text{ odd}, \quad \|\nu_j\|_1 \leq C_n \mathfrak{L}(\Omega). \quad (33)$$

It suffices to estimate N_j on $\{3/4 \leq |x| \leq 3/2\}$. Choose a smooth radial partition $\chi_{<} + \chi_0 + \chi_{>} = 1$ with

$$\text{supp } \chi_{<} \subset [0, 2/3], \quad \text{supp } \chi_0 \subset (1/2, 3), \quad \text{supp } \chi_{>} \subset (2, \infty),$$

so that the supports of the first and last pieces are separated from the compact x -annulus. If $\rho_j(z) = c_n z_j |z|^{-n-1}$ is the Riesz kernel, angular cancellation gives for the first piece

$$\int_0^{2/3} \chi_{<}(r) \int \Omega(u) [\rho_j(x - ru) - \rho_j(x)] d\sigma(u) \frac{dr}{r},$$

whose absolute value is at most $C\|\Omega\|_1$. The last piece is absolutely bounded by the same quantity, because $|x - y| \simeq |y|$ and the remaining radial integral is $\int r^{-n} dr/r$. The middle piece is controlled by (32); hence its Riesz transform is in L^1 with the same bound. Integrating over the compact x -annulus, and then using homogeneity and parity of the Riesz multiplier, proves (33). The annular calculation gives an actual locally integrable representative away from the origin. Two homogeneous extensions could differ only by a distribution supported at the origin; at degree $-n$ that distribution is a multiple of δ_0 , which is even, whereas N_j is odd. Thus there is no origin-supported ambiguity.

Define the commutator correction

$$E_j = R_j(K\psi) - \psi N_j. \quad (34)$$

Then

$$R_j(K\Phi) = \Phi N_j - E_j. \quad (35)$$

Let $\mathcal{A}_0 = x \cdot \nabla + n$, the generator of L^1 -normalized dilations. We next prove

$$\|E_j\|_1 + \|(\mathcal{A}_0 - 1)E_j\|_1 \leq C_n \mathfrak{L}(\Omega). \quad (36)$$

For $|x| < 1/8$, formula (35) reads $E_j = -R_j(K\Phi)$ and the support separation gives an absolutely convergent bound: split at $|y| = 1$ and use $|x - y| \simeq |y|$ on the unbounded tail. For $|x| > 2$, use $E_j = R_j(K\psi)$ and subtract $\rho_j(x)$ inside the angular integral. Here ψ is supported in $[0, 3/4]$ and $\int K(y)\psi(|y|)dy = 0$ in the principal-value sense, so the mean-value theorem gives $C\|\Omega\|_1|x|^{-n-1}$. On the intervening annulus, $E_j = \Phi N_j - R_j(K\Phi)$. The first term is integrable by (33). Choose $\zeta \in C_c^\infty((1/8, 8))$ equal to one on $[1/4, 4]$. The term $\zeta K\Phi$ is controlled by (32) and $R_j : H^1 \rightarrow L^1$. The remainder $(1 - \zeta)K\Phi$ is separated from the intervening x -annulus and is absolutely integrable after applying the Riesz kernel. This proves $\|E_j\|_1 \leq C\mathfrak{Q}(\Omega)$.

Since $\mathcal{A}_0$ commutes with every R_j and annihilates the homogeneous distributions K, N_j ,

$$\mathcal{A}_0 E_j = R_j(K(r\psi'(r))) - r\psi'(r)N_j.$$

The factor $r\psi'(r)$ is supported in the fixed annulus $1/4 \leq r \leq 3/4$. The first term is controlled by (32) and $R_j : H^1 \rightarrow L^1$, and the second by (33). Together with the bound for E_j , this proves (36).

For almost every u put

$$\varphi_{j,u}(t) = t^{n-1}E_j(tu).$$

The polar slicing theorem and (36) give a locally absolutely continuous representative with

$$\varphi'_{j,u}(t) = t^{n-2}(\mathcal{A}_0 - 1)E_j(tu), \quad \int_{\mathbb{S}^{n-1}} B(\varphi_{j,u}) d\sigma(u) \leq C\mathfrak{Q}(\Omega).$$

This slicing follows directly from a one-dimensional Sobolev argument. In logarithmic polar coordinates $t = e^s$, the distributional identity

$$\partial_s \varphi_{j,u}(e^s) = e^{(n-1)s}(\mathcal{A}_0 - 1)E_j(e^s u)$$

has both sides locally integrable in (s, u) by (36). Fubini and one-dimensional $W^{1,1}$ slicing give absolute continuity on almost every ray. Returning to t and integrating $t|\varphi'_{j,u}(t)|$ gives exactly the polar integral of $|(\mathcal{A}_0 - 1)E_j|$. Also $\varphi_{j,u} \in L^1(0, \infty)$ and its derivative is integrable on $(1, \infty)$, so its limit at infinity exists and equals zero. Lemma 2.5 therefore applies.

For a distribution F write $F^{(\varepsilon)}(x) = \varepsilon^{-n}F(x/\varepsilon)$. Define

$$T_\varepsilon^0 f(x) = \int_{|y| < \varepsilon} K(y)\Phi(|y|/\varepsilon)f(x-y)dy.$$

Since $(K\Phi)^{(\varepsilon)}(x) = K(x)\Phi(|x|/\varepsilon)$,

$$T_{\Omega,\varepsilon}^0 f = (K\Phi)^{(\varepsilon)} * f - T_\varepsilon^0 f. \quad (37)$$

For Schwartz f put $g_j = -R_j f$, so $f = \sum_j R_j g_j$. Riesz transforms commute with convolution and normalized dilation; hence

$$T_{\Omega,\varepsilon}^0 f = \sum_{j=1}^n (\Phi N_j)^{(\varepsilon)} * g_j - \sum_{j=1}^n E_j^{(\varepsilon)} * g_j - T_\varepsilon^0 f. \quad (38)$$

Each of the three families is now controlled without a dyadic summation. First, with $q(t) = \Phi(t)t^{-1}\mathbf{1}_{(0,1)}(t)$, $B(q) < \infty$ and

$$((\Phi N_j)^{(\varepsilon)} * g)(x) = \frac{1}{2} \int_{\mathbb{S}^{n-1}} \nu_j(u) \int_{\mathbb{R}} \Phi(|s|/\varepsilon)g(x-su) \frac{ds}{s} d\sigma(u),$$

while

$$\int_{\mathbb{R}} \Phi(|s|/\varepsilon)g(x-su) \frac{ds}{s} = H_{\varepsilon,u}g + P_{\varepsilon,u}^q g - P_{\varepsilon,-u}^q g.$$

Thus (33), Lemmas 2.3 and 2.5, and (19) control the first sum in (38) by $C\mathfrak{L}(\Omega) \sum_j \|g_j\|_p$. Polar coordinates and the profiles $\varphi_{j,u}$ control the second sum by the same quantity. Finally,

$$T_\varepsilon^0 f(x) = \int_{\mathbb{S}^{n-1}} \Omega(u) \int_0^1 q(t) f(x - \varepsilon tu) dt d\sigma(u),$$

so Lemmas 2.3 and 2.5 give $C\|\Omega\|_1 \|f\|_p$. The scalar L^p bounds for the Riesz transforms now prove (29) for Schwartz functions.

Extension in the truncation parameter and in the input.

All estimates were uniform over finite parameter sets. The exact factorization, now with ordinary L^p norms, also bounds every fixed coordinate. If $f_m \rightarrow f$ in L^p and I is finite, all coordinates on I converge in L^p ; after a subsequence they converge almost everywhere simultaneously. Lemma 2.6 shows that finite jump counts are lower semicontinuous, so Fatou applies. Nested finite sets exhaust a countable dense $D \subset (0, \infty)$, and the jump counts increase monotonically.

We make the compatibility of those coordinates explicit. Choose Schwartz functions $f_m \rightarrow f$ in L^p . For each $\varepsilon \in D$, the right side of (38) is Cauchy in L^p ; denote its limit by $T_{\Omega, \varepsilon}^0 f$. If $\varepsilon < \eta$ lie in D , then for every m

$$T_{\Omega, \varepsilon}^0 f_m - T_{\Omega, \eta}^0 f_m = \int_{\varepsilon < |z| \leq \eta} K_\Omega(z) f_m(\cdot - z) dz. \quad (39)$$

The finite-annulus kernel is in L^1 , with norm $\|\Omega\|_1 \log(\eta/\varepsilon)$. Young's inequality passes this identity in L^p to the chosen representatives. After one null set is discarded, it holds simultaneously for all pairs in the countable set D .

Fix $\varepsilon_0 \in D$. For arbitrary $\varepsilon > 0$ define the coordinate by the signed finite-annulus difference from its chosen value at ε_0 ; the preceding identity leaves all coordinates of D unchanged. On each $0 < a < b < \infty$, Fubini locally in x , with Hölder's inequality on bounded x -sets, gives, for almost every x ,

$$\int_a^b \int_{\mathbb{S}^{n-1}} |\Omega(u)| |f(x - ru)| d\sigma(u) \frac{dr}{r} < \infty.$$

Thus the resulting path is locally absolutely continuous, and its strict jump counts on D equal those on all positive parameters by Lemma 2.6. Finally use (13).

The zero-at-infinity anchor.

We also record the anchor supplied by the factorization. At the fixed truncation $\varepsilon = 1$, the one-dimensional M. Riesz theorem for the Hilbert transform [26, Chapter II, Sections 1–2], the contraction bound for one-sided averages, and Lemma 2.5 show directly from (38) that

$$\|T_{\Omega, 1}^0 f\|_p \leq C_{n,p} \mathfrak{L}(\Omega) \|f\|_p. \quad (40)$$

The constants in the jump estimate are locally bounded in the spatial exponent: this follows from the corresponding local boundedness in the one-dimensional inputs and from the normability constants in Lemma 2.3. Thus (13) and that lemma give a homogeneous V^3 bound. For compactly supported data the sharp truncation is zero, pointwise, once ε is sufficiently large; moreover

$$\sup_{\varepsilon > 0} |T_{\Omega, \varepsilon}^0 f| \leq |T_{\Omega, 1}^0 f| + V^3(T_{\Omega, \varepsilon}^0 f : \varepsilon > 0).$$

Dominated convergence proves the L^p anchor for core data. The same maximal majorant applied to differences of core functions gives the anchored L^p completion for arbitrary f . Finite variation supplies an almost-everywhere endpoint limit, and an L^p -convergent subsequence identifies it as zero. This completes the proof. $\square$

3 The polynomial phase flag and scale-covariant annular decay

This section develops the coefficient-uniform oscillatory input. Pure output and input phases do not affect operator norms or pointwise jump counts, because

$$P(x, y) = p(x) + q(y) + \tilde{P}(x, y) \implies T_{\Omega, \varepsilon}^P f = e^{ip} T_{\Omega, \varepsilon}^{\tilde{P}} (e^{iq} f). \quad (41)$$

3.1 Algebraic flag

Let $\mathcal{P}_d$ be the space of real polynomials of degree at most d . Set

$$\Delta P(y, u, h) = P(y + u, y + h) - P(y + u, y), \quad \mathfrak{D}P = \text{Mix } \Delta P,$$

where Mix retains precisely the monomials having positive degree in both y and u .

Lemma 3.1 (Kernel of the transverse map). *For every $P \in \mathcal{P}_d$, one has $\mathfrak{D}P = 0$ if and only if*

$$P(x, y) = p(x) + q(y) + Q(x - y) + x^T B y, \quad B^T = -B. \quad (42)$$

Equivalently, simultaneous translation changes P only by pure phases.

Proof. The phases in (42) are annihilated by $\mathfrak{D}$. Conversely, $\mathfrak{D}P = 0$ implies $\partial_{y_j} \partial_{u_k} \Delta P = 0$. Differentiate in h_i at $h = 0$ and put $x = y + u$. The mixed Hessian $M_{ki} = \partial_{x_k} \partial_{y_i} P$ then satisfies $(\partial_{x_j} + \partial_{y_j}) M_{ki} = 0$, so $M(x, y) = F(x - y)$. Commutation of x - and y -derivatives gives

$$\partial_{z_\ell} F_{ki} = \partial_{z_k} F_{\ell i}, \quad \partial_{z_\ell} F_{ki} = \partial_{z_i} F_{k\ell}.$$

Hence the skew part of F is a constant skew matrix B . Put $S = F - B$. The symmetric potential is produced by the following polynomial Poincaré construction. For each k the polynomial one-form $\sum_i S_{ki}(z) dz_i$ is closed, and therefore

$$v_k(z) = \int_0^1 \sum_i S_{ki}(tz) z_i dt$$

satisfies $\partial_i v_k = S_{ki}$. Symmetry of S makes $\sum_k v_k(z) dz_k$ closed, so

$$Q(z) = - \int_0^1 \sum_k v_k(tz) z_k dt$$

is a polynomial with $-\partial_k \partial_i Q = S_{ki}$. Subtracting $Q(x - y) + x^T B y$ leaves zero mixed Hessian and therefore a sum of a pure x -polynomial and a pure y -polynomial. $\square$

Order the mixed monomials in (y, u, h) first by decreasing y -degree, then by decreasing u -degree, decreasing h -degree, decreasing total degree, and finally by a fixed lexicographic order. In each homogeneous component of $\text{im } \mathfrak{D}$, take reduced row-echelon form in this order. Merge the rows from the separate homogeneous components in the global order of their pivot monomials, and call them $F_1, \dots, F_M$. Thus no row later than F_μ contains a monomial preceding the pivot of F_μ . Choose homogeneous preimages $H_1, \dots, H_M$ with $\mathfrak{D}H_\mu = F_\mu$. Replacing $H_\mu(x, y)$ by $H_\mu(x, y) - H_\mu(y, y)$, which changes only a pure input phase, arrange

$$H_\mu(y, y) = 0. \quad (43)$$

Append to this list, in decreasing homogeneous degree and fixed monomial order, the convolution monomials $(x - y)^\alpha$ with $2 \leq |\alpha| \leq d$, and, when $d \geq 2$, a fixed basis $x_i y_j - x_j y_i$ of skew bilinear forms. Denote the full finite list again by $H_1, \dots, H_{N_d}$, and write $m_\mu = \deg H_\mu$. Modulo pure phases every $P \in \mathcal{P}_d$ has a unique first nonzero flag coordinate c_ν .

Theorem 3.2 (Global polynomial phase flag). *For each $n \in \mathbb{N}$ and $d \in \mathbb{N}_0$ there is a finite ordered family of homogeneous real polynomials $H_1, \dots, H_{N_d}$ with the following properties.*

(i) *Every $P \in \mathcal{P}_d$ has a unique representation modulo pure input and output phases,*

$$P(x, y) = p(x) + q(y) + \sum_{\mu=1}^{N_d} c_\mu(P) H_\mu(x, y). \quad (44)$$

Thus every nonpure phase has a well-defined first nonzero flag coordinate.

- (ii) The coordinates lying in $\ker \mathfrak{D}$ occur after the transverse coordinates and, modulo pure phases, span exactly

$$Q(x - y) + x^T B y, \quad B^T = -B.$$

In particular, the same flag incorporates convolution and alternating-bilinear phases as its final nonterminal strata.

- (iii) If v is the first nonzero coordinate of P and $m_v = \deg H_v$, then simultaneous dilation changes that coordinate to $c_v \rho^{m_v}$. For every $a \in \mathbb{R}^n$,

$$P(a + X, a + Y) = p_a(X) + q_a(Y) + c_v H_v(X, Y) + \sum_{\mu > v} c_{\mu, a} H_\mu(X, Y). \quad (45)$$

Thus translation preserves the first coordinate and creates only later coordinates.

- (iv) Suppose that H_v is transverse and that $y^\alpha u^\beta h^\gamma$ is the pivot monomial of $\mathfrak{D}H_v$ in the reduced-row-echelon construction above. If P has first coordinate $c_v = \pm 1$, then the coefficient of $y^\alpha u^\beta$ in $\mathfrak{D}P$ is a polynomial $a_P(h)$ of degree $|\gamma|$ whose top homogeneous coefficient tensor has norm at least one. This conclusion is uniform in every later flag coefficient.

Proof. The representation and the description of the kernel strata follow from Lemma 3.1 and the choice of homogeneous preimages and bases above. Homogeneity gives the dilation statement. For a transverse row,

$$\mathfrak{D}(P(a + \cdot, a + \cdot)) = \text{Mix}[(\mathfrak{D}P)(y + a, u, h)].$$

Translation preserves the highest y -degree term and creates only lower y -degree terms, which are later in the chosen order. Reduced echelon form prevents a later row from altering the pivot coefficient. A convolution monomial is unchanged by simultaneous translation, and a skew bilinear form changes only by pure input and output phases. This proves (iii).

For (iv), project $\mathfrak{D}P$ onto the scalar tensor component indexed by $y^\alpha u^\beta$. The pivot row contributes $\pm h^\gamma$. No later row contains the pivot monomial, by reduced row-echelon form, and no later row contains a monomial with the same y - and u -component and higher h -order, because such a monomial would precede the pivot in the defining order. Later coefficients can therefore change only the remaining terms of a_P . Its degree is $|\gamma|$, and its top homogeneous coefficient tensor has norm at least one. $\square$

For packet arguments it is convenient to use anchored representatives

$$\widehat{H}_\mu(X, Y) = H_\mu(X, Y) - H_\mu(0, Y). \quad (46)$$

They have the same flag coordinates modulo pure input phases and satisfy $\widehat{H}_\mu(0, Y) = 0$.

3.2 Polynomial oscillation estimates

We give the elementary estimates needed for rough annular decay.

Lemma 3.3 (Sublevel and negative moments). *Let $k \geq 1$, let $p(t) = a_k t^k + \dots + a_0$, $a_k \neq 0$, on an interval I of length $L > 0$, and let $\eta > 0$. Then*

$$|\{t \in I : |p(t)| \leq \eta\}| \leq C_k(\eta/|a_k|)^{1/k}.$$

Consequently, if $0 < \sigma < 1/k$, then, uniformly in c and the lower coefficients,

$$\int_I |p(t) - c|^{-\sigma} dt \leq C_{k, \sigma} |a_k|^{-\sigma} L^{1-k\sigma}. \quad (47)$$

If a is a polynomial of degree k on $\mathbb{R}^n$ whose top homogeneous coefficient norm is at least one, then, for every $R > 0$ and every fixed $C_0 < \infty$,

$$\int_{|h| \leq C_0 R} |a(h)|^{-\sigma} dh \leq C R^{n-k\sigma}, \quad 0 < \sigma < 1/k. \quad (48)$$

Here the last constant may depend on C_0 . If A_k is homogeneous of degree k and $\|A_k\|_{\text{coeff}} = 1$, then

$$\int_{\mathbb{S}^{n-1}} |A_k(\theta)|^{-\beta} d\sigma(\theta) \leq C, \quad 0 < \beta < 1/k. \quad (49)$$

Proof. Choose $k+1$ points of the sublevel set separated by a constant multiple of its measure. Lagrange interpolation of the leading coefficient gives $|a_k| \leq C_k \eta |E|^{-k}$, proving the sublevel estimate. Integrating the distribution function proves (47). For (48), choose a direction on which the top homogeneous part is bounded below, split $h = w + tv$, apply (47) in t , and integrate in w . On the fixed annulus $1 \leq |x| \leq 2$, perform this directional slicing with $R = 2$; the resulting bounded negative moment, followed by polar coordinates and homogeneity of the top component, proves (49). $\square$

Lemma 3.4 (Leading-coefficient van der Corput). *Let $k \geq 1$, let I be an interval of length at most $L_0 < \infty$, and let*

$$\phi(t) = a_k t^k + \cdots + a_0, \quad a_k \neq 0.$$

For every bounded-variation amplitude ψ supported in I ,

$$\left| \int e^{i\phi(t)} \psi(t) dt \right| \leq C_{k,L_0} \min(1, |a_k|^{-1/k}) (\|\psi\|_\infty + \text{Var}(\psi)).$$

The constant is independent of all lower coefficients.

Proof. For $k = 1$ this is integration by parts. For $k \geq 2$, remove the set $\{|\phi'| \leq \mu\}$, whose measure is $O((\mu/|a_k|)^{1/(k-1)})$ by Lemma 3.3. On the complement, split at the zeros of ϕ'' and integrate by parts on each monotonicity interval. This gives $C[(\mu/|a_k|)^{1/(k-1)} + \mu^{-1}]$; take $\mu = |a_k|^{1/k}$. Integration by parts against $d\psi$ gives the bounded-variation version. $\square$

Lemma 3.5 (Compact polynomial operator). *Let $R > 0$, let Q, U be cubes of diameter at most $C_0 R$, and let $\Phi(y, u)$ be a real polynomial. Suppose the largest mixed y -degree is $m \geq 1$, and among those terms the largest u -degree is $\ell \geq 1$. If the Euclidean norm M of the coefficient tensor of bidegree (m, ℓ) is nonzero, then*

$$\left\| \mathbf{1}_Q(y) \int_U e^{i\Phi(y,u)} g(u) du \right\|_2 \leq C R^n \min\{1, (R^{m+\ell} M)^{-\epsilon_{m,\ell}}\} \|g\|_2 \quad (50)$$

for every $g \in L^2(U)$, where $\epsilon_{m,\ell} > 0$ and the constant may depend on C_0 but is independent of all pure terms and all lower mixed coefficients.

Proof. Remove pure terms by unitary multiplication. Choose unit vectors v, w so that the bidegree tensor evaluated at (v, w) has modulus at least a fixed multiple of M . Slice $y = y_\perp + sv$, $u = u_\perp + tw$. The coefficient of $s^m t^\ell$ is independent of the transverse variables and has modulus comparable to M . We give the fiber calculation, including the T^*T square root that determines the exponent.

Fix $y_\perp, u_\perp$ and denote the resulting operator from the t -line to the s -line by $T_{y_\perp, u_\perp}$. Cross-sections of Q and U are intervals of length at most CR ; their characteristic functions have variation at most two. Write $p_m(t)$ for the coefficient of s^m in the sliced phase. It is a polynomial of degree ℓ whose leading coefficient has modulus at least cM . The kernel of $T_{y_\perp, u_\perp}^* T_{y_\perp, u_\perp}$ is

$$\mathbf{1}_{I_U}(t) \mathbf{1}_{I_U}(t') \int_{I_Q} e^{i[\Phi(s,t) - \Phi(s,t')]} ds.$$

After $s = R\tau$, Lemma 3.4, including its bounded-variation amplitude form, gives

$$|K(t, t')| \leq CR \min\left\{1, (R^m |p_m(t) - p_m(t')|)^{-1/m}\right\}. \quad (51)$$

Choose

$$0 < \delta < \min\{1/m, 1/\ell\}.$$

Since $\min(1, X^{-1/m}) \leq X^{-\delta}$, the negative-moment estimate (47), applied to the polynomial in t' for fixed t , yields

$$\begin{aligned} \sup_t \int |K(t, t')| dt' &\leq CR^{1-m\delta} M^{-\delta} R^{1-\ell\delta} \\ &= CR^2 (R^{m+\ell} M)^{-\delta}. \end{aligned}$$

The same estimate with t, t' interchanged follows by symmetry. The trivial kernel estimate gives CR^2 instead, so Schur's test and then the T^*T identity imply

$$\|T_{y_\perp, u_\perp}\|_{2 \rightarrow 2} \leq CR \min\{1, (R^{m+\ell} M)^{-\delta/2}\}. \quad (52)$$

Reassemble the slices, without invoking a direct-integral identification, by Minkowski's inequality in $u_\perp$ and (52). For fixed $y_\perp$, this gives

$$\|Tg(y_\perp, \cdot)\|_{L^2_s} \leq B_R \int_{\pi_{w^\perp} U} \|g(u_\perp, \cdot)\|_{L^2_t} du_\perp, \quad B_R = CR \min\{1, (R^{m+\ell} M)^{-\delta/2}\}.$$

Cauchy-Schwarz in $u_\perp$ and then integration over the $y_\perp$ -projection of Q contribute respectively $R^{(n-1)/2}$ and $R^{(n-1)/2}$. Hence

$$\|Tg\|_2 \leq CR^{n-1} B_R \|g\|_2,$$

which is (50) with $\epsilon_{m,\ell} = \delta/2$. Every bound used only the displayed leading coefficient, so all lower mixed coefficients remain unrestricted. $\square$

For $R > 0$ and $1/2 \leq u \leq 1$, put

$$A_{R,u}^{P,\Theta} f(x) = \int_{uR < |x-y| < R} e^{iP(x,y)} K_\Theta(x-y) f(y) dy. \quad (53)$$

Proposition 3.6 (Normalized rough annular decay). *For every flag position μ there are $a_\mu > 0$ and $C_{n,d}$ such that, if the first nonzero flag coordinate of P is ± 1 at position μ , then, for $R \geq 1$ and $\Theta \in L^\infty(\mathbb{S}^{n-1})$,*

$$\sup_{1/2 \leq u \leq 1} \|A_{R,u}^{P,\Theta}\|_{2 \rightarrow 2} \leq C_{n,d} R^{-a_\mu} \|\Theta\|_\infty. \quad (54)$$

Every later coefficient is unrestricted.

Proof. We treat the three parts of the flag.

For a transverse pivot, partition the input into cubes of side comparable to R ; the corresponding output enlargements have bounded overlap. On one input cube Q , write $A_Q = A_{R,u} \mathbf{1}_Q$. Its output is supported in $X_Q = Q + B(0, R)$, and the family $\{X_Q\}$ has bounded overlap. The kernel of $A_Q^* A_Q$ is

$$L_Q(y, y+h) = \mathbf{1}_Q(y) \mathbf{1}_Q(y+h) \int e^{i[P(x,y+h)-P(x,y)]} \overline{k_{R,u}(x-y)} k_{R,u}(x-y-h) dx,$$

where $k_{R,u} = K_\Theta \mathbf{1}_{\{uR < |\cdot| < R\}}$. Put $x = y + v$ and remove the pure y - and v -terms in the phase difference. Let v be the current transverse flag position. Since its normalized coefficient is ± 1 and all earlier coordinates vanish,

$$\mathfrak{D}P(y, v, h) = \pm F_v(y, v, h) + \sum_{v < \mu \leq M} c_\mu F_\mu(y, v, h). \quad (55)$$

Let $y^\alpha v^\beta h^\gamma$ be the pivot monomial of F_v , and put

$$m = |\alpha|, \quad \ell = |\beta|, \quad k = |\gamma|.$$

Because Mix retains only terms of positive y - and v -degree, while $\Delta P(y, v, 0) = 0$, one has $m, \ell, k \geq 1$. Thus every use below of the compact-operator and negative-moment lemmas lies in their stated positive-degree range. Write $\mathsf{T}_{m,\ell}(h)$ for the coefficient tensor of bidegree (m, ℓ) in (y, v) in the mixed part of

$P(y + v, y + h) - P(y + v, y)$, and let $a(h)$ be its (α, β) -component. By Theorem 3.2(iv), a has degree k , the coefficient of h^y is exactly ± 1 , and no later flag coefficient can change that coefficient or introduce a term of higher h -degree in the same component. Consequently

$$\|\mathbf{T}_{m,\ell}(h)\|_{\text{coeff}} \geq |a(h)|, \quad (56)$$

and no mixed bidegree preceding (m, ℓ) occurs. Thus (m, ℓ) is the leading bidegree to which Lemma 3.5 applies, uniformly in all unrestricted later coefficients. For fixed h , define

$$(E_h G)(y) = \mathbf{1}_Q(y) \mathbf{1}_Q(y + h) \int_{|v| \leq R} e^{i[P(y+v, y+h) - P(y+v, y)]} G(v) dv.$$

To apply Lemma 3.5, extend $G \mathbf{1}_{\{|v| \leq R\}}$ by zero to a cube of diameter $O(R)$; this leaves the operator unchanged. Put

$$g_h(v) = \overline{k_{R,u}(v)} k_{R,u}(v - h).$$

Thus $L_Q(y, y + h) = E_h g_h(y)$. Lemma 3.5, followed by (56), gives

$$\|E_h\|_{2 \rightarrow 2} \leq CR^n \min\{1, (R^{m+\ell} |a(h)|)^{-\epsilon}\}.$$

The actual input g_h satisfies, uniformly in h and in $u \in [1/2, 1]$,

$$\|g_h\|_2 \leq \|k_{R,u}\|_\infty \|k_{R,u}\|_2 \leq CR^{-3n/2} \|\Theta\|_\infty^2. \quad (57)$$

Consequently

$$\int |L_Q(y, y + h)|^2 dy \leq CR^{-n} \|\Theta\|_\infty^4 \min\{1, (R^{m+\ell} |a(h)|)^{-2\epsilon}\}.$$

Choose $0 < \sigma < \min(2\epsilon, 1/k)$ and integrate in $|h| \lesssim R$. Using $\min(1, X^{-2\epsilon}) \leq X^{-\sigma}$ and (48), we obtain

$$\iint |L_Q(y, y + h)|^2 dy dh \leq C \|\Theta\|_\infty^4 R^{-\sigma(m+\ell+k)}. \quad (58)$$

The left side is the square of the Hilbert–Schmidt norm of $A_Q^* A_Q$. Thus

$$\|A_Q^* A_Q\|_{2 \rightarrow 2} \leq \|A_Q^* A_Q\|_{\text{HS}} \leq C \|\Theta\|_\infty^2 R^{-\sigma(m+\ell+k)/2},$$

and the $A_Q^* A_Q$ identity supplies the second square root:

$$\|A_Q\|_{2 \rightarrow 2} \leq C \|\Theta\|_\infty R^{-\sigma(m+\ell+k)/4}.$$

Finally, bounded overlap of the X_Q and orthogonality of the input pieces $\mathbf{1}_Q f$ reassemble the cubes and prove (54) with a positive exponent independent of the later coefficients.

For a convolution pivot, after pure phases are removed write $P(x, x - z) = Q(z) - x^T B z$. Let Q_s be the first nonzero homogeneous convolution component; its coefficient norm is at least one. Polar coordinates give

$$A_{R,u} f = \int_{\mathbb{S}^{n-1}} \Theta(\theta) \int_{uR}^R e^{i[Q(r\theta) - r x^T B \theta]} f(x - r\theta) \frac{dr}{r} d\sigma(\theta).$$

On lines parallel to θ this is a one-dimensional convolution multiplier; the skew modulation depends only on the transverse line since $\theta^T B \theta = 0$. After $r = R\rho$, Lemma 3.4 and Plancherel give the fiber norm

$$C \min\{1, (R^s |Q_s(\theta)|)^{-1/s}\}.$$

Choose $0 < \beta < 1/s$. The last display is bounded by $CR^{-s\beta} |Q_s(\theta)|^{-\beta}$. Minkowski in θ and (49) therefore give the required angular integral explicitly: writing $Q_s = M_s A_s$ with $\|A_s\|_{\text{coeff}} = 1$ and $M_s \geq 1$,

$$\int_{\mathbb{S}^{n-1}} |Q_s(\theta)|^{-\beta} d\sigma(\theta) = M_s^{-\beta} \int_{\mathbb{S}^{n-1}} |A_s(\theta)|^{-\beta} d\sigma(\theta) \leq C_{n,s,\beta}.$$

Thus Minkowski gives $CR^{-s\beta}\|\Theta\|_\infty$. All lower homogeneous components of Q occur only as lower coefficients in van der Corput and are unrestricted.

For a skew pivot, the skew matrix has norm at least a dimensional constant. After orthogonal block diagonalization one two-dimensional block is βJ with $|\beta| \gtrsim 1$. For the associated twisted convolution

$$Sg(u, v) = \int k(a, b) e^{i\beta(ub - va)} g(u - a, v - b) da db,$$

Fourier transformation in v makes $q = \eta + \beta u$ invariant. The fiber kernel is $K_q(u, s) = \widehat{k}^b(u - s, q - \beta(u + s))$; the change of variables $(u, s) \mapsto (a = u - s, \rho = q - \beta(u + s))$ has Jacobian $2|\beta|$. Hence $\|S\|_{2 \rightarrow 2} \leq C|\beta|^{-1/2}\|k\|_2$. Slicing the remaining $n - 2$ variables gives a two-dimensional kernel $k_{z'}$ for each $z' \in \mathbb{R}^{n-2}$. Minkowski and Cauchy–Schwarz give the explicit bound

$$\int_{|z'| \leq R} \|k_{z'}\|_2 dz' \leq CR^{(n-2)/2} \|k_{R,u}\|_2 \leq CR^{-1} \|\Theta\|_2.$$

Therefore $\|A_{R,u}\|_{2 \rightarrow 2} \leq CR^{-1} \|\Theta\|_2$, hence $CR^{-1} \|\Theta\|_\infty$. This proves the proposition in every flag stratum. There are only finitely many strata. $\square$

Lemma 3.7 (Short variation from partial-annulus decay). *Under the hypotheses of Proposition 3.6, after reducing the exponent if necessary, every $f \in L^2(\mathbb{R}^n)$ satisfies*

$$\|V^2(A_{R,u}^{P,\Theta} f : 1/2 \leq u \leq 1)\|_2 \leq CR^{-b_\mu} \|\Theta\|_\infty \|f\|_2 \quad (59)$$

for some $b_\mu > 0$.

Proof. Let $F(u) = A_{R,u} f$. Notice that $F(1) = 0$. For test functions,

$$F'(u, x) = -u^{-1} \int_{\mathbb{S}^{n-1}} \Theta(\theta) e^{iP(x, x - uR\theta)} f(x - uR\theta) d\sigma(\theta),$$

so $\sup_u \|F'(u)\|_2 \leq C\|\Theta\|_1 \|f\|_2$. If $a = \sup_u \|F(u)\|_2$ and $b = \sup_u \|F'(u)\|_2$, the fundamental theorem of calculus and the anchor at 1 give pointwise

$$\sup_{1/2 \leq u \leq 1} |F(u, x)|^2 \leq 2 \int_{1/2}^1 |F(v, x)| |F'(v, x)| dv.$$

Integration in x and Cauchy–Schwarz therefore give $\|\sup_u F(u)\|_2 \leq Ca^{1/2} b^{1/2}$. For every partition,

$$\sum |F(u_i) - F(u_{i-1})|^2 \leq 2 \sup_u |F(u)| \int_{1/2}^1 |F'(u)| du.$$

To record the exponents, put

$$M(x) = \sup_{1/2 \leq u \leq 1} |F(u, x)|, \quad A(x) = \int_{1/2}^1 |F'(u, x)| du.$$

The preceding partition estimate says $V^2(F)^2 \leq 2MA$. Minkowski gives $\|A\|_2 \leq b/2$, while the anchored fundamental theorem of calculus gives $\|M\|_2 \leq Ca^{1/2} b^{1/2}$. Consequently

$$\|V^2(F)\|_2^2 \leq 2\|M\|_2 \|A\|_2 \leq Ca^{1/2} b^{3/2},$$

and hence $\|V^2(F)\|_2 \leq Ca^{1/4} b^{3/4}$. Insert (54); since $\|\Theta\|_1 \leq C_n \|\Theta\|_\infty$, one may take $b_\mu = a_\mu/4$. $\square$

Corollary 3.8 (Scale-covariant form). *Let P be a real polynomial of degree at most d whose first flag coordinate is $c_\mu \neq 0$ of degree m_μ . Let $\Theta \in L^\infty(\mathbb{S}^{n-1})$. For a shell radius $R > 0$, packet side $q > 0$, and truncation parameter $\varepsilon > 0$, set*

$$L_\mu = |c_\mu| R^{m_\mu}, \quad t = \log_2(R/q).$$

If $L_\mu \geq 1$, then

$$\|A_{R,u}^{P,\Theta}\|_{2 \rightarrow 2} + \|V_u^2(A_{R,u}^{P,\Theta})\|_{2 \rightarrow 2} \leq C \|\Theta\|_\infty L_\mu^{-\beta_\mu} \quad (60)$$

for some $\beta_\mu > 0$. Under the normalizing dilation $x = \rho X$, $y = \rho Y$, $\rho = |c_\mu|^{-1/m_\mu}$,

$$R \mapsto \tilde{R} = R/\rho = L_\mu^{1/m_\mu}, \quad q \mapsto \tilde{q} = q/\rho, \quad \varepsilon \mapsto \tilde{\varepsilon} = \varepsilon/\rho, \quad \log_2(\tilde{R}/\tilde{q}) = t. \quad (61)$$

If $h_\rho(Y) = h(\rho Y)$, then

$$\|h_\rho\|_1 = \rho^{-n} \|h\|_1, \quad \|h_\rho\|_2 = \rho^{-n/2} \|h\|_2, \quad \|h_\rho\|_\infty = \|h\|_\infty. \quad (62)$$

Thus a normalized bound $\tilde{R}^{-a_\mu}$ becomes $L_\mu^{-a_\mu/m_\mu}$. For any packet family h_Q supported on q -cubes, put $\tilde{Q} = \rho^{-1}Q$ and $\tilde{h}_Q(Y) = h_Q(\rho Y)$. Then

$$\|\tilde{h}_Q\|_1 \leq \Lambda |\tilde{Q}| \iff \|h_Q\|_1 \leq \Lambda |Q|,$$

so the density condition is invariant under the same rescaling.

Proof. Homogeneity of K_Θ gives the exact conjugacy

$$A_{R,u}^{P,\Theta} h(\rho X) = A_{R/\rho,u}^{P_\rho,\Theta} h_\rho(X).$$

Here and below $P_\rho(X, Y) = P(\rho X, \rho Y)$ and $h_\rho(Y) = h(\rho Y)$. The displayed transformations and norm identities follow by change of variables. Apply Proposition 3.6 and Lemma 3.7 at radius $\tilde{R}$ and take $\beta_\mu = \min(a_\mu, b_\mu)/m_\mu$. $\square$

4 Far-field strong estimates and triangular induction

We convert the scale-covariant annular decay into an angular $L \log L$ estimate and close the finite induction over the phase flag. Proposition 2.9 provides the terminal unphased estimate, and Section 3 provides oscillatory decay on every nonterminal stratum.

Call a phase *normalized nonterminal* if its first nonzero flag coordinate is ± 1 . Since the flag is finite, there is a number

$$a_0 = a_0(n, d) > 0 \quad (63)$$

not exceeding the exponent in Proposition 3.6 at any nonterminal flag position. If there is no such position, this section is vacuous and we set $a_0 = 1$.

4.1 Angular interpolation and short variation

Lemma 4.1 (q' -tracked partial-annulus estimates). *Let P be a normalized nonterminal phase, let $R \geq 1$, and let $1 < q \leq 2$. Suppose that $\Theta \in L^q(\mathbb{S}^{n-1})$. Uniformly in every later coefficient of P ,*

$$\sup_{1/2 \leq u \leq 1} \|A_{R,u}^{P,\Theta}\|_{2 \rightarrow 2} \leq C_{n,d} R^{-a_0/q'} \|\Theta\|_q, \quad q' = \frac{q}{q-1}. \quad (64)$$

Moreover, for every $1 < p < \infty$ put

$$\gamma_p = \frac{a_0}{2 \max(p, p')} > 0 \quad (65)$$

and let $f \in L^p(\mathbb{R}^n)$. Then

$$\sup_{1/2 \leq u \leq 1} \|A_{R,u}^{P,\Theta} f\|_p \leq C_{n,d,p} R^{-\gamma_p/q'} \|\Theta\|_q \|f\|_p, \quad (66)$$

$$\|V^2(A_{R,u}^{P,\Theta} f : 1/2 \leq u \leq 1)\|_p \leq C_{n,d,p} R^{-\gamma_p/q'} \|\Theta\|_q \|f\|_p. \quad (67)$$

All prefactors are uniform for $1 < q \leq 2$.

Proof. The trivial annular estimate is

$$\sup_u \|A_{R,u}^{P,\Theta}\|_{2 \rightarrow 2} \leq \|K_\Theta \mathbf{1}_{\{R/2 < |\cdot| < R\}}\|_1 \leq C \|\Theta\|_1. \quad (68)$$

For fixed P, R, u , the map $\Theta \mapsto A_{R,u}^{P,\Theta}$ is linear from the angular space into $\mathcal{B}(L^2)$. Complex interpolation of (68) with Proposition 3.6 gives (64). Indeed, if ϑ denotes the weight of the $L^\infty(\mathbb{S}^{n-1})$ endpoint, then

$$\frac{1}{q} = \frac{1-\vartheta}{1} + \frac{\vartheta}{\infty}, \quad \text{so} \quad \vartheta = \frac{1}{q'}.$$

Thus R^{-a_0} becomes $R^{-a_0/q'}$, with no q -dependent interpolation loss. The interpolation constants are uniform in q .

For the short variation put $F(u) = A_{R,u}^{P,\Theta} f$. Initially take f and Θ bounded. The path is absolutely continuous, $F(1) = 0$, and polar coordinates give, for almost every $u \in [1/2, 1]$,

$$F'(u, x) = -\frac{1}{u} \int_{\mathbb{S}^{n-1}} \Theta(\theta) e^{iP(x, x-uR\theta)} f(x-uR\theta) d\sigma(\theta). \quad (69)$$

Consequently, for every $1 \leq s \leq \infty$,

$$\sup_{1/2 \leq u \leq 1} \|F'(u)\|_s \leq C \|\Theta\|_1 \|f\|_s. \quad (70)$$

Set

$$a = \sup_u \|F(u)\|_2, \quad b = \sup_u \|F'(u)\|_2.$$

Since $F(1) = 0$, the fundamental theorem of calculus yields pointwise

$$\sup_u |F(u, x)|^2 \leq 2 \int_{1/2}^1 |F(v, x) F'(v, x)| dv.$$

After integration in x ,

$$\|\sup_u |F(u)|\|_2 \leq C a^{1/2} b^{1/2}. \quad (71)$$

For every partition $1/2 \leq u_0 < \dots < u_N \leq 1$,

$$\sum_{i=1}^N |F(u_i) - F(u_{i-1})|^2 \leq 2 \sup_u |F(u)| \int_{1/2}^1 |F'(v)| dv.$$

Taking the supremum over partitions, integrating, and using (71) gives

$$\|V^2(F(u) : 1/2 \leq u \leq 1)\|_2 \leq C a^{1/4} b^{3/4}. \quad (72)$$

Equations (64), (70), and $\|\Theta\|_1 \leq C_n \|\Theta\|_q$ therefore imply

$$\|V^2(F)\|_2 \leq C R^{-a_0/(4q')} \|\Theta\|_q \|f\|_2. \quad (73)$$

The sublinear operator $f \mapsto V^2(F)$ also satisfies, by (70),

$$V^2(F) \leq V^1(F) \leq \int_{1/2}^1 |F'(u)| du,$$

and hence has nondecaying strong $(1, 1)$ and (∞, ∞) bounds with norm $C\|\Theta\|_1$. For $1 < p < 2$, interpolation with (73) uses weight $2/p'$ at the L^2 endpoint and gives decay $a_0/(2p'q')$. For $2 < p < \infty$, interpolation with the strong (∞, ∞) bound uses weight $2/p$ at L^2 and gives $a_0/(2pq')$. These exponents are exactly γ_p/q' , which proves (67). The same spatial interpolation, now starting from (64), proves the stronger corresponding bound for the annular operator; weakening its exponent to γ_p/q' gives (66). Approximation removes the boundedness assumptions on f and Θ . $\square$

4.2 The far path and the q' loss

For a normalized nonterminal phase define

$$\mathcal{F}_\varepsilon^{P,\Theta} f(x) = \int_{|x-y| > \max(\varepsilon, 1)} e^{iP(x,y)} K_\Theta(x-y) f(y) dy \quad (74)$$

on bounded compactly supported data. Equivalently, put

$$B_j^{P,\Theta} = A_{2^{j+1}, 1/2}^{P,\Theta}, \quad j \geq 0. \quad (75)$$

If $2^j \leq \varepsilon \leq 2^{j+1}$, then

$$\mathcal{F}_\varepsilon^{P,\Theta} f = A_{2^{j+1}, \varepsilon/2^{j+1}}^{P,\Theta} f + \sum_{k=j+1}^{\infty} B_k^{P,\Theta} f, \quad (76)$$

whereas for $0 < \varepsilon \leq 1$ the path is the constant $\sum_{k \geq 0} B_k^{P,\Theta} f$.

We use the elementary long-short inequality

$$V^2(F_\varepsilon : \varepsilon > 0) \leq CV^2(F_{2^j} : j \in \mathbb{Z}) + C \left(\sum_{j \in \mathbb{Z}} V^2(F_\varepsilon : 2^j \leq \varepsilon \leq 2^{j+1})^2 \right)^{1/2}. \quad (77)$$

Indeed, insert the adjacent dyadic endpoints into an arbitrary variation partition. Each increment is the sum of two within-block increments and one increment between dyadic endpoints. The between-block increments form an ordered subpartition, and every dyadic block contributes at most two within-block increments. The inequality follows from $|z_1 + z_2 + z_3|^2 \leq 3 \sum |z_i|^2$.

Proposition 4.2 (q' -tracked far V^2). *Let P be normalized and nonterminal, let $1 < p < \infty$, and let $1 < q \leq 2$. Suppose that $\Theta \in L^q(\mathbb{S}^{n-1})$; no cancellation of Θ is required. Then, for every $f \in L^p(\mathbb{R}^n)$,*

$$\|\mathcal{F}_1^{P,\Theta} f\|_p + \|V^2(\mathcal{F}_\varepsilon^{P,\Theta} f : \varepsilon > 0)\|_p \leq C_{n,d,p} q' \|\Theta\|_q \|f\|_p. \quad (78)$$

The constant is uniform in q and in every unused coefficient of P .

Proof. Lemma 4.1 gives

$$\|B_j^{P,\Theta} f\|_p \leq C_{n,d,p} 2^{-\gamma_p j/q'} \|\Theta\|_q \|f\|_p. \quad (79)$$

Thus the series in (76) converges in L^p and

$$\sum_{j \geq 0} \|B_j^{P,\Theta} f\|_p \leq C_{n,d,p} q' \|\Theta\|_q \|f\|_p, \quad (80)$$

because, for $q' \geq 2$,

$$\sum_{j \geq 0} 2^{-\gamma_p j/q'} = \frac{1}{1 - 2^{-\gamma_p/q'}} \leq C_{\gamma_p} q'.$$

This proves the asserted bound for $\mathcal{F}_1$.

At the dyadic parameters,

$$\mathcal{F}_{2^j}^{P,\Theta} f = \sum_{k=j}^{\infty} B_k^{P,\Theta} f, \quad j \geq 0,$$

and the sequence is constant for $j < 0$. Hence, pointwise,

$$V^2(\mathcal{F}_{2^j}^{P,\Theta} f : j \in \mathbb{Z}) \leq \sum_{j \geq 0} |B_j^{P,\Theta} f|. \quad (81)$$

Its L^p norm is bounded by (80).

On the block $[2^j, 2^{j+1}]$, the tail in (76) is constant and the nonconstant part is $A_{2^{j+1},u}^{P,\Theta}$, $1/2 \leq u \leq 1$. Therefore (67) gives

$$\|V^2(\mathcal{F}_\varepsilon : 2^j \leq \varepsilon \leq 2^{j+1})\|_p \leq C 2^{-\gamma_P j/q'} \|\Theta\|_q \|f\|_p.$$

Pointwise, $(\sum_j V_j^2)^{1/2} \leq \sum_j V_j$; hence the L^p norm of the short term in (77) is again bounded by the geometric sum in (80). Combining the long and short parts proves (78). $\square$

4.3 Finite-trajectory extrapolation to $L \log L$

Let

$$\Phi_0(t) = (1+t) \log(1+t) - t, \quad \Psi_0(s) = e^s - s - 1.$$

These are complementary Young functions. On the finite measure space $\mathbb{S}^{n-1}$, the Luxemburg norm for Φ_0 is equivalent, with dimensional constants, to the homogeneous gauge $\mathfrak{L}$ in (1); the Luxemburg norm for Ψ_0 will be denoted by $\|\cdot\|_{\exp L}$. The Orlicz–Hölder inequality therefore gives

$$\int_{\mathbb{S}^{n-1}} |uv| d\sigma \leq 2 \|u\|_{L \log L} \|v\|_{\exp L}. \quad (82)$$

For the quantitative direction used below the assertion is trivial when $\Theta = 0$; otherwise let $A = \|\Theta\|_1$ and $B = \mathfrak{L}(\Theta)$. Since $B \geq A$ and $\Phi_0(t) \leq Ct \log(e+t)$,

$$\int_{\mathbb{S}^{n-1}} \Phi_0\left(\frac{|\Theta|}{B}\right) d\sigma \leq \frac{C}{B} \left[A + \int_{\mathbb{S}^{n-1}} |\Theta| \log^+\left(\frac{|\Theta|}{A}\right) d\sigma \right] \leq C.$$

Convexity and $\Phi_0(0) = 0$ imply $\Phi_0(t/C_1) \leq C_1^{-1} \Phi_0(t)$ for $C_1 \geq 1$. Choosing C_1 larger than the preceding absolute constant gives

$$\|\Theta\|_{L \log L} \leq C_1 \mathfrak{L}(\Theta). \quad (83)$$

Proposition 4.3 (Angular endpoint for the far path). *Let P be normalized and nonterminal, let $1 < p < \infty$, and let $\Theta \in L \log L(\mathbb{S}^{n-1})$. No cancellation of Θ is required. Then, for every $f \in L^p(\mathbb{R}^n)$,*

$$\|\mathcal{F}_1^{P,\Theta} f\|_p + \|V^2(\mathcal{F}_\varepsilon^{P,\Theta} f : \varepsilon > 0)\|_p \leq C_{n,d,p} \mathfrak{L}(\Theta) \|f\|_p. \quad (84)$$

Consequently,

$$\mathcal{J}_{2,p}(\mathcal{F}_\varepsilon^{P,\Theta} f : \varepsilon > 0) \leq C_{n,d,p} \mathfrak{L}(\Theta) \|f\|_p. \quad (85)$$

Proof. Fix f and a finite ordered set $D \subset \mathbb{Q}_+$ containing 1. On $\mathbb{C}^D$ define

$$\|z\|_{E_D} = |z_1| + V_D^2(z_\varepsilon : \varepsilon \in D). \quad (86)$$

This is a norm: vanishing variation makes the trajectory constant, and the value at 1 then detects that constant. For bounded angular data define, after complexification if necessary, the linear map

$$\mathcal{A}_{f,D} \Theta = (\mathcal{F}_\varepsilon^{P,\Theta} f)_{\varepsilon \in D} \quad \text{in } L^p(\mathbb{R}^n; E_D).$$

Proposition 4.2, uniformly in D , says

$$\|\mathcal{A}_{f,D}\|_{L^q(\mathbb{S}^{n-1}) \rightarrow L^p(E_D)} \leq C_{n,d,p} q' \|f\|_p, \quad 1 < q \leq 2. \quad (87)$$

Let $\Lambda \in [L^p(\mathbb{R}^n; E_D)]^*$ have norm one. For $s \geq 2$ take $q = s/(s-1)$ in (87). The scalar functional $\Theta \mapsto \Lambda(\mathcal{A}_{f,D}\Theta)$ is represented by a function $h_s \in L^s(\mathbb{S}^{n-1})$ satisfying

$$\|h_s\|_s \leq C s \|f\|_p. \quad (88)$$

The representatives h_s agree almost everywhere. Indeed, two of them represent the same functional on bounded angular functions; their difference lies in L^1 , and testing against its bounded complex sign shows that the difference vanishes. Denote the common representative by h .

Put $A = \|f\|_p$. If $A = 0$ there is nothing to prove. For every integer $k \geq 2$, $\|h\|_k \leq C k A$. Hence, by Tonelli and the exponential series,

$$\int_{\mathbb{S}^{n-1}} \Psi_0\left(\frac{|h|}{LA}\right) d\sigma = \sum_{k=2}^{\infty} \frac{\|h\|_k^k}{(LA)^k k!} \leq \sum_{k=2}^{\infty} \frac{(Ck/L)^k}{k!}. \quad (89)$$

Since $k^k/k! \leq C_0 e^k \sqrt{k}$, the last series is at most one once $L = L(n, d, p)$ is sufficiently large. Thus

$$\|h\|_{\exp L} \leq C_{n,d,p} \|f\|_p. \quad (90)$$

Equations (82) and (90) give

$$|\Lambda(\mathcal{A}_{f,D}\Theta)| \leq C_{n,d,p} \|\Theta\|_{L \log L} \|f\|_p.$$

Taking the supremum over Λ proves

$$\|\mathcal{A}_{f,D}\Theta\|_{L^p(E_D)} \leq C_{n,d,p} \mathfrak{L}(\Theta) \|f\|_p, \quad (91)$$

uniformly in the finite set D .

Bounded angular functions are dense in the Luxemburg $L \log L$ norm, because Φ_0 satisfies the Δ_2 condition at infinity. Thus the finite-trajectory maps extend to all $\Theta \in L \log L$. To check consistency, if $D \subset D'$ and both contain 1, coordinate restriction is a contraction from $E_{D'}$ to E_D . The two extensions agree on bounded angular functions and hence, by density, on all of $L \log L$. Thus the rational trajectory is independent of the exhausting sequence. Let finite sets D_N increase to $\mathbb{Q}_+$. The finite-set variations increase, so Fatou and (91) give the estimate on all positive rational parameters. For $0 < a < b < \infty$, finite-shell differences satisfy

$$\|(\mathcal{F}_a - \mathcal{F}_b)f\|_p \leq \|\Theta\|_1 \log(b/a) \|f\|_p, \quad (92)$$

including when $[a, b]$ meets 1: the difference is zero if $b \leq 1$, and if $a < 1 < b$ its kernel has L^1 norm $\|\Theta\|_1 \log b \leq \|\Theta\|_1 \log(b/a)$. Moreover,

$$x \mapsto \int_{a < |z| < b} |K_{\Theta}(z)| |f(x-z)| dz$$

belongs to L^p . It follows that the finite-shell path is locally absolutely continuous for almost every x , so rational parameters recover the full variation. This proves (84). Finally (10) gives (85). $\square$

Remark 4.4 (The far anchor). The construction above is anchored at infinity. For bounded Θ , take $q = 2$ in Lemma 4.1 and interpolate spatially as in its proof. The current partial annulus and the remaining annular tail then tend to zero in L^p as $\varepsilon \rightarrow \infty$. For general $\Theta \in L \log L$, choose bounded $\Theta_m \rightarrow \Theta$ in that norm. Proposition 4.3 and

$$\sup_{\varepsilon > 0} |F_{\varepsilon}| \leq |F_1| + V^2(F_{\varepsilon} : \varepsilon > 0)$$

make the approximation uniform in L^p over ε . Hence $\mathcal{F}_{\varepsilon}^{P,\Theta} f \rightarrow 0$ in L^p . Finite V^2 supplies an almost-everywhere limit at infinity, and an L^p -convergent subsequence identifies that limit as zero.

4.4 Local triangular reduction

For a normalized phase put

$$\mathcal{L}_\varepsilon^{P,\Omega} f(x) = \begin{cases} \int_{\varepsilon < |x-y| < 1} e^{iP(x,y)} K_\Omega(x-y) f(y) dy, & 0 < \varepsilon < 1, \\ 0, & \varepsilon \geq 1. \end{cases} \quad (93)$$

We record two elementary endpoint facts. First, if a path E is zero at one endpoint and $V^1(E) \leq W$, then

$$\lambda N_\lambda(E) \leq W, \quad N_\lambda(E) > 0 \implies \lambda \leq W, \quad \lambda N_\lambda(E)^{1/2} \leq W. \quad (94)$$

Indeed, the sum of the increments in any ordered family of jump pairs is at most the total 1-variation. Second, if

$$L_\varepsilon = \begin{cases} F_\varepsilon - F_1, & 0 < \varepsilon < 1, \\ 0, & \varepsilon \geq 1, \end{cases} \quad (95)$$

and $F_\varepsilon \rightarrow F_1$ as $\varepsilon \uparrow 1$, then

$$N_\lambda(L) \leq N_\lambda(F). \quad (96)$$

To see this, retain every jump pair lying below 1; if a pair crosses 1, it is necessarily the last nonzero pair and may be replaced by the pair with upper endpoint 1.

Proposition 4.5 (Triangular local reduction in the strong range). *Let P be a real polynomial of degree at most d whose first nonzero flag coordinate is $c_\nu = \pm 1$. Let $\Omega \in L^1(\mathbb{S}^{n-1})$ and let f be bounded and compactly supported. On each unit output cube, the local path (93) is, modulo pure input and output phases, the sum of*

- (i) a localized path of the form (95) whose phase has first flag coordinate strictly later than ν ;
- (ii) an error path E satisfying

$$V^1(E)(x) \leq C_{n,d} \int_{|x-y| < 1} \frac{|\Omega((x-y)/|x-y|)|}{|x-y|^{n-1}} |f(y)| dy. \quad (97)$$

The majorant in (97) has L^p norm at most $C_{n,d} \|\Omega\|_1 \|f\|_p$ for every $1 \leq p \leq \infty$.

Proof. Let the output cube be centered at a , and write $x = a + X$, $y = a + Y$. Theorem 3.2(iii) gives

$$P(a + X, a + Y) = p_a(X) + q_a(Y) + c_\nu H_\nu(X, Y) + P_{a, > \nu}(X, Y),$$

where every flag coordinate of $P_{a, > \nu}$ is later than ν . The factors $e^{ip_a(X)}$ and $e^{iq_a(Y)}$ act by unitary multiplication and may be removed. If X lies in the reference output cube and $|X - Y| < 1$, both variables belong to a fixed bounded set. Since $H_\nu(Y, Y) = 0$,

$$|H_\nu(X, Y)| \leq C_{n,d} |X - Y|. \quad (98)$$

Thus

$$|e^{ic_\nu H_\nu(X, Y)} - 1| \leq C_{n,d} |X - Y|.$$

The total 1-variation of the difference of the two nested truncation paths is therefore bounded by (97). Its kernel has L^1 norm

$$C_{n,d} \int_0^1 \int_{\mathbb{S}^{n-1}} |\Omega(\theta)| d\sigma(\theta) dr = C_{n,d} \|\Omega\|_1,$$

which proves the asserted L^p bound. Removing this difference leaves the local truncation of $P_{a, > \nu}$, namely the localized full path in (95). $\square$

4.5 Coefficient-uniform strong jumps for every phase

Proposition 4.6 (Strong polynomial critical jumps). *Let $1 < p < \infty$, let P be any real polynomial of degree at most d , and let $\Omega \in L \log L(\mathbb{S}^{n-1})$ have mean zero. Then for every $f \in L^p(\mathbb{R}^n)$,*

$$\mathcal{J}_{2,p}(T_{\Omega,\varepsilon}^P f : \varepsilon > 0) \leq C_{n,d,p} \mathfrak{Q}(\Omega) \|f\|_p, \quad (99)$$

where the constant is independent of every coefficient of P . Consequently, for every $r > 2$,

$$\|V^r(T_{\Omega,\varepsilon}^P f : \varepsilon > 0)\|_p \leq C_{n,d,p,r} \mathfrak{Q}(\Omega) \|f\|_p. \quad (100)$$

Proof. We first take bounded compactly supported data, for which every finite-shell identity is classical. We use backward induction on the finite phase flag. If every flag coordinate vanishes, then P is a sum of a pure input phase and a pure output phase. Equation (41) and Proposition 2.9 give (99).

Assume the estimate for every phase whose first coordinate is later than ν , and suppose that the first coordinate of P is $c_\nu \neq 0$, of homogeneous degree m_ν . Under $x = \rho X$, $y = \rho Y$, where

$$\rho = |c_\nu|^{-1/m_\nu}, \quad (101)$$

the first coordinate becomes ± 1 . Work first with this normalized phase. Proposition 4.3 controls its far path.

For the local path partition the output into disjoint unit cubes Q_a . Only the restriction of f to a fixed enlargement Q_a^* contributes on Q_a . Proposition 4.5 gives a later phase $P_{a,>\nu}$ and an error E_a . By (96) and the induction hypothesis,

$$\|\lambda N_\lambda(\mathcal{L}_\varepsilon^{P_{a,>\nu},\Omega}(f \mathbf{1}_{Q_a^*}) : \varepsilon > 0)^{1/2}\|_p \leq C \mathfrak{Q}(\Omega) \|f \mathbf{1}_{Q_a^*}\|_p \quad (102)$$

uniformly in a , λ , and all translated residual coefficients. Equations (94) and (97) give the same estimate for E_a , with $\|\Omega\|_1 \leq C \mathfrak{Q}(\Omega)$.

The pointwise jump quasi-triangle inequality

$$\lambda N_\lambda(F + G)^{1/2} \leq 2 \left[\frac{\lambda}{2} N_{\lambda/2}(F)^{1/2} + \frac{\lambda}{2} N_{\lambda/2}(G)^{1/2} \right] \quad (103)$$

follows from (9). Applying it on every output cube, summing the p -th powers over the disjoint Q_a , and using bounded overlap of the Q_a^* gives

$$\mathcal{J}_{2,p}(\mathcal{L}_\varepsilon^{P,\Omega} f : \varepsilon > 0) \leq C_{n,d,p} \mathfrak{Q}(\Omega) \|f\|_p. \quad (104)$$

Adding the local and far paths and using (103) proves the normalized case.

It remains to undo the dilation. If $f_\rho(Y) = f(\rho Y)$, homogeneity of the kernel gives the exact conjugacy

$$T_{\Omega,\varepsilon}^P f(\rho X) = T_{\Omega,\varepsilon/\rho}^{P_\rho} f_\rho(X). \quad (105)$$

Both the input L^p norm and the output jump L^p norm acquire the same factor $\rho^{-n/p}$; reparameterizing $\varepsilon \mapsto \varepsilon/\rho$ leaves the jump number unchanged. Thus the constant does not depend on c_ν . The finite backward induction proves (99) for every phase.

The estimates just proved hold for every spatial exponent in $(1, \infty)$, with constants locally bounded in that exponent; this is immediate from the finitely many interpolation and summation constants in the proof. Applying (13) and the strong jump-to-variation implication in Lemma 2.3 on a compact interval whose interior contains both p and 2 gives (100).

We next construct the anchored trajectory and pass to general data. For a normalized nonterminal phase, Proposition 4.3 defines the far path on L^p , including its value at 1; add the finite-shell local path (93). Their sum has the classical finite-shell increments, agrees with the far path for $\varepsilon \geq 1$, and is anchored

at infinity by Remark 4.4. For a nonnormalized nonterminal phase use the conjugacy (105). If every flag coordinate vanishes, use the anchored factorization of Proposition 2.9 and the pure modulations (41).

Choose bounded compactly supported $f_m \rightarrow f$ in L^p and bounded mean-zero angular functions

$$\Omega_m = \Omega \mathbf{1}_{\{|\Omega| \leq m\}} - \frac{1}{m_n} \int_{\mathbb{S}^{n-1}} \Omega \mathbf{1}_{\{|\Omega| \leq m\}} d\sigma, \quad m_n = \sigma(\mathbb{S}^{n-1}).$$

The Δ_2 property and $L \log L \hookrightarrow L^1$ give $\Omega_m \rightarrow \Omega$ in $L \log L$. For a normalized nonterminal phase, Proposition 4.3 makes the far coordinates for (Ω_m, f_m) converge in L^p to those for (Ω, f) , while the local coordinates converge by Young's inequality. The same conclusion follows after dilation, and in the pure case from the factorization estimates in Proposition 2.9. The finite-shell bound used here is, for $0 < a < b < \infty$,

$$\left\| \int_{a < |z| < b} e^{iP(x, x-z)} K_\Theta(z) g(x-z) dz \right\|_p \leq \|\Theta\|_1 \log(b/a) \|g\|_p.$$

Fix a finite $D \subset \mathbb{Q}_+$. A subsequence of the coordinatewise L^p convergence is almost everywhere simultaneous on D . Every finite collection of strict λ -jumps of the limiting trajectory has a positive margin and therefore persists for sufficiently large m . Hence N_λ is lower semicontinuous under this convergence, and Fatou transfers (99) and (100) to the limit, uniformly in D . Increase D to $\mathbb{Q}_+$. The finite-shell integrability used in (92) gives local absolute continuity in the truncation parameter almost everywhere, so the rational estimates recover all positive parameters. The construction above supplies the L^p and almost-everywhere anchor at infinity. $\square$

5 Separated-packet estimates for the terminal phase

All packet families in this and the following packet sections are finite; we suppress this qualifier in individual statements. Rankings and randomizations are always performed at this finite level. In the endpoint arguments, cube, shell, and parameter cutoffs are imposed explicitly and removed only after the estimates have been established, by coordinatewise convergence on finite annuli and lower semicontinuity of strict jump counts. Thus no infinite randomization or infinite-rank argument is used.

For $n \geq 2$, we use the following two microlocal estimates of Seeger [25, Lemmas 2.1 and 2.2]. Let H_j be supported where $2^{j-2} \leq |x| \leq 2^{j+2}$ and suppose

$$M_N = \sup_{j, \theta, r} \sup_{0 \leq \ell \leq N} r^{n+\ell} |\partial_r^\ell H_j(r\theta)| < \infty. \quad (106)$$

For an integer $s > 3$ and a fixed $0 < \kappa < 1$, Seeger's construction [25, equation (2.2)] is $H_j = \Gamma_j^s + (H_j - \Gamma_j^s)$. Put $L(Q) = m$ when $2^{m-1} < \ell(Q) \leq 2^m$. If the cubes have disjoint interiors, $F_m = \sum_{L(Q)=m} f_Q$, and $\text{supp } f_Q \subset Q$, $\|f_Q\|_1 \leq \Lambda|Q|$, then, for $0 < \kappa < 1$,

$$\left\| \sum_j \Gamma_j^s * F_{j-s} \right\|_2^2 \leq C M_0^2 2^{-s(1-\kappa)} \Lambda \sum_Q \|f_Q\|_1. \quad (107)$$

If additionally $\text{supp } b_Q \subset Q$, $\int b_Q = 0$, $\ell(Q) = 2^{j-s}$, $N \geq n+1$, and $0 \leq \epsilon \leq 1$, then

$$\|(H_j - \Gamma_j^s) * b_Q\|_1 \leq C_N \left[M_0 2^{-s\epsilon} + M_N 2^{s(n+(\epsilon-\kappa)N)} \right] \|b_Q\|_1. \quad (108)$$

The constants may depend on the fixed value of κ and on Seeger's fixed auxiliary cutoffs, but not on j, s , the cubes, or the functions. These are precisely the support hypotheses in [25, Lemmas 2.1–2.2]: the first lemma requires no cancellation, whereas the second uses the displayed mean-zero condition. In dimension one, the one-shell consequences below follow directly from cancellation and the radial derivative bounds; the unphased weak endpoint is supplied by Lemma 2.4.

We also need a geometric observation.

Lemma 5.1 (Thin radial layers). *Fix $0 < c_- < c_+ < \infty$. Let $\Theta \in L^\infty(\mathbb{S}^{n-1})$, let $R > 0$, let $t \in \mathbb{N}_0$, put $q = R2^{-t}$, and let $\Lambda \geq 0$. Suppose $d_U \in L^1$ are supported on pairwise disjoint q -cubes and satisfy $\|d_U\|_1 \leq \Lambda|U|$. If $E \subset [c_-R, c_+R]$ is a union of M_E intervals of total length $|E|$, then*

$$\left\| K_\Theta \mathbf{1}_{\{|z| \in E\}} * \sum_U d_U \right\|_\infty \leq C_{n, c_-, c_+} \Lambda \|\Theta\|_\infty \left(\frac{|E|}{R} + M_E 2^{-t} \right), \quad (109)$$

$$\left\| K_\Theta \mathbf{1}_{\{|z| \in E\}} * \sum_U d_U \right\|_1 \leq C_{n, c_-, c_+} \|\Theta\|_1 \frac{|E|}{R} \sum_U \|d_U\|_1. \quad (110)$$

Here and in the proof the implicit constants depend only on n, c_-, c_+ .

Proof. For fixed x , every contributing cube lies in a Cq -neighborhood of the translated radial set. That neighborhood has volume at most $CR^{n-1}(|E| + M_E q)$. Disjointness and the mass-density bound prove (109). Equation (110) is Young's inequality and $\int_E dr/r \leq C|E|/R$. $\square$

Lemma 5.2 (Variational Rademacher–Menshov). *For a finite sequence $g_1, \dots, g_M$ in L^2 ,*

$$\left\| V^2 \left(\sum_{\ell \leq m} g_\ell : 0 \leq m \leq M \right) \right\|_2 \leq C \log(2 + M) \sup_{\varepsilon_\ell \in \{-1, 0, 1\}} \left\| \sum_\ell \varepsilon_\ell g_\ell \right\|_2. \quad (111)$$

Proof. Pad the sequence with zeros to length 2^L and let $\mathcal{D}_r$ be the fixed partition of $\{1, \dots, 2^L\}$ into dyadic index intervals of length 2^r , $0 \leq r \leq L$. Put $G_I = \sum_{\ell \in I} g_\ell$. The canonical dyadic decomposition of an arbitrary index interval uses at most two members of each $\mathcal{D}_r$. If a collection of index intervals is pairwise disjoint, then all dyadic pieces appearing in their canonical decompositions are pairwise disjoint. Consequently, for every fixed variation partition, Minkowski in the outer finite ℓ^2 sum gives the pointwise estimate

$$\left(\sum_j \left| \sum_{\ell \in J_j} g_\ell \right|^2 \right)^{1/2} \leq \sqrt{2} \sum_{r=0}^L \left(\sum_{I \in \mathcal{D}_r} |G_I|^2 \right)^{1/2}.$$

The right side is independent of the variation partition, so taking the supremum proves the deterministic domination

$$V^2 \left(\sum_{\ell \leq m} g_\ell : 0 \leq m \leq 2^L \right) \leq \sqrt{2} \sum_{r=0}^L \left(\sum_{I \in \mathcal{D}_r} |G_I|^2 \right)^{1/2}. \quad (112)$$

For a fixed level, independent Rademacher signs ϵ_I and Fubini give

$$\left\| \left(\sum_{I \in \mathcal{D}_r} |G_I|^2 \right)^{1/2} \right\|_2^2 = \mathbb{E} \left\| \sum_{I \in \mathcal{D}_r} \epsilon_I G_I \right\|_2^2.$$

Each randomized block sum is $\sum_\ell \epsilon_\ell g_\ell$ with $\epsilon_\ell \in \{-1, 1\}$ fixed on its dyadic block; hence its norm is bounded by the supremum in (111). Take L^2 norms in (112) and sum the $L+1 \leq C \log(2+M)$ levels. $\square$

Because Seeger's microlocal kernels Γ_j^s have noncompact spatial support, localization and ranking are carried out with the original compactly supported kernels H_j .

Lemma 5.3 (Fixed-sign full-kernel estimate). *Let $n \geq 2$, $N = 4n + 1$, and let H_j be supported where $2^{j-2} \leq |x| \leq 2^{j+2}$ and satisfy (106). Let $\mathcal{Q}$ be a finite family of cubes with disjoint interiors, and suppose*

$$\text{supp } b_Q \subset Q, \quad \int b_Q = 0, \quad \|b_Q\|_1 \leq \Lambda|Q|.$$

For $s \in \mathbb{N}$, $s \geq 5$, pair H_j with the cubes of side 2^{j-s} . Then, for every fixed choice $|c_Q| \leq 1$,

$$\left\| \sum_j H_j * \sum_{\ell(Q)=2^{j-s}} c_Q b_Q \right\|_2^2 \leq C_n M_N^2 2^{-s/24} \Lambda \sum_Q \|b_Q\|_1. \quad (113)$$

Proof. By homogeneity assume $M_N = 1$ and put $\Sigma = \sum_Q \|b_Q\|_1$. Apply Seeger's decomposition with $\kappa = 1/2$ and write the left-hand sum as $F = A + R$. Equation (107) gives

$$\|A\|_2^2 \leq C2^{-s/2}\Lambda\Sigma. \quad (114)$$

Summing (108) with $\epsilon = 1/4$ gives

$$\|R\|_1 \leq C2^{-s/4}\Sigma, \quad (115)$$

because $n - (4n + 1)/4 = -1/4$. The coefficients c_Q preserve cancellation and the mass-density estimate.

Choose a fixed smooth radial cutoff equal to one on $\{2^{-2} \leq |x| \leq 2^2\}$ and dilate $C2^{-jn}$ times that cutoff. The resulting nonnegative radial majorants v_j satisfy

$$|H_j| \leq v_j, \quad \|v_j\|_1 \leq C, \quad \|\nabla v_j\|_\infty \leq C2^{-j(n+1)},$$

and set

$$\mathcal{P} = \sum_j v_j * \sum_{\ell(Q)=2^{j-s}} |b_Q|.$$

Then $|F| \leq \mathcal{P}$ and $\|\mathcal{P}\|_1 \leq C\Sigma$. It remains to prove

$$\|\mathcal{P}\|_{\text{BMO}} \leq C\Lambda. \quad (116)$$

Fix a cube R_0 . For $2^j < \ell(R_0)$, every Q contributing to the integral over R_0 lies in a fixed enlargement of R_0 : compact support of v_j confines the relevant cube mass to a ball of volume $O(2^{jn})$ about each $x \in R_0$. The cubes in Q are disjoint across all scales, so after integrating in x the total mass of all such cubes is at most the volume of that enlargement. The integral of these terms is therefore at most $C\Lambda|R_0|$. For $2^j \geq \ell(R_0)$ subtract the value at the center x_0 of R_0 . A cube which contributes to either convolution at x or x_0 lies in a ball of radius $C2^j$ about x_0 ; disjointness and $\|b_Q\|_1 \leq \Lambda|Q|$ therefore bound the relevant total mass by $C\Lambda 2^{jn}$. The mean-value theorem and the displayed gradient bound then give

$$\sup_{x \in R_0} \left| v_j * \sum_{\ell(Q)=2^{j-s}} |b_Q|(x) - v_j * \sum_{\ell(Q)=2^{j-s}} |b_Q|(x_0) \right| \leq C\Lambda\ell(R_0)2^{-j}.$$

Summation in j proves (116). The Fefferman–Stein L^1 –BMO interpolation inequality [14, Section III.5, equation (5.1)] gives

$$\|h\|_3^3 \leq C\|h\|_1\|h\|_{\text{BMO}}^2, \quad h \in L^1 \cap \text{BMO}. \quad (117)$$

Consequently,

$$\|F\|_3^3 \leq \|\mathcal{P}\|_3^3 \leq C\Lambda^2\Sigma. \quad (118)$$

Put $u = \Lambda 2^{-s/8}$ and $E = \{|R| > u\}$. By (115),

$$|E| \leq C\Lambda^{-1}2^{-s/8}\Sigma.$$

On E^c , equations (114) and (115) give

$$\int_{E^c} |F|^2 \leq 2\|A\|_2^2 + 2u\|R\|_1 \leq C2^{-3s/8}\Lambda\Sigma.$$

On E , Hölder's inequality and (118) give

$$\int_E |F|^2 \leq \|F\|_3^2 |E|^{1/3} \leq C2^{-s/24}\Lambda\Sigma.$$

This proves (113) when $M_N = 1$; rescaling the kernels restores M_N^2 . $\square$

Proposition 5.4 (Selector-free terminal packet trajectory). *There are $t_0 \in \mathbb{N}$ and $\delta_T > 0$, depending only on n , such that the following holds. Let $\Theta \in L^\infty(\mathbb{S}^{n-1})$, let $R > 0$, $t \in \mathbb{N}$, $q = R2^{-t}$, $t \geq t_0$, and let $\{U\}$ be pairwise disjoint q -cubes. Suppose*

$$\text{supp } d_U \subset U, \quad \int d_U = 0, \quad \|d_U\|_1 \leq \Lambda|U|, \quad M = \sum_U \|d_U\|_1.$$

For fixed complex coefficients $|a_U| \leq 1$, define

$$F_\varepsilon(x) = \int_{\varepsilon < |x-y| \leq 2R} K_\Theta(x-y) \sum_U a_U d_U(y) dy, \quad R \leq \varepsilon \leq 2R.$$

Then

$$\|V^2(F_\varepsilon : R \leq \varepsilon \leq 2R)\|_2^2 \leq C\|\Theta\|_\infty^2 2^{-\delta_T t} \Lambda M. \quad (119)$$

The same estimate holds after multiplication of the entire output path by $e^{i\xi \cdot x}$, for any fixed $\xi \in \mathbb{R}^n$. It also holds after replacing each $d_U(y)$ by $e^{i\eta_U \cdot y} d_U(y)$, with arbitrary $\eta_U \in \mathbb{R}^n$, provided $\int e^{i\eta_U \cdot y} d_U(y) dy = 0$ for every U . In the phase-adapted application below, the input modulation is frozen packet by packet and the resulting error is estimated separately.

Proof. Scale to $R = 1$. First consider $n = 1$. For a radially smoothed one-shell kernel H at relative scale h , cancellation gives

$$H * d_U(x) = \int_U [H(x-y) - H(x-c_U)] d_U(y) dy.$$

Since $\|\tau_z H - H\|_1 \leq |z| \|H'\|_1$ and $\|H'\|_1 \leq C\|\Theta\|_\infty h^{-1}$, summing over U yields

$$\left\| H * \sum_U a_U d_U \right\|_1 \leq C\|\Theta\|_\infty 2^{-t} h^{-1} M.$$

Lemma 5.1 gives the simultaneous L^∞ bound $C\Lambda\|\Theta\|_\infty$; hence the squared L^2 norm is at most $C\|\Theta\|_\infty^2 \Lambda M 2^{-t} h^{-1}$. Partition $[1, 2]$ into $M_t \simeq 2^{\gamma t}$ fixed cells of length $\delta = 2^{-\gamma t}$, where $0 < \gamma < \eta < 1/2$, and take $h = 2^{-\eta t}$. Within a cell $V^2 \leq V^1$. If W_ℓ is the positive convolution over that cell, Lemma 5.1 gives

$$\sup_\ell \|W_\ell\|_\infty \leq C\Lambda\|\Theta\|_\infty(\delta + 2^{-t}), \quad \sum_\ell \|W_\ell\|_1 \leq C\|\Theta\|_\infty M.$$

Consequently,

$$\int_{\mathbb{R}} \sum_\ell V^2(F_\varepsilon : \varepsilon \in I_\ell)^2 dx \leq C\Lambda\|\Theta\|_\infty^2 M 2^{-\gamma t}.$$

For a fixed choice of signs on the cells, smooth the resulting radial step function at scale h . The cancellation estimate above and the L^∞ part of Lemma 5.1 give

$$\|H * \sum_U a_U d_U\|_2^2 \leq C\Lambda\|\Theta\|_\infty^2 M 2^{-(1-\eta)t}.$$

The difference between the step function and its smoothing is supported in $O(M_t)$ intervals of total length $O(M_t h)$. Applying both bounds of Lemma 5.1 and $\|G\|_2^2 \leq \|G\|_\infty \|G\|_1$ gives

$$\|G_{\text{bdry}}\|_2^2 \leq C\Lambda\|\Theta\|_\infty^2 M(M_t h + M_t 2^{-t}) M_t h.$$

This also decays geometrically. Lemma 5.2 costs $O(t^2)$ after squaring, which is absorbed by reducing the exponent. Together with the within-cell estimate this proves (119) in dimension one.

Now assume $n \geq 2$ and choose $N = 4n + 1$, $\eta = (16N)^{-1}$ and $\gamma = \eta/2$. Partition $[1, 2]$ into $M_t \simeq 2^{\gamma t}$ fixed cells I_ℓ . Inserting the cell endpoints into a variation partition gives

$$V^2(F_\varepsilon : 1 \leq \varepsilon \leq 2) \leq CV^2 \left(\sum_{\ell \leq m} G_\ell : 0 \leq m \leq M_t \right) + C \left(\sum_\ell V^2(F_\varepsilon : \varepsilon \in I_\ell)^2 \right)^{1/2}, \quad (120)$$

where $G_\ell = K_\Theta \mathbf{1}_{\{|z| \in I_\ell\}} * \sum_U a_U d_U$.

Inside one cell, $V^2 \leq V^1$ and Lemma 5.1 gives

$$\int \sum_\ell V^2(F : I_\ell)^2 \leq C \|\Theta\|_\infty^2 \Lambda M 2^{-\gamma t}.$$

Indeed, the supremum of one absolute cell convolution is $C\Lambda \|\Theta\|_\infty (2^{-\gamma t} + 2^{-t})$, while the sum of its L^1 norms is $C\|\Theta\|_1 M$.

It remains to estimate the grid partial sums. Fix signs $\varepsilon_\ell \in \{-1, 0, 1\}$ and let $a(r) = \sum \varepsilon_\ell \mathbf{1}_{I_\ell}(r)$. Mollify a at radial scale $h = 2^{-\eta t}$. The smoothed annular kernel H satisfies $M_N(H) \leq C\|\Theta\|_\infty h^{-N}$. Apply (107)–(108) with $\kappa = 1/2$, $\epsilon = 1/4$, embedding the one-shell kernel into Seeger's family by taking $H_0 = H$ and $H_j = 0$ for $j \neq 0$. The microlocal part A obeys

$$\|A\|_2^2 \leq C\|\Theta\|_\infty^2 2^{-t/2} \Lambda M,$$

while the remainder obeys

$$\|R\|_1 \leq C\|\Theta\|_\infty 2^{-(1/4-\eta N)t} M.$$

The full smoothed convolution $S = A + R$ has $\|S\|_\infty \leq C\Lambda \|\Theta\|_\infty$ by Lemma 5.1. Put $u = \Lambda \|\Theta\|_\infty 2^{-(1/8-\eta N/2)t}$ and $E = \{|R| > u\}$. On E^c , use $|S|^2 \leq 2|A|^2 + 2u|R|$; on E , use $\int_E |S|^2 \leq \|S\|_\infty^2 |E|$ and $|E| \leq u^{-1} \|R\|_1$. This yields

$$\|S\|_2^2 \leq C\|\Theta\|_\infty^2 2^{-c_1 t} \Lambda M \quad (121)$$

for a dimensional $c_1 > 0$.

The difference between a and its mollification is supported in $O(M_t)$ intervals of total length $O(M_t h)$. Lemma 5.1 and $\|H\|_2^2 \leq \|H\|_\infty \|H\|_1$ give an additional bound $C\|\Theta\|_\infty^2 \Lambda M 2^{-2(\eta-\gamma)t}$. Therefore the right side of (121) holds for the unsmoothed signed sum with some $c_2 > 0$, uniformly in the signs. Lemma 5.2 contributes a factor $O(t^2)$ after squaring; absorb it by reducing the exponent. Together with the within-cell estimate and (120), this proves (119). $\square$

6 An anchored phase–packet induction

After removal of the parent-dependent input modulation, the phase-adapted decomposition recovers a mean-zero packet structure. We formulate the corresponding estimate in this anchored normalization.

Fix a real polynomial $P \in \mathcal{P}_d$ and $\Theta \in L^\infty(\mathbb{S}^{n-1})$. Let $t \in \mathbb{N}$, and let $R > 0$ and $q = R2^{-t}$ be dyadic. Let Q be a family of pairwise disjoint dyadic cubes, each contained in a dyadic R -cube, and for $Q \in \mathcal{Q}$ let c_Q denote its center. Suppose d_U are mean-zero descendants, supported on pairwise disjoint q -cubes $U \subset Q$, with

$$\|d_U\|_\infty \leq \Lambda, \quad M = \sum_U \|d_U\|_1. \quad (122)$$

Define the phase-adapted shell trajectory

$$\mathcal{P}_\varepsilon(x) = \sum_{Q \in \mathcal{Q}} \sum_{U \subset Q} \int_{\varepsilon < |x-y| \leq 2R} e^{i[P(x,y)-P(c_Q,y)]} K_\Theta(x-y) d_U(y) dy, \quad R \leq \varepsilon \leq 2R. \quad (123)$$

Lemma 6.1 (Fourier expansion of a low pivot). *Let H be one of the fixed anchored flag polynomials, and let X, Y, ξ range in fixed bounded cubes. Fix smooth cutoffs $\chi, \tilde{\chi}$ which equal one on the relevant cubes. Then, for every $L \in \mathbb{R}$,*

$$\chi(X) \tilde{\chi}(Y) e^{iL[H(X,Y)-H(\xi,Y)]} = \sum_{u,v \in \mathbb{Z}^n} a_{u,v}(L, \xi) e^{iu \cdot X} e^{iv \cdot Y}.$$

There exist numbers $A_{u,v}(L)$ independent of ξ such that $|a_{u,v}(L, \xi)| \leq A_{u,v}(L)$ and

$$\sum_{u,v} A_{u,v}(L)(1 + |v|) \leq C(1 + |L|)^{C_F}. \quad (124)$$

The constants are uniform in L and ξ apart from the displayed power, and may depend on the fixed cubes and cutoffs.

Proof. Extend the cutoff function periodically on a fixed torus. Repeated integration by parts in X and Y gives, uniformly in ξ , $|a_{u,v}| \leq C_N(1 + |L|)^N(1 + |u| + |v|)^{-N}$. Take the right side as $A_{u,v}(L)$, choose $N > 2n + 2$, and sum (increasing C_F if needed). $\square$

Proposition 6.2 (Anchored phase–packet principle). *There are $c_P, c_E > 0$ and t_0 , depending only on n, d , such that every trajectory (123) under the finite-packet setup above, with $P \in \mathcal{P}_d$, $\Theta \in L^\infty(\mathbb{S}^{n-1})$, and $t \geq t_0$, admits a decomposition $\mathcal{P} = G + E$ satisfying*

$$\|V^2(G_\varepsilon : R \leq \varepsilon \leq 2R)\|_2 \leq C\|\Theta\|_\infty 2^{-c_P t} (\Lambda M)^{1/2}, \quad (125)$$

$$\|V^1(E_\varepsilon : R \leq \varepsilon \leq 2R)\|_1 \leq C\|\Theta\|_1 2^{-c_E t} M. \quad (126)$$

The constants are independent of R, q , every coefficient of P , the locations of the cubes, and the parent labels Q .

Proposition 6.2 is included as a self-contained bounded-density model; it is not used in the weak-endpoint induction. Its purpose is to isolate the phase–packet mechanism from the additional concentration difficulty. The actual Calderón–Zygmund bad functions satisfy only cancellation and an average mass bound. They are handled instead by the concentration-stable estimates of Section 7.

Proof. Partition the input into dyadic R -cubes J and the output into dyadic R -cubes I . Only $O_n(1)$ relative positions (I, J) interact with the annulus. For a fixed pair, let a be the center of J and scale $x = a + RX, y = a + RY$. Put

$$M_{I,J} = \sum_{\substack{Q \subset J \\ U \subset Q}} \|d_U\|_1.$$

Each input block has only $O_n(1)$ interacting output blocks, and hence

$$\sum_{I,J} M_{I,J} \leq C_n M. \quad (127)$$

For $U \subset J$, set

$$\tilde{U} = R^{-1}(U - a), \quad \tilde{d}_U(Y) = d_U(a + RY), \quad \tilde{I} = R^{-1}(I - a).$$

Then $\tilde{U}$ has side 2^{-t} and

$$\|\tilde{d}_U\|_\infty = \|d_U\|_\infty \leq \Lambda, \quad \|\tilde{d}_U\|_1 = R^{-n} \|d_U\|_1, \quad \|\tilde{d}_U\|_1 \leq \Lambda |\tilde{U}|, \quad \sum_U \|\tilde{d}_U\|_1 = R^{-n} M_{I,J}. \quad (128)$$

Homogeneity, $K_\Theta(RZ)R^n = K_\Theta(Z)$, cancels the Jacobian, so the scaled trajectory has the unit-annulus kernel. If $\tilde{G}$ and $\tilde{E}$ are paths in the scaled variables, then

$$\|\mathbf{1}_I V^2(G)\|_{L_x^2} = R^{n/2} \|\mathbf{1}_{\tilde{I}} V^2(\tilde{G})\|_{L_x^2}, \quad \|\mathbf{1}_I V^1(E)\|_{L_x^1} = R^n \|\mathbf{1}_{\tilde{I}} V^1(\tilde{E})\|_{L_x^1}. \quad (129)$$

Thus unit-scale factors $(\Lambda R^{-n} M_{I,J})^{1/2}$ and $R^{-n} M_{I,J}$ become $(\Lambda M_{I,J})^{1/2}$ and $M_{I,J}$ in physical variables. Put

$$\Phi(X, Y) = P(a + RX, a + RY) - P(a, a + RY).$$

Then $\Phi(0, Y) = 0$. If $Q \subset J$ and $\xi_Q = (c_Q - a)/R$, the phase in (123) is exactly

$$\Phi(X, Y) - \Phi(\xi_Q, Y). \quad (130)$$

Modulo a pure output polynomial, write

$$\Phi(X, Y) = p(X) + \sum_{\mu=1}^{N_d} C_\mu \hat{H}_\mu(X, Y). \quad (131)$$

All C_μ are dimensionless; they may depend arbitrarily on a and R . The list and its length depend only on n, d .

If $N_d = 0$, the block phase is a pure output modulation and Proposition 5.4 gives the conclusion directly. We therefore assume $N_d \geq 1$ below.

Let $\beta_* = \min_\mu \beta_\mu > 0$, where the β_μ are from Corollary 3.8. Let C_F be from Lemma 6.1. Choose $B > 1 + 8C_F/\beta_*$, then choose $\varepsilon_* > 0$ so small that

$$C_F N_d \varepsilon_* < \min(\delta_T/8, 1/8), \quad \kappa_\mu = \varepsilon_* B^{\mu - N_d}. \quad (132)$$

Thus the κ_μ increase with μ and

$$C_F \sum_{r < \mu} \kappa_r \leq \frac{1}{4} \beta_* \kappa_\mu. \quad (133)$$

Starting at the first flag coordinate, call a coordinate low if $|C_\mu| \leq 2^{\kappa_\mu t}$. Expand every low factor

$$e^{iC_\mu[\hat{H}_\mu(X, Y) - \hat{H}_\mu(\xi_Q, Y)]}$$

by Lemma 6.1. To record the product estimate explicitly, let S be any set of low positions already expanded and index a term by tuples $\mathbf{u} = (u_r)_{r \in S}$ and $\mathbf{v} = (v_r)_{r \in S}$. Its parent-dependent coefficient and combined frequencies are

$$a_Q(\mathbf{u}, \mathbf{v}) = \prod_{r \in S} a_{u_r, v_r}^{(r)}(C_r, \xi_Q), \quad U(\mathbf{u}) = \sum_{r \in S} u_r, \quad V(\mathbf{v}) = \sum_{r \in S} v_r.$$

With $A(\mathbf{u}, \mathbf{v}) = \prod_{r \in S} A_{u_r, v_r}^{(r)}(C_r)$, the inequality $1 + |\sum_{r \in S} v_r| \leq \prod_{r \in S} (1 + |v_r|)$ and Lemma 6.1 give

$$\begin{aligned} \sum_{\mathbf{u}, \mathbf{v}} A(\mathbf{u}, \mathbf{v}) (1 + |V(\mathbf{v})|) &\leq \prod_{r \in S} \sum_{u_r, v_r} A_{u_r, v_r}^{(r)}(C_r) (1 + |v_r|) \\ &\leq C \prod_{r \in S} (1 + |C_r|)^{C_F} \leq C 2^{C_F t \sum_{r \in S} \kappa_r}. \end{aligned} \quad (134)$$

The same estimate without the weight controls the ordinary Fourier loss. Keeping the tuple indexing and applying the triangle inequality yields the same bound while preserving uniformity in every ξ_Q . In particular, the product of the expansions preceding position μ has coefficient ℓ^1 norm at most

$$C 2^{C_F t \sum_{r < \mu} \kappa_r}. \quad (135)$$

Suppose first that μ is the first high coordinate. After fixing a Fourier mode from the preceding low factors, the remaining common operator phase is $\sum_{r \geq \mu} C_r \hat{H}_r(X, Y)$; its first flag coordinate is C_μ . The factor obtained by evaluating the remaining phase at $X = \xi_Q$ is a unimodular input multiplier on the descendants of Q . After division by $A(\mathbf{u}, \mathbf{v})$, the parent-dependent coefficients have modulus at most one. Absorb them and the preceding unimodular factor into input multipliers $\vartheta_{Q, \mathbf{u}, \mathbf{v}}$. Disjointness of the descendants and (128) give, in scaled variables,

$$\left\| \sum_{Q, U} \vartheta_{Q, \mathbf{u}, \mathbf{v}}(Y) \tilde{d}_U(Y) \right\|_2^2 \leq \sum_U \|\tilde{d}_U\|_\infty \|\tilde{d}_U\|_1 \leq \Lambda R^{-n} M_{I, J}. \quad (136)$$

Thus Corollary 3.8 applies to one common operator phase. The fixed output modulation $e^{iU(\mathbf{u}) \cdot X}$ does not change variation. Summing the majorants produces exactly the loss in (135), and (129) restores physical scale. With $L_\mu = |C_\mu|$, we obtain

$$C\|\Theta\|_\infty |C_\mu|^{-\beta_\mu} (\Lambda M_{I,J})^{1/2} \leq C\|\Theta\|_\infty 2^{-\beta_* \kappa_\mu t} (\Lambda M_{I,J})^{1/2}$$

for the entire short trajectory. Multiplication by the loss (135) and (133) leave decay at least $2^{-\beta_* \kappa_\mu t/2}$. Square-summing over disjoint output cubes and using bounded overlap of the input R -cubes proves (125) in every high branch, with $E = 0$.

If every coordinate is low, expand all factors. For a combined tuple and a descendant U , let ζ_U be the center of $\tilde{U}$. Write

$$e^{iV(\mathbf{v}) \cdot Y} \tilde{d}_U(Y) = e^{iV(\mathbf{v}) \cdot \zeta_U} \tilde{d}_U(Y) + \tilde{r}_{U,\mathbf{v}}(Y), \quad \|\tilde{r}_{U,\mathbf{v}}\|_1 \leq C|V(\mathbf{v})|2^{-t} \|\tilde{d}_U\|_1. \quad (137)$$

The first term is still mean zero. For each fixed mode, its coefficients are fixed before evaluation in x . More precisely, divide the parent-dependent coefficient by $A(\mathbf{u}, \mathbf{v})$; the resulting packet coefficients have modulus at most one, so Proposition 5.4 applies to the scaled unit-annulus packets. Equation (134) makes the summation uniform in every parent label ξ_Q . The total Fourier loss is at most $2^{C_F N_d \varepsilon_* t}$, which by (132) leaves a positive fraction of the terminal decay. The remainders in (137) are controlled by total radial variation and Young's inequality. Using (134) and (128), their total L^1 variation in scaled variables is

$$C\|\Theta\|_1 2^{-t} 2^{C_F N_d \varepsilon_* t} R^{-n} M_{I,J}.$$

After (129) and (132), this is at most $C\|\Theta\|_1 2^{-3t/4} M_{I,J}$. The good term similarly returns with the factor $(\Lambda M_{I,J})^{1/2}$. Summing the squared L^2 bounds over disjoint output cubes and the L^1 errors by (127) proves (125)–(126). $\square$

The proof is a simultaneous phase–packet induction. At every later phase, regardless of the sizes of its coefficients, one either reaches a high coordinate and obtains scale-covariant oscillatory decay, or expands a low coordinate and proceeds. The finite hierarchy (133) ensures that Fourier losses accumulated before a later high coordinate are strictly smaller than its oscillatory gain. Translation-generated coefficients enter as unrestricted later coordinates in (131), and the flag hierarchy supplies their control.

7 Concentration-stable degree-block packet estimates

The original phase-adapted Calderón–Zygmund blocks do not satisfy the pointwise packet-density hypothesis of Proposition 6.2. This section replaces that hypothesis by estimates using only cancellation and an average L^1 mass bound. Throughout, coefficient norms are fixed Euclidean norms on the relevant bounded-degree polynomial spaces.

7.1 Polynomial incidence and rational finite type

Lemma 7.1 (Multivariable sublevels and discrete smearing). *Let $I \subset \mathbb{R}^N$ be a fixed compact box with nonempty interior, and let $\mathbf{p}$ be a nonzero vector of polynomials of degree at most D on I . Put $A = \|\mathbf{p}\|_{\text{coeff}}$. There are $c, C > 0$, depending only on I, N, D and the vector length, such that*

$$|\{x \in I : |\mathbf{p}(x)| \leq \tau A\}| \leq C\tau^c, \quad (138)$$

$$\int_I \min \left\{ 1, \left(\frac{L|\mathbf{p}(x)|}{A} \right)^{-\sigma} \right\} dx \leq CL^{-\delta_\sigma}, \quad \delta_\sigma := \frac{c\sigma}{c + \sigma} > 0, \quad (139)$$

for $0 < \tau \leq 1$, $L \geq 1$, and $\sigma > 0$. At a zero of $\mathbf{p}$, the integrand in (139) is defined to be 1.

Let $0 < q \leq 1$, let $\mathcal{Q}$ be a family of pairwise disjoint q -cubes in I , choose $y_Q \in Q$ for every $Q \in \mathcal{Q}$, and attach weights $0 \leq w_Q \leq \Lambda q^N$. If $M = \sum_{Q \in \mathcal{Q}} w_Q$, then every polynomial p on $\mathbb{R}^N \times \mathbb{R}^N$ of degree at most D and coefficient norm one satisfies, for $0 < \tau \leq 1$,

$$\sum_{Q, Q' \in \mathcal{Q}} w_Q w_{Q'} \mathbf{1}_{\{|p(y_Q, y_{Q'})| \leq \tau\}} \leq C \Lambda M (\tau + q)^c. \quad (140)$$

The same conclusions hold on a fixed finite family of rational charts whose Jacobians and denominators are bounded above and away from zero; the constants may additionally depend on this fixed chart data.

Proof. We first prove (138). A scalar polynomial of degree at most D on a fixed interval satisfies a uniform sublevel estimate. Indeed, when a sublevel set E has positive measure, one may choose $D + 1$ of its points with consecutive separations bounded below by a constant multiple of $|E|$; Lagrange interpolation then bounds the coefficient norm by $C\tau|E|^{-D}$. The case $D = 0$ is immediate, so assume henceforth that $D \geq 1$. For several variables, write $p(x', t) = \sum_{j=0}^D a_j(x')t^j$ and choose a_j with coefficient norm bounded below. If the $(N - 1)$ -variable exponent already obtained is c_{N-1} , split the base according to $|a_j(x')| \leq s$. The induction hypothesis bounds the first part by $Cs^{c_{N-1}}$, while on the complement the one-dimensional estimate bounds every fibre by $C(\tau/s)^{1/D}$. Taking $s = \tau^{1/(1+Dc_{N-1})}$ gives a positive exponent depending only on N and D , and proves (138). For a polynomial vector, one component has coefficient norm at least a fixed multiple of A , and normalization of that component completes the proof. Choose c in the statement no larger than this exponent and no larger than one half of the corresponding scalar sublevel exponent in dimension $2N$; for the fixed finite chart family, take the minimum of the finitely many resulting exponents.

For (139), set $\theta = \sigma/(c + \sigma)$ and

$$E_L = \{x \in I : |\mathbf{p}(x)| \leq AL^{-\theta}\}.$$

Equation (138) gives $|E_L| \leq CL^{-c\theta} = CL^{-\delta\sigma}$. On $I \setminus E_L$ the integrand is at most $L^{-\sigma(1-\theta)} = L^{-\delta\sigma}$. This proves (139), with a constant uniform in σ .

For (140), join two cubes when their distance is less than $10q$. Since the cubes have pairwise disjoint interiors and equal volume, a packing argument in a fixed enlargement of any one cube bounds the degree of this proximity graph by a constant depending only on N . It therefore has a bounded proper coloring. Within one color the selected points are cq -separated. Put disjoint cq -balls around them and spread each weight uniformly on its ball. The resulting density on a fixed enlargement of I is bounded by $C\Lambda$ and has integral at most M . A normalized bounded-degree polynomial has bounded gradient on a fixed enlargement of I^2 , so every bad selected pair smears into the $C(\tau + q)$ -sublevel. For any ordered pair of colors, if F, G are the two smeared densities and E is this sublevel, then

$$\iint_E F(x)G(y) dx dy \leq \min\{M^2, C\Lambda^2|E|\} \leq C\Lambda M|E|^{1/2}.$$

If $\tau + q$ is bounded below by a fixed constant, the desired estimate follows directly from $M \leq C_I \Lambda$. Otherwise apply the scalar sublevel estimate just proved, now in $2N$ variables, to E and sum the finite color pairs. The choice of c made above proves (140). Clearing bounded denominators proves the chart version. $\square$

We first fix the rational quotient and its norm. This is needed because all later rational phases have parameters which may approach degenerate values, and a qualitative assertion that a phase is not separated does not by itself give a uniform oscillatory constant.

Fix compact intervals I, J and slightly larger intervals I^*, J^* . For fixed $D_0 \in \mathbb{N}$ and $0 < \eta < 1 < M$, call

$$\Psi(u, v) = \frac{N(u, v)}{Q(u, v)} \quad (141)$$

an admissible rational presentation if N, Q are real polynomials of degree at most D_0 ,

$$\|Q\|_{\text{coeff}} \leq M, \quad \eta \leq |Q(u, v)| \leq M \quad ((u, v) \in I^* \times J^*). \quad (142)$$

The denominator in (141) is part of the presentation. In every application below it is the displayed common denominator produced by the coordinate change.

For an admissible presentation define the polynomial

$$\mathfrak{C}_Q \Psi(u, u', v) = \{Q(u, v)Q(u', v)\}^2 \partial_v \{\Psi(u, v) - \Psi(u', v)\} \quad (143)$$

and set

$$\|[\Psi]\|_{\text{rat}, Q} := \|\mathfrak{C}_Q \Psi\|_{\text{coeff}, (u, u', v)}. \quad (144)$$

This definition is invariant under addition of $F(u) + G(v)$. Conversely, if (143) vanishes, then $\partial_v \Psi(u, v)$ is independent of u , and integration on the rectangle gives $\Psi(u, v) = F(u) + G(v)$. Thus (144) is a genuine norm on the quotient by separated phases. If the same rational function is written using two admissible common denominators of bounded degree, the resulting norms are uniformly equivalent. Indeed the two difference numerators differ pointwise by a bounded, nowhere-vanishing rational factor; coefficient norms and supremum norms are equivalent on the fixed bounded-degree polynomial spaces, while (142) bounds that factor and its reciprocal.

We will also use the mixed numerator

$$\mathfrak{M}_Q \Psi(u, v) = Q(u, v)^3 \partial_u \partial_v \Psi(u, v). \quad (145)$$

It is a polynomial of uniformly bounded degree and satisfies

$$\|[\Psi]\|_{\text{rat}, Q} \geq c \|\mathfrak{M}_Q \Psi\|_{\text{coeff}}. \quad (146)$$

To verify this, differentiate (143) in u' and then put $u' = u$. Since the phase difference and its v derivative vanish on the diagonal, derivatives falling on the prefactor make no contribution, and one obtains exactly

$$\partial_{u'} \mathfrak{C}_Q \Psi(u, u', v) \Big|_{u'=u} = -Q(u, v)^4 \partial_u \partial_v \Psi(u, v) = -Q(u, v) \mathfrak{M}_Q \Psi(u, v).$$

Differentiation and diagonal restriction are bounded maps on coefficient spaces. Moreover $\|QH\|_{\text{coeff}} \geq c\|H\|_{\text{coeff}}$ for every polynomial H of the relevant bounded degree: this follows from $\|QH\|_{L^\infty(I^* \times J^*)} \geq \eta\|H\|_{L^\infty(I^* \times J^*)}$ and finite-dimensional norm equivalence. This proves (146).

Lemma 7.2 (Rational van der Corput with a BV amplitude). *Let $\psi = N/q$ on a fixed compact interval, where N, q have degree at most D_0 , $\|q\|_{\text{coeff}} \leq M$, and $\eta \leq |q| \leq M$ on a fixed enlargement. Put*

$$G = \|q^2 \psi'\|_{\text{coeff}} = \|N'q - Nq'\|_{\text{coeff}}.$$

If a is a zero-extended bounded-variation function supported in the interval, then

$$\left| \int e^{i\psi(v)} a(v) dv \right| \leq C(\|a\|_\infty + \text{Var}(a)) \min\{1, G^{-c}\}. \quad (147)$$

When $G = 0$, the minimum on the right is understood to be 1. The same assertion holds after a bounded partition into intervals, with the sum of the sectional $L^\infty + \text{BV}$ norms on the right. The constants and the positive exponent depend only on D_0, η, M and the fixed interval and its enlargement.

Proof. Only $G > 1$ requires proof; the case $G \leq 1$ is the trivial L^1 bound. Write $p = q^2 \psi'$. By Lemma 7.1, applied to p/G , the set

$$E = \{v : |p(v)| \leq G^{1/2}\}$$

has measure at most CG^{-c_0} . Its contribution is bounded by $C\|a\|_\infty G^{-c_0}$. On the complement, $|\psi'| \geq cG^{1/2}$. The numerator of ψ'' has bounded degree. Apart from the two endpoints of the ambient interval, the boundary of E is contained in the union of the zero sets of the two bounded-degree polynomials $p - G^{1/2}$ and $p + G^{1/2}$; hence E has only $O_{D_0}(1)$ components. Partition its complement at those boundary points and at the zeros of the numerator of ψ'' ; the number of intervals is therefore bounded in terms of D_0 . On each remaining interval ψ' is monotone and bounded away from zero. The first derivative van der Corput estimate, proved for a BV amplitude by one Stieltjes integration by parts, gives $C(\|a\|_\infty + \text{Var}(a))G^{-1/2}$. Summing the boundedly many pieces and decreasing the exponent proves (147). If the numerator of ψ'' vanishes identically, ψ' is already monotone and the same argument applies without this partition. $\square$

Lemma 7.3 (Coefficient-norm polynomial van der Corput). *Let I be a fixed compact interval, let $D \geq 1$, let p be a real polynomial of degree at most D , and put*

$$\Lambda_p = \|p - p(0)\|_{\text{coeff}}.$$

If a is supported in I and is of bounded variation, then

$$\left| \int_I e^{ip(t)} a(t) dt \right| \leq C_D \{\|a\|_\infty + \text{Var}(a)\} \min\{1, \Lambda_p^{-c_D}\}. \quad (148)$$

The value $p(0)$ may be replaced by any constant; equivalently, Λ_p may be any fixed coefficient norm on the space of real polynomials of degree at most D , modulo constants. When $\Lambda_p = 0$, the minimum in (148) is understood to be 1.

Proof. Apply Lemma 7.2 with denominator identically one. Differentiation is injective from the finite-dimensional quotient of real polynomials of degree at most D by the constants into the space of polynomials of degree at most $D - 1$, and therefore $\|p'\|_{\text{coeff}} \geq c_D \Lambda_p$ by finite-dimensional norm equivalence. Decreasing the exponent in (147) proves (148). $\square$

Lemma 7.4 (Uniform rational oscillatory operator). *Let I, J be fixed compact intervals, let $A : I \times J \rightarrow \mathbb{C}$ be measurable, and define $S : L^2(J) \rightarrow L^2(I)$ initially on bounded functions by*

$$Sf(u) = \int_J e^{i\Psi(u,v)} A(u,v) f(v) dv.$$

Assume that (141)–(142) hold with fixed data. Suppose $B \geq 1$ and

$$\sup_{u \in I} \int_J |A(u,v)| dv \leq B, \quad \sup_{v \in J} \int_I |A(u,v)| du \leq B.$$

Assume also that, for every $u \in I$, there is a partition of J into at most M_0 intervals on each of which

$$\|A(u, \cdot)\|_\infty + \text{Var}(A(u, \cdot)) \leq B.$$

The endpoints may depend on u . Put $L = \|\Psi\|_{\text{rat}, Q}$. Then

$$\|S\|_{2 \rightarrow 2} \leq CB^C \min\{1, L^{-c}\}. \quad (149)$$

When $L = 0$, the minimum is again understood to be 1. The constants depend only on the fixed intervals, D_0, η , the admissibility bound M , and M_0 .

Proof. The trivial part follows from Schur's test, so assume $L \geq 1$. The kernel of SS^* is

$$K(u, u') = \int_J e^{i[\Psi(u,v) - \Psi(u',v)]} A(u,v) \overline{A(u',v)} dv.$$

For fixed (u, u') , an admissible common denominator of the phase difference is $q_{u,u'}(v) = Q(u,v)Q(u',v)$; its degree, coefficient norm, and upper and lower bounds are controlled by the fixed admissibility data. Its derivative numerator is precisely (143). If

$$G(u, u') = \|\mathfrak{C}_Q \Psi(u, u', \cdot)\|_{\text{coeff}, v},$$

then refine the two partitions of J belonging to u and u' . There are at most $2M_0$ resulting intervals, and

$$\|A(u, \cdot) \overline{A(u', \cdot)}\|_{L^\infty + \text{BV}} \leq CB^2$$

on each of them. Lemma 7.2, summed over this partition, gives

$$|K(u, u')| \leq CB^2 \min\{1, G(u, u')^{-c_0}\}. \quad (150)$$

The entries of the coefficient vector defining G are polynomials in (u, u') of uniformly bounded degree, and their joint coefficient norm is exactly L . Consequently (139), applied on $I \times I$, yields

$$\iint_{I \times I} |K(u, u')|^2 du du' \leq CB^C L^{-c_1}.$$

There is no deletion of the diagonal or of any algebraically degenerate branch: all such points are included in this negative-moment integral. Finally,

$$\|S\|_{2 \rightarrow 2}^2 = \|SS^*\|_{2 \rightarrow 2} \leq \|K\|_{L^2(I \times I)}.$$

Taking the two square roots and decreasing c_1 proves (149). $\square$

7.2 Rough two-ray correlation and a flat high block

Lemma 7.5 (Universal rough polynomial correlation). *Let $n \geq 1$, let $D \geq 1$, let $K = [r_-, r_+] \Subset (0, \infty)$, and let a, b be radial BV profiles supported in K with $L^\infty + \text{BV}$ norm at most $B \geq 1$. Let $\Theta \in L^\infty(\mathbb{S}^{n-1})$ and put*

$$k_a(z) = |z|^{-n} \Theta(z/|z|) a(|z|), \quad k_b(z) = |z|^{-n} \Theta(z/|z|) b(|z|).$$

Let S be a real polynomial of degree at most D , and let Λ be the coefficient norm of its nonconstant part. Then, for every $h \in \mathbb{R}^n$, if $n \geq 2$,

$$\left| \int e^{iS(z)} \overline{k_a(z)} k_b(z-h) dz \right| \leq CB^C \|\Theta\|_\infty^2 \min\{1, (\Lambda|h|^D)^{-c}\}. \quad (151)$$

For $n \geq 2$ the trivial bound is understood when $h = 0$ or $\Lambda = 0$. In dimension one, for every h , the precise unit-scale analogue is

$$\left| \int e^{iS(z)} \overline{k_a(z)} k_b(z-h) dz \right| \leq CB^C \|\Theta\|_\infty^2 \min\{1, \Lambda^{-c}\}.$$

Lemma 7.6 (Quantitative equal-shell pullback). *Let $D \geq 1$ and let $\phi(t, r) = \sum_{m=0}^D \phi_m(t, r)$, where ϕ_m is homogeneous of degree m , and put*

$$\Gamma = \|\phi - \phi_0\|_{\text{coeff}}.$$

Let $0 < \alpha \leq 1$, $0 < H \leq C_0$, $|U_0| \leq C_0$, let $\Delta \in \mathbb{R}$, and let $q(X, Y) = \Delta + X - Y$ satisfy $c_0 \leq |q| \leq C_0$ on a fixed enlargement of a fixed rectangle. Define

$$\Psi(X, Y) = \phi\left(\frac{U_0 + HX}{q(X, Y)}, \frac{\alpha}{q(X, Y)}\right) = \sum_{m=0}^D \frac{\phi_m(U_0 + HX, \alpha)}{q(X, Y)^m}. \quad (152)$$

Use the canonical common denominator $Q = q^D$. Then

$$\|[\Psi]\|_{\text{rat}, Q} \geq c(\alpha H)^D \Gamma. \quad (153)$$

The constant depends only on D, c_0, C_0 and the fixed rectangles.

Proof of Lemma 7.6. Put $A(X) = U_0 + HX$ and use (X, q) rather than (X, Y) as polynomial coordinates; this is an invertible affine change with a fixed inverse. For $m \geq 1$, direct differentiation gives

$$q^{D+2} \partial_X \partial_Y \{\phi_m(A(X), \alpha) q^{-m}\} = q^{D-m} \{mHq \partial_t \phi_m(A(X), \alpha) - m(m+1)\phi_m(A(X), \alpha)\}. \quad (154)$$

This identity also identifies all possible cancellation between adjacent homogeneous degrees. After summing (154), the coefficient of q^{D-m} is

$$-m(m+1)\phi_m(A, \alpha) + (m+1)Hq \partial_t \phi_{m+1}(A, \alpha), \quad \phi_{D+1} = 0. \quad (155)$$

Starting with $m = D$ and descending, the triangular system (155) recovers every $\phi_m(A, \alpha)$, $1 \leq m \leq D$. Its inverse norm is bounded in terms of D and the upper bound for H . Hence

$$\|q^{D+2}\partial_X\partial_Y\Psi\|_{\text{coeff}} \geq c \left(\sum_{m=1}^D \|\phi_m(U_0 + HX, \alpha)\|_{\text{coeff}, X}^2 \right)^{1/2}. \quad (156)$$

In particular, no choice of a largest homogeneous term is being made: all homogeneous degrees are recovered simultaneously.

The affine map $t = U_0 + HX$ and evaluation at $r = \alpha$ are quantitatively injective on homogeneous two-variable polynomials. Indeed, recover $\phi_m(t, \alpha)$ from its composition with $U_0 + HX$ and then recover the coefficient of $t^j r^{m-j}$ by division by α^{m-j} . Since $|U_0| \leq C_0$, $0 < \alpha \leq 1$, and $0 < H \leq C_0$, finite-dimensional norm equivalence gives

$$\|\phi_m(U_0 + HX, \alpha)\|_{\text{coeff}, X} \geq c(\alpha H)^D \|\phi_m\|_{\text{coeff}}. \quad (157)$$

Indeed, the inverse of the affine substitution $t = U_0 + HX$ has norm $O_{m, C_0}(H^{-m})$, while evaluation at $r = \alpha$ multiplies the coefficient of $t^j r^{m-j}$ by α^{m-j} . The left side is thus at least $cH^m \alpha^m \|\phi_m\|_{\text{coeff}}$. If $\alpha H \leq 1$, this is at least the right side of (157); if $\alpha H > 1$, the bounded upper ranges of α and H allow the remaining factor to be absorbed into c . Combining (156) and (157) gives the desired lower bound for the reduced mixed numerator $q^{D+2}\Psi_{XY}$.

For the presentation with denominator $Q = q^D$, the mixed numerator $Q^3\Psi_{XY}$ differs from the reduced numerator by multiplication by q^{2D-2} . This factor is bounded above and away from zero on the enlarged rectangle, so their coefficient norms are uniformly equivalent. Equation (146) now proves (153). $\square$

Proof of Lemma 7.5. Assume $n \geq 2$. Since the radial profiles are supported in a fixed compact subset of $(0, \infty)$, kernel overlap implies $0 < |h| \leq C$. The cases $h = 0$ and $\Lambda = 0$ are covered by the stated trivial bound. Put $H = |h|$, rotate $h = He_1$, and write

$$z = (t, \rho\omega), \quad u = t/\rho, \quad v = (t - H)/\rho, \quad \omega \in \mathbb{S}^{n-2}.$$

Then exactly

$$\rho = \frac{H}{u - v}, \quad z = \frac{H(u, \omega)}{u - v}, \quad z - h = \frac{H(v, \omega)}{u - v}. \quad (158)$$

The two rough factors are respectively functions of (u, ω) and (v, ω) , and the Jacobian times the two homogeneous kernels is

$$H^{-n}|u - v|^{n-1}(1 + u^2)^{-n/2}(1 + v^2)^{-n/2} du dv d\omega. \quad (159)$$

Thus the rough angular factors are input and output L^2 functions and are never differentiated.

Decompose the transverse radius into a bounded-overlap dyadic family $\rho \simeq \alpha$, $0 < \alpha \leq C$. The finitely many bands with $\alpha \simeq 1$ are absorbed into the constants, and hence we may write $0 < \alpha \leq 1$. Put $U = \alpha u, V = \alpha v$. In these variables (159) becomes

$$H^{-n}\alpha^{n-1}|U - V|^{n-1}(\alpha^2 + U^2)^{-n/2}(\alpha^2 + V^2)^{-n/2} dU dV d\omega. \quad (160)$$

Annular support gives $|U - V| \simeq H$ in a fixed bounded set and also $\alpha^2 + U^2 \simeq \alpha^2 + V^2 \simeq 1$. Thus, after the factor α^{n-1} is removed, the kernel in (U, V) has size $O(H^{-1})$ and is supported in a strip of width $O(H)$.

Choose a fixed $\varepsilon_0 > 0$, smaller than one tenth of the lower constant in $|U - V| \simeq H$, and partition the U - and V -axes into half-open intervals of length $\varepsilon_0 H$. Each input interval interacts with only boundedly many output intervals and conversely. On an interacting pair write

$$U = U_0 + HX, \quad V = V_0 + HY, \quad \Delta = \frac{U_0 - V_0}{H};$$

then X, Y range over fixed intervals of length ε_0 and

$$q(X, Y) = \Delta + X - Y$$

is bounded above and away from zero on a fixed enlargement of their full product. This last assertion concerns the whole product rectangle, not only the support of the amplitude, and is the reason for the small fixed choice of ε_0 .

The phase on this block is exactly

$$\Psi_{\omega, U_0, V_0, H, \alpha}(X, Y) = S \left(\frac{(U_0 + HX)e_1 + \alpha\omega}{q(X, Y)} \right). \quad (161)$$

The factor $dV = H dY$ cancels the H^{-1} in the normalized kernel. The two angular factors are, up to the fixed sign of q ,

$$\Theta \left(\frac{Ue_1 + \alpha\omega}{\sqrt{U^2 + \alpha^2}} \right), \quad \Theta \left(\frac{Ve_1 + \alpha\omega}{\sqrt{V^2 + \alpha^2}} \right);$$

they are respectively functions of the output and input variables and are never differentiated. The radial factors are

$$a \left(\frac{\sqrt{U^2 + \alpha^2}}{|q|} \right), \quad b \left(\frac{\sqrt{V^2 + \alpha^2}}{|q|} \right).$$

Here is the uniformity in this assertion. The sign of q is fixed on the enlarged block. Put $s_U = (U^2 + \alpha^2)^{1/2}$ and $s_V = (V^2 + \alpha^2)^{1/2}$. Up to nonzero factors of fixed sign, the four sectional derivatives of $s_U/|q|$ and $s_V/|q|$ are therefore controlled by those of s_U/q and s_V/q . Directly,

$$\begin{aligned} \partial_X \frac{s_U}{q} &= \frac{HUq - s_U^2}{s_U q^2}, & \partial_Y \frac{s_U}{q} &= \frac{s_U}{q^2}, \\ \partial_X \frac{s_V}{q} &= -\frac{s_V}{q^2}, & \partial_Y \frac{s_V}{q} &= \frac{HVq + s_V^2}{s_V q^2}. \end{aligned}$$

Thus a derivative can vanish only through

$$HUq - s_U^2 = 0 \quad \text{or} \quad HVq + s_V^2 = 0;$$

the other two derivatives never vanish. Since $U = U_0 + HX$, $V = V_0 + HY$, and $q = \Delta + X - Y$, the two displayed critical-point equations are polynomial equations of degree at most two. Their zero sets therefore split every section into $O(1)$ monotonicity intervals (with an identically vanishing derivative requiring no split). On an interval I on which a profile argument γ is monotone, the one-dimensional BV pullback inequality gives

$$\text{Var}_I(a \circ \gamma) \leq \text{Var}_{\gamma(I)}(a) \leq \text{Var}(a),$$

and the same conclusion is immediate if γ is constant. Summing over the uniformly bounded number of intervals gives

$$\text{Var}(a \circ \gamma) + \text{Var}(b \circ \tilde{\gamma}) \leq C_D \{\text{Var}(a) + \text{Var}(b)\}.$$

The annular support gives $\sqrt{U^2 + \alpha^2} \simeq \sqrt{V^2 + \alpha^2} \simeq 1$ and $|q| \simeq 1$, so all remaining algebraic factors in (160) and their sectional derivatives are uniformly bounded. Moreover, on the block $\rho/\alpha = H/(U - V) = 1/q(X, Y)$, so a smooth dyadic cutoff in ρ/α is a smooth function of q (and a sharp half-open cutoff would contribute only a bounded number of jumps on every section). Thus the amplitude in each normalized block satisfies Lemma 7.4 with bound CB^C .

For completeness, the block synthesis has no hidden dependence on H . If T_{ij} is the operator between an interacting V -interval and U -interval and

$$(\mathcal{U}_i g)(X) = H^{1/2} g(U_i + HX), \quad (\mathcal{V}_j f)(Y) = H^{1/2} f(V_j + HY),$$

then $\mathcal{U}_i T_{ij} \mathcal{V}_j^{-1}$ is precisely the normalized operator above: the factor $dV = H dY$ cancels the kernel factor H^{-1} . If d_{in} and d_{out} are the two degrees of the block interaction graph, Cauchy–Schwarz gives

$$\|T\|_{2 \rightarrow 2} \leq (d_{\text{in}} d_{\text{out}})^{1/2} \sup_{i,j} \|T_{ij}\|_{2 \rightarrow 2}.$$

Both graph degrees are absolute constants. The same calculation with Schur’s test proves the trivial band bound $CB^C \alpha^{n-1}$.

For fixed ω , put

$$\phi_\omega(t, r) = S(te_1 + r\omega), \quad \Gamma(\omega) = \|\phi_\omega - \phi_\omega(0, 0)\|_{\text{coeff}}.$$

Formula (161) is (152). Lemma 7.6 gives, for its canonical denominator q^D ,

$$\|[\Psi_{\omega, U_0, V_0, H, \alpha}]\|_{\text{rat}, q^D} \geq c\Gamma(\omega)(\alpha H)^D.$$

Lemma 7.4 gives the fixed- ω block gain, and the preceding almost-orthogonal synthesis preserves it. We now spell out the angular synthesis. Let $I_0 \subset \mathbb{R}$ be a fixed compact interval containing every possible U - and V -coordinate on the band, and let β denote the fixed zero-extended cutoff defining this band in $\rho/\alpha = H/(U - V)$. After the factor α^{n-1} in (160) is removed, put

$$\begin{aligned} A_{\alpha, H}(U, V) &= \beta \left(\frac{H}{U - V} \right) H^{-n} (U - V)^{n-1} (\alpha^2 + U^2)^{-n/2} (\alpha^2 + V^2)^{-n/2} \\ &\quad \times a \left(\frac{H\sqrt{U^2 + \alpha^2}}{U - V} \right) b \left(\frac{H\sqrt{V^2 + \alpha^2}}{U - V} \right), \end{aligned}$$

extended by zero off the band support. Here $U - V > 0$ by (158). Define

$$\begin{aligned} (T_{\alpha, \omega} f)(U) &= \int_{I_0} e^{iS(H(Ue_1 + \alpha\omega)/(U-V))} A_{\alpha, H}(U, V) f(V) dV, \\ f_{\alpha, \omega}(V) &= \Theta \left(\frac{Ve_1 + \alpha\omega}{\sqrt{V^2 + \alpha^2}} \right) \mathbf{1}_{I_0}(V), \\ g_{\alpha, \omega}(U) &= \Theta \left(\frac{Ue_1 + \alpha\omega}{\sqrt{U^2 + \alpha^2}} \right) \mathbf{1}_{I_0}(U). \end{aligned}$$

With the convention $\langle F, G \rangle = \int F \overline{G}$, the contribution $\mathcal{I}_\alpha$ of this transverse band is exactly

$$\mathcal{I}_\alpha = \alpha^{n-1} \int_{\mathbb{S}^{n-2}} \langle T_{\alpha, \omega} f_{\alpha, \omega}, g_{\alpha, \omega} \rangle_{L^2(I_0)} d\omega. \quad (162)$$

The functions in this formula are jointly measurable, and $\omega \mapsto f_{\alpha, \omega}, g_{\alpha, \omega}$ are strongly measurable $L^2(I_0)$ -valued maps. Moreover

$$\|f_{\alpha, \omega}\|_2 + \|g_{\alpha, \omega}\|_2 \leq C\|\Theta\|_\infty.$$

The block argument above, Lemma 7.6, and Lemma 7.4 give, uniformly in ω, α, H ,

$$\|T_{\alpha, \omega}\|_{2 \rightarrow 2} \leq CB^C \min\{1, [\Gamma(\omega)(\alpha H)^D]^{-c_0}\}. \quad (163)$$

The trivial Schur bound supplies an integrable majorant in (162); hence ordinary Fubini and Cauchy–Schwarz yield

$$|\mathcal{I}_\alpha| \leq CB^C \|\Theta\|_\infty^2 \alpha^{n-1} \int_{\mathbb{S}^{n-2}} \min\{1, [\Gamma(\omega)(\alpha H)^D]^{-c_0}\} d\omega. \quad (164)$$

No measurable-field or abstract direct-integral assertion is being used.

It remains to estimate the scalar angular integral. When $n = 2$, $\mathbb{S}^0 = \{-1, 1\}$, and the two restrictions are related by $r \mapsto -r$; either restriction has coefficient norm comparable to Λ , so the required bound follows by summing the two points. Assume henceforth in this paragraph that $n \geq 3$. Split $\mathbb{S}^{n-2}$ by a fixed measurable partition subordinate to a finite rational atlas $\omega = \omega_\nu(\zeta)$; the chart Jacobians are bounded above and away from zero on fixed enlargements. In the ν -th chart let $\mathbf{g}_\nu(\zeta)$ be the vector of all nonconstant (t, r) coefficients of $S(te_1 + r\omega_\nu(\zeta))$, after multiplication by one fixed common chart denominator. The denominator and its reciprocal are bounded on the enlargement, so pointwise

$$|\mathbf{g}_\nu(\zeta)| \simeq \Gamma(\omega_\nu(\zeta)). \quad (165)$$

For each nonempty chart, the restriction map

$$S - S(0) \mapsto \mathbf{g}_\nu$$

is injective: if its nonconstant (t, r) coefficients vanish throughout the chart, then $S - S(0)$ vanishes on the corresponding open cone and therefore identically. With $A_\nu = \|\mathbf{g}_\nu\|_{\text{coeff}, \zeta}$, finite-dimensional norm equivalence and the finite atlas give

$$A_\nu \geq c\Lambda \quad \text{for every } \nu. \quad (166)$$

If $(\alpha H)^D A_\nu \geq 1$, apply (139) to $\mathbf{g}_\nu/A_\nu$, with $L = (\alpha H)^D A_\nu$ and $\sigma = c_0$; if $(\alpha H)^D A_\nu < 1$, use the trivial bound. In view of (165)–(166), summing the finitely many charts gives

$$\int_{\mathbb{S}^{n-2}} \min\{1, [\Gamma(\omega)(\alpha H)^D]^{-c_0}\} d\omega \leq C \min\{1, [\Lambda(\alpha H)^D]^{-c_1}\}. \quad (167)$$

All constants here are uniform as $\alpha, H \downarrow 0$: the normalized rectangles, denominator bounds, block-graph degrees, and numbers of sectional monotonicity intervals are fixed. Combining (164) and (167) proves the band bound

$$CB^C \|\Theta\|_\infty^2 \alpha^{n-1} \min\{1, [\Lambda(\alpha H)^D]^{-c}\}.$$

For clarity, write $A = \Lambda H^D$. The dyadic band sum is bounded by

$$\sum_{\alpha \in 2^{-\mathbb{N}_0}} \alpha^{n-1} \min\{1, (A\alpha^D)^{-c}\}.$$

If $A \leq 1$, this is $O(1)$. If $A > 1$, choose $\alpha_0 \simeq A^{-1/D}$. The bands below the threshold satisfy

$$\sum_{\alpha \leq \alpha_0} \alpha^{n-1} \lesssim A^{-(n-1)/D}.$$

After decreasing the oscillatory exponent so that $Dc < n - 1$, the remaining bands satisfy

$$\sum_{\alpha > \alpha_0} A^{-c} \alpha^{n-1-Dc} \lesssim A^{-c}.$$

Thus the complete sum is $O(\min\{1, A^{-c'}\})$, where $c' = \min\{c, (n-1)/D\} > 0$, which is the right side of (151).

When $n = 1$, split the support into the finitely many signed intervals on which the two homogeneous kernel factors and radial profiles are of bounded variation. Their product has $L^\infty + \text{BV}$ norm at most $CB^C \|\Theta\|_\infty^2$. Lemma 7.3 gives the asserted unit-scale bound uniformly in h . $\square$

For a homogeneous polynomial $H(x, y)$, let $\mathcal{D}H$ denote the vector of all nonconstant-in- z coefficients of

$$H(y + z, y') - H(y + z, y). \quad (168)$$

Lemma 7.7 (Same-shell degree-block injectivity). *For every integer $m \geq 2$, on homogeneous degree- m polynomials modulo pure phases, the map $[H] \mapsto \mathcal{D}H$ is injective. Hence $\|\mathcal{D}H\|_{\text{coeff}} \geq c_{n,m} \|[H]\|_{\text{coeff}}$.*

Proof. If (168) is independent of z , then $\partial_x H(x, y') - \partial_x H(x, y) = 0$. Thus $\partial_x H$ is independent of its second variable, and integration makes H pure. Quantitative injectivity is finite-dimensional norm equivalence. $\square$

Proposition 7.8 (Flat-measure high block). *Fix a translate B_0 of a prescribed bounded unit-scale cube, and assume that every cube below is contained in B_0 ; the constants are uniform in the location of B_0 . Let $t \geq 0$, let $\mathcal{Q}$ be a finite family of pairwise disjoint q -cubes with $q = 2^{-t}$, and, for each $Q \in \mathcal{Q}$, let ν_Q be a complex finite measure supported on Q . Assume $\Lambda \geq 0$ and*

$$|\nu_Q|(Q) \leq \Lambda|Q|, \quad M = \sum_Q |\nu_Q|(Q). \quad (169)$$

Suppose P is a real polynomial of degree at most d , its highest nonpure homogeneous block has coefficient norm $L \geq 1$, and $\Theta \in L^\infty(\mathbb{S}^{n-1})$. For every $|e_Q| \leq 1$ and every radial BV profile a , supported in a fixed compact subinterval of $(0, \infty)$ and of $L^\infty + \text{BV}$ norm at most B , where $B \geq 1$, write

$$A_a^{P, \Theta} \nu(x) = \int e^{iP(x, y)} k_a(x - y) d\nu(y).$$

Then

$$\left\| A_a^{P, \Theta} \sum_Q e_Q \nu_Q \right\|_2^2 \leq CB^C \|\Theta\|_\infty^2 \Lambda M (L^{-c} + 2^{-ct}). \quad (170)$$

Proof. After simultaneously translating the input and output variables by the center of B_0 , we may assume that B_0 lies in a fixed reference cube. This translation preserves the kernel geometry and the output L^2 norm; it also preserves the highest nonpure homogeneous block of P and its coefficient norm, since higher homogeneous blocks are pure and remain pure under translation. Put $w_Q = |\nu_Q|(Q)$. Since the cubes are disjoint and contained in B_0 , the data in (169) give

$$M = \sum_Q w_Q \leq \Lambda \sum_Q |Q| \leq \Lambda |B_0| \leq C\Lambda, \quad \sum_{Q, Q'} w_Q w_{Q'} = M^2 \leq C\Lambda M.$$

Write $d\nu_Q = e^{i\theta_Q} d|\nu_Q|$ and independently choose Y_Q with law $d|\nu_Q|/|\nu_Q|(Q)$. Convexity of the squared Hilbert norm reduces the assertion to the atomic measures $|\nu_Q|(Q) e^{i\theta_Q(Y_Q)} \delta_{Y_Q}$. Colour the cubes by the bounded proximity coloring used in Lemma 7.1.

If $1 \leq L < 2$, the trivial correlation bound and $\sum_{Q, Q'} w_Q w_{Q'} \leq C\Lambda M$ on a fixed unit block prove (170). Hence assume $L \geq 2$.

After expanding the square, the correlation phase at a selected pair is $P(y + z, y') - P(y + z, y)$. By Lemma 7.7, its nonconstant z -coefficient vector $\mathbf{C}(y, y')$ has coefficient norm $N := \|\mathbf{C}\|_{\text{coeff}} \geq cL$; lower-degree blocks cannot alter its top total-degree coefficient. Put $\mathbf{p} = \mathbf{C}/N$, so $\|\mathbf{p}\|_{\text{coeff}} = 1$, and discard pairs on which $|\mathbf{p}(y, y')| \leq L^{-1/3}$. Choose a scalar component p_i with $\|p_i\|_{\text{coeff}} \geq c$ and apply (140) to $p_i/\|p_i\|_{\text{coeff}}$. The discarded vector sublevel is contained in the resulting scalar $CL^{-1/3}$ -sublevel, so that equation bounds its weighted mass by $C\Lambda M(L^{-c} + q^c)$. On the complementary pairs the actual nonconstant z -coefficient norm is at least $NL^{-1/3} \geq cL^{2/3}$, uniformly even when lower blocks make N arbitrarily large.

If $n = 1$, the one-dimensional conclusion of Lemma 7.5 has no separation factor. It therefore gives $CB^C \|\Theta\|_\infty^2 L^{-c}$ directly on these complementary pairs. Together with the preceding bad-pair mass bound and $\sum_{Q, Q'} w_Q w_{Q'} \leq C\Lambda M$, this proves (170) in dimension one. Hence assume $n \geq 2$.

The weighted mass of pairs with $|y - y'| \leq r$ is at most $C\Lambda M(r + q)^n$ by the same smearing argument. Take $r = L^{-1/(6d_*)}$, where $d_* = \max(d, 1)$. On every remaining pair the product in (151) is at least $L^{1/2}$. The remaining correlation is therefore $CB^C \|\Theta\|_\infty^2 L^{-c}$. The total weight of all pairs in a fixed block is at most $C\Lambda M$. This proves (170), and Jensen returns to the original measures. $\square$

7.3 A one-shell phase-adapted packet theorem

Lemma 7.9 (Fourier expansion of a low homogeneous block). *Let $\mathbf{H} = (H_1, \dots, H_N)$ be a fixed basis of one homogeneous degree block modulo pure phases. Fix smooth cutoffs $\chi, \tilde{\chi}$ supported on fixed bounded cubes and equal to one on the relevant smaller cubes. Then, for every $\mathbf{C} \in \mathbb{R}^N$ and every ξ in the specified fixed cube,*

$$\chi(X)\tilde{\chi}(Y)e^{i\mathbf{C} \cdot [\mathbf{H}(X,Y) - \mathbf{H}(\xi,Y)]} = \sum_{u,v \in \mathbb{Z}^n} a_{u,v}(\mathbf{C}, \xi) e^{iu \cdot X} e^{iv \cdot Y},$$

and there are numbers $A_{u,v}(\mathbf{C})$, independent of ξ , such that $|a_{u,v}(\mathbf{C}, \xi)| \leq A_{u,v}(\mathbf{C})$ and

$$\sum_{u,v} A_{u,v}(\mathbf{C})(1 + |v|) \leq C(1 + \|\mathbf{C}\|)^{C_F}. \quad (171)$$

Proof. Extend the cutoff expression periodically to a fixed torus and integrate by parts more than $2n + 2$ times. Every derivative is bounded by a fixed power of $1 + \|\mathbf{C}\|$, uniformly on the compact set of ξ . More precisely, for $N > 2n + 2$ the integration-by-parts estimate is

$$|a_{u,v}(\mathbf{C}, \xi)| \leq C_N(1 + \|\mathbf{C}\|)^N(1 + |u| + |v|)^{-N}.$$

Taking the right side as $A_{u,v}(\mathbf{C})$ and summing proves the common-majorant assertion. $\square$

Proposition 7.10 (Phase-adapted one-shell packet principle). *There are $c_P, c_E > 0$ and t_0 , depending only on n, d , such that the following holds. Let P be a real polynomial of degree at most d , let $\Theta \in L^\infty(\mathbb{S}^{n-1})$, and let $R > 0$, $t \in \mathbb{N}$, $q = R2^{-t}$, $t \geq t_0$, and let $\mathcal{Q}$ be a finite family of pairwise disjoint dyadic q -cubes (with R and q dyadic scales), with center c_Q for each $Q \in \mathcal{Q}$. Suppose*

$$\text{supp } d_Q \subset Q, \quad \int d_Q = 0, \quad \|d_Q\|_1 \leq \Lambda|Q|, \quad M = \sum_{Q \in \mathcal{Q}} \|d_Q\|_1. \quad (172)$$

For $R \leq \varepsilon \leq 2R$ put

$$\mathcal{P}_\varepsilon(x) = \sum_{Q \in \mathcal{Q}} \int_{\varepsilon < |x-y| \leq 2R} e^{i[P(x,y) - P(c_Q,y)]} K_\Theta(x-y) d_Q(y) dy.$$

Then $\mathcal{P} = G + E$ with

$$\|V^2(G_\varepsilon : R \leq \varepsilon \leq 2R)\|_2^2 \leq C\|\Theta\|_\infty^2 2^{-c_P t} \Lambda M, \quad (173)$$

$$\|V^1(E_\varepsilon : R \leq \varepsilon \leq 2R)\|_1 \leq C\|\Theta\|_1 2^{-c_E t} M. \quad (174)$$

No pointwise bound on d_Q is assumed. Both estimates remain uniform after replacing every d_Q by $e_Q d_Q$, where $|e_Q| \leq 1$.

Proof. The optional factors e_Q may be absorbed into the packets, so suppress them. Partition input into dyadic R -cubes J and output into dyadic R -cubes I ; only $O_n(1)$ relative positions interact. For one such pair put

$$M_{I,J} = \sum_{Q \subset J} \|d_Q\|_1.$$

Every input block has boundedly many interacting output blocks, and hence

$$\sum_{I,J} M_{I,J} \leq C_n M. \quad (175)$$

Let a be the center of J , set $x = a + RX$, $y = a + RY$, and define

$$\Phi(X, Y) = P(a + RX, a + RY) - P(a, a + RY), \quad \xi_Q = \frac{c_Q - a}{R}. \quad (176)$$

Then $\Phi(0, Y) = 0$, and the packet phase is exactly $\Phi(X, Y) - \Phi(\xi_Q, Y)$. The rescaled packets $\tilde{d}_Q(Y) = d_Q(a + RY)$ are supported on disjoint dyadic 2^{-t} -cubes and satisfy

$$\|\tilde{d}_Q\|_1 = R^{-n}\|d_Q\|_1, \quad \|\tilde{d}_Q\|_1 \leq \Lambda|\tilde{Q}|, \quad \tilde{M}_{I,J} := \sum_{Q \subset J} \|\tilde{d}_Q\|_1 = R^{-n}M_{I,J}. \quad (177)$$

All normalized packet cubes lie in the fixed normalized input block $B_0 = R^{-1}(J - a)$, as required in Proposition 7.8. We prove the unit-scale estimate and suppress tildes on the packets, but retain $\tilde{M}_{I,J}$, until the last paragraph. Insert fixed smooth cutoffs which equal one on the normalized interacting input and output blocks; these are the cutoffs in Lemma 7.9.

Write the pure part of Φ as $p(X) + q(Y)$. Its mixed representative vanishes at $X = 0$, so $\Phi(0, Y) = 0$ forces q to be constant. In the anchored difference $p(X) - p(\xi_Q)$ is a common output modulation times a packet scalar and may be removed. Modulo these factors, write the common block phase in descending nonpure homogeneous degrees as

$$\Phi(X, Y) = H_1(X, Y) + \cdots + H_N(X, Y), \quad \deg H_1 > \cdots > \deg H_N \geq 2,$$

so the anchored phase is $\Phi(X, Y) - \Phi(\xi_Q, Y)$. Put $C_r = \|H_r\|_{\text{coeff}}$.

If $N = 0$, only the removed output modulation and packet scalars remain; Proposition 5.4 gives the required unit-block bound with its positive terminal decay exponent and $E = 0$. The rescaling and bounded-overlap summation in the last paragraph then give (173). Hence assume $N \geq 1$ below.

Let c_F, C_B be respectively the decay exponent and the BV power in Proposition 7.8, let δ_T be from Proposition 5.4, and let C_F be enlarged to cover the Fourier losses in Lemma 7.9. Decrease c_F if necessary so that $0 < c_F \leq 1$. Set

$$\theta = \frac{c_F}{16(C_B + 1)}.$$

Choose $B_* > 1$ so large that

$$\frac{2C_F}{B_* - 1} \leq \min\{\theta/2, c_F/8\}, \quad (178)$$

and then choose $0 < \kappa_* \leq 1$ so small that, with

$$\kappa_r = \kappa_* B_*^{r-N}, \quad S_r = \sum_{i < r} \kappa_i, \quad \gamma_r = \theta \kappa_r, \quad (179)$$

one has

$$2C_F \sum_{r=1}^N \kappa_r \leq \min\{\delta_T/2, 1/2\}. \quad (180)$$

Then $S_r \leq \kappa_r/(B_* - 1)$ and

$$\gamma_r - 2C_F S_r \geq \gamma_r/2, \quad (181)$$

$$c_F \kappa_r - C_B \gamma_r - 2C_F S_r \geq c_F \kappa_r/2, \quad (182)$$

$$c_F - C_B \gamma_r - 2C_F S_r \geq c_F/2. \quad (183)$$

Call block r low if $C_r \leq 2^{\kappa_r t}$ and expand it by Lemma 7.9. The product of the preceding low expansions has weighted coefficient ℓ^1 norm at most

$$C 2^{C_F S_r t}. \quad (184)$$

Suppose r is the first high block. After a preceding Fourier tuple is fixed, denote its parent-dependent Fourier coefficient by $a_Q(\mathbf{u}, \mathbf{v})$ and the product of the corresponding common majorants in Lemma 7.9 by $A(\mathbf{u}, \mathbf{v})$. Then $|a_Q/A| \leq 1$, and summation of A gives exactly (184). Absorb this normalized scalar together with the factor

$$e^{-i \sum_{i \geq r} H_i(\xi_Q, Y)} e^{i \mathbf{v} \cdot Y}$$

into the complex measure on packet Q . This multiplier need not be constant; total variation and (172) are unchanged. The remaining common operator phase has H_r as its highest nonpure block, and all lower blocks are unrestricted.

Partition $[1, 2]$ into $K_r \simeq 2^{\gamma_r t}$ radial cells. Inside a cell use $V^2 \leq V^1$. Lemma 5.1, whose proof uses only the average mass cap, gives

$$\sup_{\ell} \|W_{\ell}\|_{\infty} \leq C \Lambda \|\Theta\|_{\infty} (2^{-\gamma_r t} + 2^{-t}), \quad \sum_{\ell} \|W_{\ell}\|_1 \leq C \|\Theta\|_1 \tilde{M}_{I,J}.$$

Thus the squared L^2 norm of the within-cell square function is at most $C \|\Theta\|_{\infty}^2 \Lambda \tilde{M}_{I,J} 2^{-\gamma_r t}$.

For the grid partial sums, a choice of signs on the cells produces a sharp radial profile with $L^{\infty} + BV$ norm at most CK_r . Proposition 7.8 and Lemma 5.2 therefore give

$$Ct^2 \|\Theta\|_{\infty}^2 2^{C_B \gamma_r t} \Lambda \tilde{M}_{I,J} \{2^{-C_F K_r t} + 2^{-C_F t}\}. \quad (185)$$

Inserting cell endpoints into an arbitrary variation partition bounds the full variation by the grid variation plus the within-cell square function. After the square of (184) is included, (181)–(183) leave geometric decay $2^{-c_r t}$ for some $c_r > 0$. The factor t^2 is absorbed after decreasing the exponent. This proves every high branch with $E = 0$.

If every block is low, expand them all. For a combined input frequency ν and the packet center ζ_Q , write

$$e^{i\nu \cdot Y} d_Q(Y) = e^{i\nu \cdot \zeta_Q} d_Q(Y) + r_{Q,\nu}(Y), \quad \|r_{Q,\nu}\|_1 \leq C |\nu| 2^{-t} \|d_Q\|_1.$$

The frozen term remains mean zero, so Proposition 5.4 applies with constant packet coefficients of modulus at most one: divide the parent-dependent Fourier coefficient by its product common majorant and absorb the normalized scalar into the packet. Summation of the remaining majorants is governed by (171), and the output Fourier factor is one fixed modulation. Equation (180) leaves at least $2^{-\delta_T t/2}$ after the squared Fourier loss. Young's inequality and the weighted Fourier sum give total L^1 variation at most $C \|\Theta\|_1 2^{-t/2} \tilde{M}_{I,J}$ for the remainders.

At unit scale the mass in the fixed block pair is $\tilde{M}_{I,J} = R^{-n} M_{I,J}$. Returning to x , an L^2 square and an L^1 norm both acquire the Jacobian R^n , cancelling this rescaling of the mass. Square-sum the disjoint output blocks and use (175). This restores physical scale and proves (173)–(174). $\square$

7.4 Exact oriented algebra and the collective high-shell estimate

Let

$$\mathcal{V}_m = \{\text{homogeneous degree-}m \text{ polynomials in } (x, y)\} / \{p(x) + q(y)\}.$$

Each class has a unique representative consisting only of monomials which have positive degree in both x and y . We use this representative whenever an element of $\mathcal{V}_m$ is evaluated. Let $\mathcal{W}_m$ be the space of homogeneous degree- m polynomials in (W, X) . For such a representative put

$$\mathcal{E}H(W, X) = \text{Mix}_{W,X} \{H(W + X, 0) - H(W + X, W)\}. \quad (186)$$

Lemma 7.11 (Oriented graph injectivity). *For every integer $m \geq 2$, the map $\mathcal{E} : \mathcal{V}_m \rightarrow \mathcal{W}_m$ is injective and*

$$\|\mathcal{E}H\|_{\text{coeff}} \geq c_{n,m} \|H\|_{\text{coeff}}. \quad (187)$$

Moreover

$$\mathcal{E}H = \sum_{\substack{A+B=m \\ A,B \geq 1}} E_{A,B}(W, X), \quad (188)$$

where $E_{A,B}$ is bihomogeneous of bidegree (A, B) .

Proof. If the mixed part in (186) vanishes, the expression in braces is $A(W) + B(X)$. At $W = 0$ it vanishes, and hence B is constant. Substituting $x = W + X$ and $y = W$ shows that $H(x, 0) - H(x, y)$ is a function of y alone. Thus H is pure. Quantitative injectivity gives (187) by norm equivalence on the fixed finite-dimensional spaces. Homogeneity and the identities at $W = 0$ and $X = 0$ give (188). $\square$

We next record the exact projective calculation used below. Its purpose is to identify ordinary coefficients of one common numerator; no formal asymptotic expansion in the scale ratio is used.

Lemma 7.12 (Exact projective numerator tags). *Fix $D_0 \geq 1$, a multi-index γ with $q = |\gamma|$, and a polynomial $Q(s) = \sum_{p=0}^P c_p s^p$, where $P \in \mathbb{N}_0$ and $P + q \leq D_0 - 1$. Let $\delta > 0$, $0 < \alpha \leq 1$, and let $\omega \in \mathbb{R}^k$, where k is the length of γ . On the set $D := V - \delta u \neq 0$, put*

$$t = \frac{\delta u}{D}, \quad r_\perp = \frac{\delta \alpha \omega}{D},$$

and put

$$F_{q,Q}(u, V, \omega) = r_\perp^\gamma Q(t).$$

Then

$$D^{D_0+2} \partial_u \partial_V F_{q,Q} = \alpha^q \omega^\gamma \sum_{j=0}^{D_0-q} \delta^{q+1+j} \tau_{q,j}(Q) u^j V^{D_0-q-j}, \quad (189)$$

where the functionals $\tau_{q,j}$ depend only on D_0, q, j . If $q = 0$ and $P \geq 1$, the map

$$(c_1, \dots, c_P) \mapsto (\tau_{0,0}(Q), \dots, \tau_{0,P-1}(Q))$$

is invertible. If $q \geq 1$, the map

$$(c_0, \dots, c_P) \mapsto (\tau_{q,0}(Q), \dots, \tau_{q,P}(Q))$$

is invertible. The inverse norms are bounded in terms of D_0 only. When $q = P = 0$, the first map is the empty map and the assertion is vacuous. In particular, every detecting tag in the case $q = 0$, $P \geq 1$ has order $\kappa = q + 1 + j \leq P$, and in the case $q \geq 1$ it has order $\kappa = q + 1 + j \leq q + P + 1$. After multiplication by D^{D_0+2} , the displayed equality is a polynomial identity. Although the derivation below uses the dense open set on which $DV \neq 0$, the cleared identity therefore extends across both $D = 0$ and $V = 0$.

Proof. Set $z = \delta u/V$ and

$$H_{q,Q}(z) = (1 - z)^{-q} Q\left(\frac{z}{1 - z}\right).$$

Apart from the factor $\alpha^q \omega^\gamma$, one has

$$F_{q,Q} = \delta^q V^{-q} H_{q,Q}(z).$$

Consequently

$$\partial_u \partial_V F_{q,Q} = -\alpha^q \omega^\gamma \delta^{q+1} V^{-q-2} [(q+1) H'_{q,Q}(z) + z H''_{q,Q}(z)]. \quad (190)$$

Multiplication by $D^{D_0+2} = V^{D_0+2} (1 - z)^{D_0+2}$ gives a polynomial. The coefficient of z^j becomes the ordinary coefficient of $u^j V^{D_0-q-j}$, with the exact factor δ^{q+1+j} . This proves (189).

Write $H_{q,Q}(z) = \sum_{s \geq 0} h_s z^s$. The coefficient of z^{s-1} in the bracket in (190) is $s(q+s)h_s$. Multiplication by $(1 - z)^{D_0+2}$ is triangular with diagonal one. It therefore suffices to inspect $(h_1, \dots, h_P)$ when $q = 0$, and $(h_1, \dots, h_{P+1})$ when $q \geq 1$. For $q = 0$, $Q(z/(1 - z))$ is triangular with diagonal one on $(c_1, \dots, c_P)$. For $q \geq 1$, if the latter $P + 1$ coefficients vanished, then, after $s = z/(1 - z)$,

$$(1 + s)^q Q(s) = c_0 + O(s^{P+2}).$$

The first P equations make Q equal to c_0 times the degree- P Taylor polynomial of $(1 + s)^{-q}$, while the coefficient of s^{P+1} is a nonzero multiple of $c_0 \binom{q+P}{P+1}$. Hence $c_0 = 0$, and then all $c_p = 0$. Finite-dimensional norm equivalence gives the quantitative assertion. $\square$

We next define the unequal-shell phase appearing in the correlation. Throughout the remainder of this subsection assume that P has a highest nonpure homogeneous block P_m ; necessarily $m \geq 2$. Write, modulo pure phases,

$$P(a + X, a + Y) = p_a(X) + q_a(Y) + \sum_{\ell=2}^m H_{\ell,a}(X, Y), \quad (191)$$

where every $H_{\ell,a}$ is homogeneous of degree ℓ , and where $H_{m,a} = P_m$. For $W, X \in \mathbb{R}^n$, set

$$\Phi_{a,S}(W, X) = P(a + S(W + X), a) - P(a + S(W + X), a + SW). \quad (192)$$

Modulo a function independent of X ,

$$\Phi_{a,S}(W, X) \equiv \sum_{\ell=2}^m S^\ell \mathcal{E} H_{\ell,a}(W, X). \quad (193)$$

Indeed, the pure output phase cancels, the pure input phase is independent of X , and a mixed representative has no pure- X term in the difference defining $\mathcal{E}$.

Fix once and for all annular constants $0 < c_* < C_* < \infty$ containing all values of $|W|$ allowed by the two radial supports, and then choose C_0 sufficiently large, depending only on those fixed supports and annular constants. Thus $R \leq S/C_0$ makes $\delta = R/(S\rho)$ uniformly small throughout the overlap region. Let $W = \rho\theta$, with $c_* \leq \rho \leq C_*$. Choose and order a fixed finite rational atlas with nested chart domains $U_\chi \subseteq U_\chi^*$, where the smaller U_χ still cover the sphere. Write $\theta = \theta_\chi(\vartheta)$ in rational chart coordinates ϑ , with uniformly bounded Jacobian and inverse Jacobian on U_χ^* , and make the smaller cover disjoint by putting

$$E_1 = U_1, \quad E_\chi = U_\chi \setminus \bigcup_{\nu < \chi} U_\nu.$$

These are measurable, disjoint, and cover the sphere. Write $\chi(W)$ for the unique active index satisfying $W/|W| \in E_{\chi(W)}$. On each U_χ^* , choose a rational orthogonal frame $O_\chi(\vartheta)$ satisfying $O_\chi(\vartheta)e_1 = \theta_\chi(\vartheta)$. All frame denominators are bounded above and away from zero. The annular coordinate map $(\rho, \vartheta) \mapsto W = \rho\theta_\chi(\vartheta)$ is uniformly bi-Lipschitz on $c_* \leq \rho \leq C_*$. In frame coordinates define

$$G_{\ell,a,\chi}(\vartheta; T) = \rho^{-\ell} \mathcal{E} H_{\ell,a}(\rho O_\chi e_1, \rho O_\chi T). \quad (194)$$

This is independent of ρ , is a polynomial in T with no constant term, and has degree at most $\ell - 1$.

Proposition 7.13 (Oriented unequal-shell coefficient). *Assume $n \geq 2$. Let P_m be the highest nonpure homogeneous block of a real polynomial P of degree at most d , put*

$$L_R = R^m \|P_m\|_{\text{coeff}},$$

and suppose $0 < R \leq S/C_0$. For $y = a + SW$, assume $c_ \leq |W| \leq C_*$, where $a, W \in \mathbb{R}^n$; put $\rho = |W|$. On every chart χ there is a bounded-degree chart-polynomial vector $\mathbf{p}_{a,R,S,\chi}(W)$, independent of the polar band parameter α , such that*

$$\|\mathbf{p}_{a,R,S,\chi}\|_{\text{coeff}} \geq 1. \quad (195)$$

On the disjoint active-chart partition define

$$\mathbf{p}_{a,R,S}(W) := \mathbf{p}_{a,R,S,\chi(W)}(W). \quad (196)$$

Thus $\mathbf{p}$ is piecewise chart-polynomial, with a unique value at every W . Every later incidence argument is applied after splitting the atoms into the finitely many active sets E_χ . Here $\|\mathbf{p}\|_{\text{coeff}}$ denotes the fixed Euclidean norm of the chart-polynomial coefficient vector, whereas $|\mathbf{p}(W)|$ denotes the Euclidean norm in its fixed finite-dimensional value space. The coefficient lower bound in (195) is not asserted to be a pointwise lower bound; small values of $|\mathbf{p}(W)|$ are handled by the incidence estimate.

More precisely, put $\delta = R/(S\rho)$. For $0 < \alpha \leq 1$, $\omega \in \mathbb{S}^{n-2}$, and on each member of the fixed finite family of (u, V) -rectangles arising from the radial supports, use variables

$$D = V - \delta u, \quad T = T_{\delta, \alpha, \omega}(u, V) = \frac{(\delta u, \delta \alpha \omega)}{D}. \quad (197)$$

Every such rectangle has a fixed enlargement on which $D = V - \delta u$ is uniformly bounded above and away from zero. The actual unequal-shell phase is

$$\begin{aligned} \Psi_{a, R, S, W, \alpha, \omega}(u, V) &= \Phi_{a, S}(W, \rho O_\chi T_{\delta, \alpha, \omega}(u, V)) \\ &= P\left(a + S\rho O_\chi \frac{(V, \delta \alpha \omega)}{D}, a\right) \\ &\quad - P\left(a + S\rho O_\chi \frac{(V, \delta \alpha \omega)}{D}, a + S\rho O_\chi e_1\right). \end{aligned} \quad (198)$$

Since $\deg P \leq d$, the phase in (198) has the canonical common denominator

$$Q(u, V) = D^d = (V - \delta u)^d.$$

Define its literal mixed numerator by

$$\mathcal{N}_\Psi(u, V) := D^{d+2} \partial_u \partial_V \Psi(u, V). \quad (199)$$

This is a polynomial of bounded degree. For the canonical presentation,

$$\mathfrak{M}_{D^d} \Psi = D^{3d} \partial_u \partial_V \Psi = D^{2d-2} \mathcal{N}_\Psi.$$

Multiplication by D^{2d-2} is uniformly bounded below in coefficient norm on these fixed polynomial spaces: this follows from the equivalence of coefficient and supremum norms and the lower bound for $|D|$ on the enlarged rectangle. Equation (146) therefore gives

$$\|[\Psi]\|_{\text{rat}, D^d} \geq c \|\mathcal{N}_\Psi\|_{\text{coeff}(u, V)}. \quad (200)$$

For every α there is a bounded-degree vector $\mathbf{q}_{\alpha, W}(\omega)$, made from selected ordinary coefficients of $\mathcal{N}_\Psi$, such that

$$\|[\Psi_{a, R, S, W, \alpha, \omega}]\|_{\text{rat}, D^d} \geq c |\mathbf{q}_{\alpha, W}(\omega)|, \quad (201)$$

$$\|\mathbf{q}_{\alpha, W}\|_{\text{coeff}, \omega} \geq c \alpha^d L_R |\mathbf{p}_{a, R, S, \chi}(W)|. \quad (202)$$

The constants are independent of a and of all coefficients of total degree less than m .

Proof. By (193) and (194), modulo $F(u) + G(V)$ the phase in (198) is

$$\sum_{\ell=2}^m S^\ell \rho^\ell G_{\ell, a, \chi} \left(\vartheta; \frac{(\delta u, \delta \alpha \omega)}{D} \right). \quad (203)$$

Expand

$$G_{\ell, a, \chi}(\vartheta; t, r_\perp) = \sum_{q, |\gamma|=q} r_\perp^\gamma Q_{\ell, a, \chi, \gamma}(\vartheta; t).$$

Here $\deg Q_{\ell, a, \chi, \gamma} + q \leq \ell - 1$, and the constant coefficient of $Q_{\ell, a, \chi, 0}$ is zero because $G_{\ell, a, \chi}(\vartheta; 0) = 0$. Lemma 7.12, used with $D_0 = d$, shows that the coefficient of $\omega^\gamma u^j V^{d-q-j}$ in $\mathcal{N}_\Psi$ is exactly

$$C_{q, \gamma, j, \alpha}(\rho, \vartheta) = \alpha^q \sum_{\ell=2}^m R^\ell S^{\ell-\kappa} \rho^{\ell-\kappa} \tau_{q, j}(Q_{\ell, a, \chi, \gamma}), \quad \kappa = q + 1 + j. \quad (204)$$

Thus different tag orders are different ordinary numerator monomials; there is no appeal to a formal expansion in δ .

We make the finite-dimensional coefficient step explicit. Fix the mixed monomial basis of $\mathcal{V}_m$, and use ordinary monomial coefficients in the chart variable ϑ . Choose once and for all a nonvanishing polynomial $\Delta_\chi(\vartheta)$ which clears every denominator introduced by the frame O_χ in degrees at most d ; both Δ_χ and its reciprocal are bounded on U_χ^* . For $T = (t, r_\perp) \in \mathbb{R} \times \mathbb{R}^{n-1}$ define the cleared restriction map

$$(\mathbf{R}_\chi H)(\vartheta; T) = \Delta_\chi(\vartheta) \mathcal{E}H(O_\chi(\vartheta)e_1, O_\chi(\vartheta)T). \quad (205)$$

By homogeneity this is precisely $\Delta_\chi G_{m,a,\chi}$ when $H = P_m$; in particular it is independent of a and ρ . Write

$$\mathbf{R}_\chi H = \sum_\gamma r_\perp^\gamma \tilde{Q}_{\chi,\gamma}(\vartheta; t), \quad q = |\gamma|.$$

The $q = 0$ polynomial has zero constant coefficient, and $\deg_t \tilde{Q}_{\chi,\gamma} + q \leq m - 1$.

Let

$$\mathfrak{S}_m = \{(0, j) : 0 \leq j \leq m - 2\} \cup \{(\gamma, j) : 1 \leq |\gamma| \leq m - 1, 0 \leq j \leq m - 1 - |\gamma|\}, \quad (206)$$

where 0 denotes the zero multi-index, and put $\kappa(\gamma, j) = |\gamma| + 1 + j$. Let D_χ be a fixed degree bound for the cleared chart polynomials and set

$$\mathcal{Z}_{m,\chi} = \bigoplus_{(\gamma,j) \in \mathfrak{S}_m} \mathcal{P}_{\leq D_\chi}(\vartheta),$$

with the Euclidean norm of all ordinary monomial coefficients. All tag functionals below are those of Lemma 7.12 with $D_0 = d$, extended coefficientwise in ϑ . Define

$$\mathbf{T}_\chi : \mathcal{V}_m \longrightarrow \mathcal{Z}_{m,\chi}, \quad \mathbf{T}_\chi H = (\tau_{|\gamma|,j}(\tilde{Q}_{\chi,\gamma}(\vartheta; \cdot)))_{(\gamma,j) \in \mathfrak{S}_m}. \quad (207)$$

Multiplication by Δ_χ commutes with the tag functionals because it is independent of t and $r_\perp$. Lemma 7.12 therefore recovers, separately for each γ , all coefficients of $\tilde{Q}_{\chi,\gamma}$: for $q = 0$ it recovers the nonconstant t -coefficients, and for $q \geq 1$ it recovers all of them. Thus vanishing of $\mathbf{T}_\chi H$ implies $\mathbf{R}_\chi H = 0$.

The restriction map $\mathbf{R}_\chi$ is injective. Indeed, its vanishing implies $\mathcal{E}H(\theta, X) = 0$ for every X and every direction θ in the open chart. In the bidegree decomposition (188), distinct X -degrees separate the terms $E_{A,B}$; each is homogeneous in W , so vanishing on an open patch of the unit sphere extends to the corresponding open cone. Hence $\mathcal{E}H = 0$, and Lemma 7.11 gives $H = 0$ in $\mathcal{V}_m$. Consequently $\mathbf{T}_\chi$ is injective. Its uniform constant is the genuine least singular value

$$\sigma_\chi = \min_{\|H\|_{\text{coeff}}=1} \|\mathbf{T}_\chi H\|_{\text{coeff}} > 0, \quad \sigma_* := \min_\chi \sigma_\chi > 0. \quad (208)$$

The minimum is taken on the unit sphere of the fixed finite-dimensional space $\mathcal{V}_m$; the maps depend only on n, m, d and the fixed atlas, not on a, R, S, α or any coefficient of a lower homogeneous block.

Now retain exactly the coordinates in (206) and multiply (204) by ρ^κ . This is a uniformly invertible diagonal change on $c_* \leq \rho \leq C_*$. For $\alpha = 1$, after the fixed denominator is cleared, the selected polynomial vector is

$$\widehat{C}_{q,\gamma,j}(\rho, \vartheta) = \Delta_\chi(\vartheta) \sum_{\ell=2}^m R^\kappa S^{\ell-\kappa} \rho^\ell \tau_{q,j}(Q_{\ell,a,\chi,\gamma}). \quad (209)$$

Let π_m be ordinary coefficient projection onto radial degree ρ^m . Since the chart denominator and the tag selection are independent of ρ , both commute with π_m , and therefore

$$\pi_m \widehat{\mathbf{C}}_{a,R,S,\chi} = D_{R,S} \mathbf{T}_\chi P_m, \quad (D_{R,S} v)_{\gamma,j} = R^{\kappa(\gamma,j)} S^{m-\kappa(\gamma,j)} v_{\gamma,j}. \quad (210)$$

This identity also shows directly why no translated lower-degree term can enter a protected coordinate: the block of total degree ℓ carries exactly the radial monomial ρ^ℓ , while $H_{m,a} = P_m$. For every retained coordinate, $1 \leq \kappa(\gamma, j) \leq m$; hence, since $R \leq S$,

$$\|D_{R,S}v\|_{\text{coeff}} \geq R^m \|v\|_{\text{coeff}}. \quad (211)$$

Combining (208)–(211) with the bounded coefficient projection gives

$$\|\widehat{\mathbf{C}}_{a,R,S,\chi}\|_{\text{coeff}} \geq c_0 R^m \|P_m\|_{\text{coeff}} = c_0 L_R, \quad (212)$$

uniformly in all lower blocks. For orientation, the retained orders are $\{1, 2\}$ when $m = 2$ and $\{1, 2, 3\}$ when $m = 3$, giving respectively the scale factors RS, R^2 and RS^2, R^2S, R^3 . Define

$$\mathbf{p}_{a,R,S,\chi} = \frac{\widehat{\mathbf{C}}_{a,R,S,\chi}}{c_0 L_R}.$$

This proves (195).

For fixed W , group the selected coefficients in (204) according to (q, j) , leaving the homogeneous degree- q polynomial in ω unexpanded; after the harmless factors $\rho^\kappa \Delta_\chi$ are inserted, call the resulting vector $\mathbf{q}_{\alpha,W}(\omega)$. It consists of selected coefficients of the literal numerator $\mathcal{N}_\Psi$, up to a uniformly invertible diagonal change, and therefore (201) follows from (200). The dependence on α is exactly diagonal:

$$\widehat{\mathbf{C}}_{q,\gamma,j,\alpha} = \alpha^q \widehat{\mathbf{C}}_{q,\gamma,j,1}.$$

Since $0 < \alpha \leq 1$ and $0 \leq q \leq d$, the least singular value of this diagonal map is at least α^d . Consequently

$$\|\mathbf{q}_{\alpha,W}\|_{\text{coeff},\omega} \geq c\alpha^d |\widehat{\mathbf{C}}_{a,R,S,\chi}(W)| \geq c\alpha^d L_R |\mathbf{p}_{a,R,S,\chi}(W)|.$$

This proves (202) and also proves that the same $\mathbf{p}$ works on every polar band. $\square$

Corollary 7.14 (Rough unequal-shell alternative). *Let P be a real polynomial of degree at most d with highest nonpure homogeneous block P_m , let $\Theta \in L^\infty(\mathbb{S}^{n-1})$, and let $0 < R \leq S/C_0$. Put $L_R = R^m \|P_m\|_{\text{coeff}}$, and let b_R, b_S be supported in a fixed compact subinterval of $(0, \infty)$, with $L^\infty + \text{BV}$ norm at most B , where $B \geq 1$. Put*

$$k_R(z) = |z|^{-n} \Theta(z/|z|) b_R(|z|/R), \quad k_S(z) = |z|^{-n} \Theta(z/|z|) b_S(|z|/S).$$

Fix $a, W \in \mathbb{R}^n$, put $y = a + SW$, and assume $c_ \leq |W| \leq C_*$. Use the active-chart vector $\mathbf{p}(W) = \mathbf{p}_{a,R,S,\chi(W)}(W)$ from (196). Then, for $n \geq 2$,*

$$\left| \int e^{i[P(y+z,a)-P(y+z,y)]} \overline{k_R(z)} k_S(z+y-a) dz \right| \leq CB^C \|\Theta\|_\infty^2 S^{-n} \min\{1, (L_R |\mathbf{p}(W)|)^{-c}\}. \quad (213)$$

If $\mathbf{p}(W) = 0$, the last factor is understood to be 1. For $n = 1$ the same conclusion holds, with S^{-1} , for a bounded-degree polynomial vector $\mathbf{p}(W)$ satisfying $\|\mathbf{p}\|_{\text{coeff}} \geq 1$.

Proof. If the two radial supports have no overlap after the translation by SW , the integral is zero in every dimension. Assume that the overlap is nonempty, and first let $n \geq 2$. Then $|z| \simeq R$, $|z + SW| \simeq S$, and the contributing set has measure $O(R^n)$; hence

$$\int |k_R(z) k_S(z + SW)| dz \leq CB^2 \|\Theta\|_\infty^2 S^{-n}. \quad (214)$$

This proves (213) when $\mathbf{p}(W) = 0$. Henceforth assume that $\mathbf{p}(W) \neq 0$. Write $W = \rho\theta$ and decompose $z = t\theta + r\omega$, where $r > 0$ and $\omega \in \mathbb{S}^{n-2}$. Put

$$\xi = \frac{t}{r}, \quad \eta = \frac{t + S\rho}{r}.$$

Thus the two rays are $z = r(\xi\theta + \omega)$ and $z + SW = r(\eta\theta + \omega)$. Use a bounded-variation dyadic partition in the transverse radius. After combining its boundedly many largest pieces, its parameters may be taken to satisfy $\alpha \in \{2^{-k} : k \geq 0\}$, and the corresponding band is $r \simeq \alpha R$. Set

$$\delta = \frac{R}{S\rho}, \quad u = \alpha\xi, \quad V = \alpha\delta\eta, \quad D = V - \delta u. \quad (215)$$

Then, exactly,

$$t = \frac{Ru}{D}, \quad r = \frac{\alpha R}{D}, \quad z = \frac{R(u\theta + \alpha\omega)}{D}, \quad z + SW = \frac{S\rho(V\theta + \alpha\delta\omega)}{D}. \quad (216)$$

Here $D = \alpha R/r > 0$. Moreover $z/S = \rho O_\chi T_{\delta, \alpha, \omega}(u, V)$, so the phase in these variables is exactly (198). Indeed, cylindrical measure is $r^{n-2} dt dr d\omega$, and

$$\left| \frac{\partial(t, r)}{\partial(u, V)} \right| = \frac{\alpha R^2}{D^3}.$$

Since

$$|z| = \frac{R\sqrt{u^2 + \alpha^2}}{D}, \quad |z + SW| = \frac{S\rho\sqrt{V^2 + \alpha^2\delta^2}}{D},$$

the cylindrical Jacobian and the two homogeneous kernels give the exact factor

$$(S\rho)^{-n} \alpha^{n-1} D^{n-1} (\alpha^2 + u^2)^{-n/2} (V^2 + \alpha^2\delta^2)^{-n/2} du dV d\omega. \quad (217)$$

The radial profiles become

$$\overline{b_R \left(\frac{\sqrt{u^2 + \alpha^2}}{D} \right)}, \quad \overline{b_S \left(\frac{\rho\sqrt{V^2 + \alpha^2\delta^2}}{D} \right)}, \quad (218)$$

while the two rough factors are, respectively,

$$\overline{\Theta \left(\frac{u\theta + \alpha\omega}{\sqrt{u^2 + \alpha^2}} \right)}, \quad \overline{\Theta \left(\frac{V\theta + \alpha\delta\omega}{\sqrt{V^2 + \alpha^2\delta^2}} \right)}. \quad (219)$$

Thus neither rough angular factor is differentiated.

On the support of one transverse band and of the two radial profiles, u, V lie in fixed compact sets and D is bounded above and away from zero. The profile support also gives $\sqrt{u^2 + \alpha^2} \gtrsim 1$ and $\sqrt{V^2 + \alpha^2\delta^2} \gtrsim 1$; hence the two possibly singular weights in (217) and all of their sectional derivatives are uniformly bounded. After a bounded number of sign and monotonicity pieces, the remaining amplitude in (217)–(218), including the transverse band cutoff, has uniform row and column L^1 bounds and sectional $L^\infty + \text{BV}$ norm at most CB^C . This follows from the one-dimensional BV chain rule on each monotonicity piece. Explicitly, the two profile arguments are

$$\gamma_1(u, V) = \frac{\sqrt{u^2 + \alpha^2}}{V - \delta u}, \quad \gamma_2(u, V) = \frac{\rho\sqrt{V^2 + \alpha^2\delta^2}}{V - \delta u}.$$

Writing $s_1 = (u^2 + \alpha^2)^{1/2}$, $s_2 = (V^2 + \alpha^2\delta^2)^{1/2}$, and $D = V - \delta u$, direct differentiation gives

$$\partial_u \gamma_1 = \frac{uV + \delta\alpha^2}{s_1 D^2}, \quad \partial_V \gamma_1 = -\frac{s_1}{D^2}, \quad \partial_u \gamma_2 = \frac{\rho\delta s_2}{D^2}, \quad \partial_V \gamma_2 = -\frac{\rho\delta(uV + \delta\alpha^2)}{s_2 D^2}.$$

Thus the only nontrivial sectional critical-point equation is the linear equation $uV + \delta\alpha^2 = 0$ in the varying variable. Each section therefore has a uniformly bounded number of monotonicity pieces; on each piece $\text{Var}(b_R \circ \gamma_1) \leq \text{Var}(b_R)$ and $\text{Var}(b_S \circ \gamma_2) \leq \text{Var}(b_S)$. More explicitly, on an interval I on which γ is monotone,

$$\text{Var}_I(b \circ \gamma) \leq \text{Var}_{\gamma(I)}(b) \leq \text{Var}(b),$$

and the conclusion is trivial if γ is constant. The numerator $uV + \delta\alpha^2$ is affine in either varying coordinate, so each section requires at most one additional cut, independently of α, δ, ρ . Products are controlled by

$$\text{Var}(fg) \leq \|f\|_\infty \text{Var}(g) + \|g\|_\infty \text{Var}(f).$$

After extracting the explicit factor α^{n-1} in (217), the displayed lower bounds for D and the two square roots give uniformly bounded sectional $W^{1,\infty}$ norms for all remaining algebraic factors. Hence the sectional $L^\infty + \text{BV}$ bounds and the Schur row and column bounds remain uniform as $\alpha, \delta \downarrow 0$. Cover this support by a fixed bounded number of rectangles on slightly enlarged versions of which D is still bounded away from zero. The interaction graph of these rectangles has bounded degree, so their L^2 operator bounds synthesize by Cauchy–Schwarz with no scale-dependent block factor (equivalently, here even the finite triangle inequality has only a fixed loss).

For fixed ω , apply Lemma 7.4 to the phase (198), treating the two factors in (219) as the output and input functions. Their L^2 norms on the fixed u - and V -intervals are at most $C\|\Theta\|_\infty$. Proposition 7.13, followed by (139) on the finite rational atlas of $\mathbb{S}^{n-2}$, therefore gives the band bound below. Here one uses that restriction of a homogeneous degree- q polynomial in ω to the sphere is injective; clearing the fixed atlas denominators preserves its coefficient norm. Thus the coefficient lower bound in (202) is exactly the normalization needed for (139). We obtain

$$CB^C \|\Theta\|_\infty^2 S^{-n} \alpha^{n-1} \min\{1, (L_R \alpha^d |\mathbf{p}(W)|)^{-c}\}. \quad (220)$$

Here ρ is confined to a fixed annulus and has been absorbed into the constant. When $n = 2$, $\omega \in \mathbb{S}^0 = \{-1, 1\}$. A homogeneous degree- q polynomial in the one transverse variable is $c\omega^q$, so its value at either point has modulus $|c|$; the same estimate follows by summing these two points directly. If $A = L_R |\mathbf{p}(W)| \leq 1$, summability of α^{n-1} gives the trivial alternative. If $A > 1$, split the dyadic sum at $\alpha \simeq A^{-1/d}$; the two resulting geometric series are $O(A^{-c'})$ after decreasing the exponent. This proves (213).

For $n = 1$, work on one of the finitely many signed overlap intervals and write $z = Rt$. Let $\mathbf{C}(W)$ be the vector of all nonconstant-in- t coefficients of the complete translated correlation phase. Its degree- m part is

$$\sum_{\substack{A+B=m \\ A,B \geq 1}} c_{A,B} S^A R^B W^A t^B. \quad (221)$$

For a fixed t^B coefficient, every degree $\ell < m$ contribution has W -degree at most $\ell - B < m - B$, and cannot alter the coefficient of W^{m-B} . Since $S^A R^B \geq R^m$, Lemma 7.11 gives

$$\|\mathbf{C}\|_{\text{coeff}, W} \geq cR^m \|P_m\|_{\text{coeff}} = cL_R.$$

Put $\mathbf{p} = \mathbf{C}/(cL_R)$. The product of the two kernels and dz is $S^{-1}b(t)dt$, on finitely many intervals, with $\|b\|_{L^\infty + \text{BV}} \leq CB^C \|\Theta\|_\infty^2$. Lemma 7.3 gives the asserted estimate when $\mathbf{p}(W) \neq 0$; when $\mathbf{p}(W) = 0$, the same conclusion is the trivial S^{-1} amplitude bound. $\square$

Lemma 7.15 (One-sided variable-mesh incidence). *Let $0 < \varepsilon \leq 1$, let $q_R = \varepsilon R$, let the scale- R packet cubes form a finite disjoint family $\mathcal{Q}_R$ of dyadic cubes in one aligned grid, where $R > 0$. For each $Q \in \mathcal{Q}_R$ choose a point $y_Q \in Q$, and attach a weight $0 \leq w_Q \leq \Lambda q_R^n$, where $\Lambda \geq 0$. Fix $a \in \mathbb{R}^n$ and a large shell scale $S \geq C_0 R$. If $\mathbf{p}$ is as in Proposition 7.13 for $n \geq 2$, or as constructed in Corollary 7.14 for $n = 1$, then, for the fixed annular-overlap constants $0 < c_* < C_* < \infty$ and every $0 < \tau \leq 1$,*

$$\sum_{\substack{Q: c_* S \leq |y_Q - a| \leq C_* S \\ |\mathbf{p}((y_Q - a)/S)| \leq \tau}} w_Q \leq C \Lambda S^n \{\tau^c + (\varepsilon R/S)^c\}. \quad (222)$$

Proof. For $n \geq 2$, first split the sum into the finitely many disjoint active sets E_χ from (196); on the χ -th piece use $\mathbf{p}_{a,R,S,\chi}$. (For $n = 1$ no chart split is needed.) Choose on that piece a scalar component

and clear its bounded chart denominator; call the resulting polynomial p_0 . Bounded invertibility of this operation and the fixed vector length allow the choice to satisfy $A_0 = \|p_0\|_{\text{coeff}} \geq c > 0$. Put $\tilde{p}_0 = p_0/A_0$. This last polynomial has coefficient norm one. At a bad point it satisfies $|\tilde{p}_0| \leq \tau/A_0 \leq C\tau$, regardless of how large the translated lower blocks make A_0 . After division by S , color the q_R grid and put disjoint balls of radius $c\varepsilon R/S$ around the selected points. Work in the fixed enlarged radial annulus $c_*/2 < |W| < 2C_*$. If this radius exceeds a fixed fraction of either the radial collar or, when $n \geq 2$, the angular chart collar $\text{dist}(U_\chi, \partial U_\chi^*)$, the mesh term on the right of (222) is already bounded below by a fixed constant. Disjointness and the weight cap give total weight at most $C\Lambda S^n$ in the fixed annulus, so the assertion is then trivial. Otherwise the balls stay in the enlarged radial annulus and, for $n \geq 2$, in the enlarged chart U_χ^* . Uniform bi-Lipschitz equivalence between physical W -balls and chart-coordinate balls (ordinary W -intervals when $n = 1$) changes their radii and volumes by fixed factors only. The normalized polynomial $\tilde{p}_0$ has uniformly bounded gradient on the enlarged coordinate domain, so a ball centered at a bad point lies in the $C(\tau + \varepsilon R/S)$ -sublevel. Lemma 7.1 bounds the union of these balls. Divide by one ball volume, multiply by the weight cap, and reduce the exponent. The large atom was fixed before smearing, so no mesh term of size $\varepsilon S/S$ occurs. $\square$

Proposition 7.16 (Collective fixed-sign high-shell estimate). *Let P be a real polynomial of degree at most d , let P_m be its highest nonpure homogeneous block, let $\Theta \in L^\infty(\mathbb{S}^{n-1})$, fix $\kappa > 0$, let $s \in \mathbb{N}$, let $\varepsilon = 2^{-s}$, and suppose that $\mathcal{R}$ is a finite set of positive dyadic radii such that every $R \in \mathcal{R}$ satisfies*

$$L_R = R^m \|P_m\|_{\text{coeff}} \geq \varepsilon^{-\kappa}. \quad (223)$$

At radius $R \in \mathcal{R}$, let $\mathcal{Q}_R$ be a finite family of disjoint dyadic cubes of side εR in one aligned grid; assume the cubes over all radii are globally disjoint and each complex measure ν_Q is supported on Q and satisfies $|\nu_Q|(Q) \leq \Lambda|Q|$, where $\Lambda \geq 0$. Let

$$T_R \nu(x) = \int e^{iP(x,y)} K_\Theta(x-y) b_R(|x-y|/R) d\nu(y),$$

where $B \geq 1$ and all normalized radial profiles are supported in one fixed compact subinterval of $(0, \infty)$ with $L^\infty + \text{BV}$ norm at most B . Then, uniformly for arbitrary complex coefficients $|e_Q| \leq 1$,

$$\left\| \sum_{R \in \mathcal{R}} T_R \sum_{Q \in \mathcal{Q}_R} e_Q \nu_Q \right\|_2^2 \leq C B^C \|\Theta\|_\infty^2 \Lambda M 2^{-c_\kappa s}, \quad M = \sum_{R \in \mathcal{R}} \sum_{Q \in \mathcal{Q}_R} |\nu_Q|(Q). \quad (224)$$

Here $c_\kappa > 0$, and the implicit constant may depend on κ (in addition to the fixed structural data).

Proof. If $\mathcal{R} = \emptyset$, the assertion is immediate; hence assume $\mathcal{R} \neq \emptyset$. Put $w_Q = |\nu_Q|(Q)$ and write the polar decomposition $d\nu_Q = h_Q d|\nu_Q|$, with $|h_Q| = 1$ $|\nu_Q|$ -almost everywhere; cubes with $w_Q = 0$ may be omitted. On the finite product probability space

$$(\mathfrak{X}, \mathbb{P}) = \prod_{R \in \mathcal{R}} \prod_{Q \in \mathcal{Q}_R} \left(Q, \frac{|\nu_Q|}{w_Q} \right),$$

let Y_Q be the coordinate random variables. Then, in the weak sense of complex measures,

$$\mathbb{E}[w_Q h_Q(Y_Q) \delta_{Y_Q}] = \nu_Q; \quad (225)$$

indeed, testing against a bounded Borel function gives the defining polar integral. Set

$$c_Q(\mathbf{Y}) = e_Q h_Q(Y_Q), \quad \mathcal{F}(\mathbf{Y}) = \sum_{R \in \mathcal{R}} T_R \sum_{Q \in \mathcal{Q}_R} c_Q(\mathbf{Y}) w_Q \delta_{Y_Q}.$$

The radial support is separated from zero, and hence, uniformly in y ,

$$\|T_R \delta_y\|_2^2 \leq C B^2 \|\Theta\|_\infty^2 R^{-n}. \quad (226)$$

Since all index sets are finite, this makes $\mathcal{F}$ a strongly measurable Bochner-integrable L^2 -valued random variable and justifies all exchanges of expectation, finite sums, and integration. By (225),

$$\mathbb{E}\mathcal{F} = \sum_R T_R \sum_Q e_Q v_Q \quad \text{in } L^2.$$

Hilbert-space Jensen therefore gives

$$\left\| \sum_R T_R \sum_Q e_Q v_Q \right\|_2^2 = \|\mathbb{E}\mathcal{F}\|_2^2 \leq \mathbb{E}\|\mathcal{F}\|_2^2. \quad (227)$$

It therefore suffices to prove the desired bound for every fixed selection $y_Q \in Q$, with weights $w_Q \leq \Lambda|Q|$ and arbitrary packet coefficients c_Q satisfying $|c_Q| \leq 1$. Fix such a selection and put

$$F_R = T_R \sum_{Q \in Q_R} c_Q w_Q \delta_{y_Q}.$$

Partition physical input and output space into R -cubes; annular support leaves only boundedly many relative input–output positions. To make the scale normalization explicit, fix one interacting output–input pair (I, J) , let x_I be the chosen center of I , and let $Q_{R,I,J}$ be the packet cubes in J which contribute to this pair. Put

$$M_{R,I,J} = \sum_{Q \in Q_{R,I,J}} w_Q, \quad \sigma_{x_I,R}(y) = \frac{y - x_I}{R},$$

and, for each such Q , set

$$\tilde{v}_Q = R^{-n}(\sigma_{x_I,R})_\#(w_Q \delta_{y_Q}), \quad \tilde{Q} = \sigma_{x_I,R}(Q).$$

Then $\tilde{Q}$ has side length ε and, exactly,

$$|\tilde{v}_Q|(\tilde{Q}) = R^{-n} w_Q \leq R^{-n} \Lambda |Q| = \Lambda |\tilde{Q}|, \quad \sum_{Q \in Q_{R,I,J}} |\tilde{v}_Q|(\tilde{Q}) = R^{-n} M_{R,I,J}.$$

All cubes $\tilde{Q}$ lie in the fixed normalized input block $B_0 = \sigma_{x_I,R}(J)$. Homogeneity of the kernel gives

$$T_R \sum_{Q \in Q_{R,I,J}} c_Q w_Q \delta_{y_Q}(x_I + RX) = \tilde{T}_1 \sum_{Q \in Q_{R,I,J}} c_Q \tilde{v}_Q(X),$$

where $\tilde{T}_1$ has phase $P(x_I + RX, x_I + RY)$ and normalized radial profile $b_R(|X - Y|)$. Modulo pure phases, the highest nonpure homogeneous block of the normalized phase is $R^m P_m$, whose coefficient norm is L_R . Proposition 7.8 therefore produces a normalized bound proportional to $\Lambda R^{-n} M_{R,I,J}$. Since $dx = R^n dX$, the output L^2 -square Jacobian cancels this R^{-n} factor exactly. Apply Proposition 7.8 on every interacting block, then square-sum the disjoint output blocks and use the bounded overlap of the input blocks. If M_R is the total packet mass at radius R , this gives

$$\|F_R\|_2^2 \leq C B^C \|\Theta\|_\infty^2 \Lambda M_R (L_R^{-c} + \varepsilon^c).$$

The finite squared-norm expansion is

$$\left\| \sum_{R \in \mathcal{R}} F_R \right\|_2^2 = \sum_R \|F_R\|_2^2 + 2 \operatorname{Re} \sum_{\substack{R, S \in \mathcal{R} \\ R < S}} \langle F_R, F_S \rangle. \quad (228)$$

For $R < S$, split the pairs into the comparable range $S < C_0 R$ and the separated range $R \leq S/C_0$; these ranges are disjoint and exhaustive. If R and S are comparable, then $2|\langle F_R, F_S \rangle| \leq \|F_R\|_2^2 + \|F_S\|_2^2$ reduces

the cross term to those two diagonal bounds. Each radius has only boundedly many comparable dyadic neighbours, so their total is bounded by the right side of (224).

For $R \leq S/C_0$, fix a large-shell atom y_A of weight w_A and put

$$\tau_R = L_R^{-1/2}. \quad (229)$$

For this fixed atomic selection, the scale- R contribution to the inner product with the fixed scale- S atom has the form, up to packet coefficients of modulus at most one,

$$|\langle F_R, F_{S,A} \rangle| \leq w_A \sum_{Q \in Q_R} w_Q |C_{R,S}(y_Q, y_A)|,$$

where w_Q is the fixed weight attached to Q and $C_{R,S}$ is precisely the correlation estimated in Corollary 7.14. On the bad set, Lemma 7.15 and the trivial S^{-n} correlation bound give, before multiplication by w_A ,

$$C\Lambda\{L_R^{-c} + (\varepsilon R/S)^c\}.$$

On its complement, Corollary 7.14 gives $S^{-n}L_R^{-c}$. At one fixed scale the total small-atom weight in the interacting S -ball is at most $C\Lambda S^n$, by packing the disjoint εR -cubes. Thus the good row has the same bound $C\Lambda L_R^{-c}$.

Let R_0 be the smallest retained dyadic radius. Since $L_R = L_{R_0}(R/R_0)^m$ and $L_{R_0} \geq \varepsilon^{-\kappa}$,

$$\sum_{\substack{R \in \mathcal{R} \\ R \geq R_0}} L_R^{-c} \leq \varepsilon^{c\kappa} \sum_{\ell \geq 0} 2^{-m c \ell} \lesssim \varepsilon^{c\kappa}.$$

Likewise,

$$\sum_{R \leq S/C_0} (\varepsilon R/S)^c \lesssim \varepsilon^c \sum_{\ell \geq 0} 2^{-c \ell} \lesssim \varepsilon^c.$$

Thus the complete small-shell row is at most $C\Lambda 2^{-c\kappa S}$. Multiply by w_A , sum the large atoms and then S . Equivalently, if $F_{S,A}$ denotes the contribution of the fixed large-shell atom, the preceding row estimate says

$$\sum_{R \leq S/C_0} |\langle F_R, F_{S,A} \rangle| \leq C\Lambda 2^{-c\kappa S} w_A.$$

Summing first in the atoms at scale S and then in S gives $\sum_{S,A} w_A = M$, so no additional large-shell counting factor occurs. The result is $C\Lambda M 2^{-c\kappa S}$ for every fixed atomic selection. Together with (228), this proves the deterministic atomic estimate; taking expectation in (227) proves (224) for the original complex measures. Every separated unordered pair has been oriented by its unique smaller and larger radius and counted once; the conjugate orientation is exactly the factor two in (228). All estimates used absolute values, so the coefficients may be fixed independently for every packet, as required by the later rank argument. $\square$

8 The unphased weak endpoint

We begin with the unphased weak endpoint. When $n = 1$, every mean-zero function on $\mathbb{S}^0$ is a scalar multiple of the odd Hilbert kernel, and the result follows from Lemma 2.4. Henceforth assume $n \geq 2$. The good part of the Calderón–Zygmund decomposition is controlled by the $p = 2$ case of Proposition 2.9:

$$\sup_{\lambda > 0} \|\lambda N_\lambda(T_{\Omega, \varepsilon}^0 h : \varepsilon > 0)^{1/2}\|_2 \leq C_n \mathfrak{L}(\Omega) \|h\|_2, \quad \int \Omega = 0. \quad (230)$$

Proposition 8.1 (Unphased weak critical-jump inequality). *For mean-zero $\Omega \in L \log L(\mathbb{S}^{n-1})$ and every $f \in L^1(\mathbb{R}^n)$,*

$$\sup_{\lambda > 0} \|\lambda N_\lambda(T_{\Omega, \varepsilon}^0 f : \varepsilon > 0)^{1/2}\|_{L^{1, \infty}} \leq C_n \mathfrak{L}(\Omega) \|f\|_1. \quad (231)$$

Proof. Finite reduction and Calderón–Zygmund decomposition.

We first treat bounded compactly supported f ; Section 12 completes the general L^1 trajectory. We impose the finite restrictions needed for every later ranking argument at the outset. After forming the full Calderón–Zygmund decomposition, fix a finite subfamily Q_0 of its bad cubes, a finite interval of shell indices $j_- \leq j \leq j_+$, and a finite ordered set of positive rational truncation parameters, enlarged by the finitely many dyadic and cell endpoints used below. The bad sums, localized families, and trajectories are initially interpreted with these restrictions. In particular, every family $\mathcal{R}_s$ ranked below is finite when Lemma 5.2 is invoked. All bounds are uniform in the three cutoffs, which are suppressed from the notation.

If $\mathfrak{L}(\Omega) = 0$, then $\Omega = 0$ and the assertion is immediate. By homogeneity normalize $\mathfrak{L}(\Omega) = 1$. Fix $\lambda, \alpha > 0$ and perform the ordinary dyadic Calderón–Zygmund decomposition at height α as follows. Take the maximal dyadic cubes Q for which $|Q|^{-1} \int_Q |f| > \alpha$. They are disjoint, $\sum_Q |Q| \leq \alpha^{-1} \|f\|_1$, and dyadic differentiation gives $|f| \leq \alpha$ off their union. Maximality makes the average over the dyadic parent at most α , hence $|Q|^{-1} \int_Q |f| \leq 2^n \alpha$. Put

$$g = f \mathbf{1}_{(\cup Q)^c} + \sum_Q f_Q \mathbf{1}_Q, \quad b_Q = (f - f_Q) \mathbf{1}_Q, \quad f_Q = |Q|^{-1} \int_Q f.$$

The preceding facts, followed by $\|g\|_2^2 \leq \|g\|_\infty \|g\|_1$, give

$$\|g\|_2^2 \leq C\alpha \|f\|_1, \quad \int b_Q = 0, \quad \|b_Q\|_1 \leq C\alpha |Q|, \quad \Sigma := \sum_Q \|b_Q\|_1 \leq C\|f\|_1. \quad (232)$$

The good part is bounded by (230). Choose $s_0 \geq \max\{5, t_0^{\text{term}}\}$, where t_0^{term} denotes the constant t_0 in Proposition 5.4, and enlarge every bad cube by a fixed factor large enough that, outside the resulting set E_0 , an annulus of radius 2^j can meet a cube of side 2^m only if $s = j - m \geq s_0$. This factor depends only on n , and $|E_0| \leq C\alpha^{-1} \|f\|_1$. Put $b = \sum_Q b_Q$ and write $B_m = \sum_{\ell(Q)=2^m} b_Q$.

Angular decomposition and short variation.

For every s , make the mean-zero angular split (27) with a parameter $\delta_a > 0$. By Lemma 2.7, the sum of the L^1 norms of the tails is bounded. If Tail_s denotes the trajectory formed with $\Omega_s^\#$, then Young's inequality on each annulus and the triangle inequality for V^1 give

$$\sum_s \|V^1(\text{Tail}_s)\|_1 \lesssim \sum_s \|\Omega_s^\#\|_1 \sum_j \sum_{\ell(Q)=2^{j-s}} \|b_Q\|_1 \lesssim \mathfrak{L}(\Omega) \Sigma.$$

Under the present normalization this is at most $C\Sigma$.

For the bounded pieces ω_s , the short square function is controlled by Proposition 5.4:

$$\left\| \left(\sum_j V^2(K_{\omega_s} \mathbf{1}_{\{\varepsilon < |z| \leq 2^{j+1}\}} * B_{j-s} : 2^j \leq \varepsilon \leq 2^{j+1})^2 \right)^{1/2} \right\|_2^2 \leq C 2^{2\delta_a s} 2^{-\delta_T s} \alpha \Sigma. \quad (233)$$

Choose $\delta_a < \delta_T/8$ and sum the square roots over s .

Fixed-rank control of the long variation.

It remains to control the dyadic long variation. Smooth each sharp annular kernel at relative radial scale $h_s = 2^{-\eta_0 s}$, where $N = 4n + 1$, $\eta_0 = 1/(192N)$, and choose

$$0 < \delta_a < \min(\delta_T/8, 1/192). \quad (234)$$

The boundary errors have total L^1 variation at most $C\|\omega_s\|_1 h_s \Sigma$ and are summable in s . Denote the resulting compactly supported smooth annular kernels by H_j^s ; they obey

$$M_N(H^s) \leq C\|\omega_s\|_\infty h_s^{-N} \leq C 2^{(\delta_a + N\eta_0)s}. \quad (235)$$

We organize the long variation by a fixed global rank. This selector-free construction converts each point-dependent shell tail into a suffix of one finite ranked sequence. For a bad cube Q with $\ell(Q) = 2^{j-s}$, let $A_j(Q)$ be its ancestor of side 2^j in the original dyadic grid. Since H_j^s is supported in a fixed annular enlargement of radius $C2^j$, the function $H_j^s * b_Q$ is supported in a fixed enlargement of $A_j(Q)$. We use the following nested adjacent dyadic systems. For $a \in \{0, 1/3, 2/3\}^n$, put

$$\mathcal{D}^a = \{2^{-k}([0, 1]^n + m + (-1)^k a) : k \in \mathbb{Z}, m \in \mathbb{Z}^n\}. \quad (236)$$

These systems are nested. In one coordinate, if $s = (-1)^k$, the left endpoint of a child differs from a level- $(k-1)$ endpoint, in units of 2^{-k} , by $m - 2m' + 3sa$. Since $N = m + 3sa$ is integral, take $m' = \lfloor N/2 \rfloor$; then $N - 2m' \in \{0, 1\}$. Taking products proves nesting. They also have the adjacent covering property: choose a dyadic length L with $3\ell(I) < L \leq 6\ell(I)$ for a given interval I . At scale L , the three boundary classes are separated by $L/3$, so at least one class has no boundary in I and its length- L interval contains I . Choose this class in every coordinate. Thus every cube is contained in a member of some $\mathcal{D}^a$ whose side is at most six times larger.

Applying this property to the fixed enlargement gives one of the 3^n systems and a cube $R(Q)$ in that system such that the enlargement is contained in $R(Q)$ and $\ell(R(Q)) \leq C_n 2^j$. Splitting into the finitely many systems and the finitely many possible generation offsets, fix one class. All cubes below belong to one dyadic system and $\ell(R(Q)) = 2^{j+\iota_{\text{gen}}}$ for a fixed integer ι_{gen} .

For R in this class define the localized output

$$F_R = H_j^s * \sum_{\substack{Q: R(Q)=R \\ \ell(Q)=2^{j-s}}} b_Q, \quad \ell(R) = 2^{j+\iota_{\text{gen}}}. \quad (237)$$

It is essential that $\text{supp } F_R \subset R$. Let $\mathcal{R}_s$ be the finite family of nonzero R . Each Q is assigned once, and each nonempty R contains at least one assigned Q of side 2^{j-s} . Choosing one such cube for each R proves

$$\sum_{R \in \mathcal{R}_s} |R| \leq C 2^{ns} \sum_Q |Q|. \quad (238)$$

Indeed $|R| \leq C 2^{ns} |Q|$, and the selected Q are distinct. Set

$$\mathcal{N}_s(x) = \sum_{R \in \mathcal{R}_s} \mathbf{1}_R(x), \quad D_s = \lceil 2^{(n+1)s} \rceil, \quad X_s = \{\mathcal{N}_s > D_s\}.$$

Chebyshev, (238), and the Calderón–Zygmund packing bound give

$$|X_s| \leq C 2^{-s} \alpha^{-1} \|f\|_1. \quad (239)$$

Because F_R is supported in R , cubes $R \subset X_s$ contribute nothing on X_s^c and may be discarded. Put

$$\mathcal{R}_s^{\text{ret}} = \{R \in \mathcal{R}_s : R \not\subset X_s\}.$$

Every chain in $\mathcal{R}_s^{\text{ret}}$ has length at most D_s : if $R_1 \supsetneq \cdots \supsetneq R_L$, choose $x \in R_L \setminus X_s$; then $L \leq \mathcal{N}_s(x) \leq D_s$.

Give a remaining cube the recursive rank

$$\text{rk}(R) = 1 + \max\{\text{rk}(R') : R' \in \mathcal{R}_s^{\text{ret}}, R' \subsetneq R\}, \quad (240)$$

with the maximum of the empty set equal to zero. Thus $1 \leq \text{rk}(R) \leq D_s$, equal-rank cubes form an antichain, and strict inclusion strictly increases rank. Put

$$g_r = \sum_{\substack{R \in \mathcal{R}_s^{\text{ret}} \\ \text{rk}(R)=r}} F_R, \quad 1 \leq r \leq D_s.$$

For a fixed $x \notin X_s$, the cubes containing x form a chain. Their generation order is the shell order, and their ranks strictly increase in that order. More explicitly, list them as

$$R_1(x) \subsetneq \cdots \subsetneq R_L(x).$$

Then $\text{rk}(R_1(x)) < \cdots < \text{rk}(R_L(x))$. Every term arising from a cube not containing x vanishes because $\text{supp } F_R \subset R$. Therefore, for every shell threshold k , there is an integer $r = r(k, x)$ such that

$$\sum_{j \geq k} \sum_{\ell(R)=2^{j+t_{\text{gen}}}} F_R(x) = \sum_{q \geq r} g_q(x). \quad (241)$$

Conversely every distinct suffix of the active ranks is obtained by placing k between the corresponding consecutive generations. Missing ranks merely insert repeated values. Repetitions do not affect variation, and inclusion of the zero and total endpoints shows that the shell-tail 2-variation at x equals the 2-variation of the rank suffixes. Subtracting each suffix from the fixed total sum identifies suffix variation with prefix variation exactly.

Apply Lemma 5.2 to $g_1, \dots, g_{D_s}$. For a fixed choice of rank signs, (237) becomes

$$\sum_r \epsilon_r g_r = \sum_j H_j^s * \sum_{\ell(Q)=2^{j-s}} c_Q b_Q, \quad c_Q = \epsilon_{\text{rk}(R(Q))} \in \{-1, 0, 1\}.$$

These coefficients are fixed before x is evaluated and therefore satisfy the fixed-sign hypothesis of Lemma 5.3; the x -dependent threshold acts only through the pointwise rank-suffix identity (241). Lemma 5.3 and (235) give

$$\|\mathbf{1}_{X_s^c} V^2(\sum_{j \geq k} H_j^s * B_{j-s} : k)\|_2^2 \leq C s^2 2^{-[1/24 - 2\delta_a - 2N\eta_0]s} \alpha \Sigma. \quad (242)$$

The exponent in brackets is positive by (234). The finite number of adjacent systems changes only the dimensional constant. Equations (239)–(242) are summable in s .

Let $X = \bigcup_s X_s$. Combining the short and long estimates, the bounded angular bad trajectory outside $E_0 \cup X$ is the sum of a path G with $\|V^2(G)\|_2^2 \leq C\alpha\Sigma$ and a path W with $\|V^1(W)\|_1 \leq C\Sigma$. Lemma 2.2 and Chebyshev give

$$|\{x \notin E_0 \cup X : \lambda N_\lambda(T_{\Omega, \varepsilon}^0 b : \varepsilon > 0)^{1/2} > C\alpha\}| \leq C\alpha^{-1}\Sigma.$$

Add the good part and the exceptional sets.

Removal of the auxiliary cutoffs.

We remove the cutoffs in their declared order. With shells and parameters fixed, increase Q_0 to the full bad family. Writing

$$f_{Q_0} = g + \sum_{Q \in Q_0} b_Q,$$

we have $f_{Q_0} \rightarrow f$ in L^1 , since $\sum_Q \|b_Q\|_1 < \infty$. Every coordinate has a finite-annulus L^1 kernel, so Young's inequality gives the corresponding coordinatewise L^1 convergence. After a subsequence, convergence is pointwise simultaneously on the finite parameter set. Lemma 2.6, Fatou's lemma, and monotone convergence preserve the jump distribution and exceptional-set bounds.

Next send $j_- \rightarrow -\infty$ and $j_+ \rightarrow \infty$. For core data, each fixed spatial point x and positive truncation q sees only finitely many shells: if $2^{j+1} \leq q$, the shell lies below the truncation, while for all sufficiently large j the annulus centered at x no longer meets the compact support of the input. Thus every coordinate is eventually constant in the shell cutoff. The truncated coordinates converge pointwise, and Lemma 2.6 and Fatou apply again. Finally increase the finite rational parameter sets to $\mathbb{Q}_+$. For the present bounded compactly supported data, Fubini on every finite annulus gives local absolute continuity of the truncation path (as recorded again in Section 12); hence rational and full-parameter jump counts agree. Every Rademacher–Menshov ranking is therefore finite, and the cutoff limits recover the full trajectory from these finite fixed-rank sums. The resulting distribution bound is $C\|f\|_1/\alpha$, uniformly in λ , and hence gives (231) after restoring the homogeneous angular scale. The general L^1 estimate follows from the completion in Section 12. $\square$

9 Phase-adapted Calderón–Zygmund decomposition

We modify the Calderón–Zygmund decomposition so that each bad block cancels against the input modulation generated by its parent cube. This is exactly the cancellation condition required by the packet estimates in the next section.

Fix a polynomial P and take the maximal dyadic family $\mathcal{Q}$ constructed in the proof of Proposition 8.1, at height α for f . For each $Q \in \mathcal{Q}$, let c_Q be its center and put

$$a_Q = |Q|^{-1} \int_Q e^{iP(c_Q, y)} f(y) dy, \quad g = f \mathbf{1}_{(\cup \mathcal{Q})^c} + \sum_Q a_Q e^{-iP(c_Q, \cdot)} \mathbf{1}_Q, \quad b_Q = (f - a_Q e^{-iP(c_Q, \cdot)}) \mathbf{1}_Q. \quad (243)$$

Then

$$\|g\|_\infty \leq C\alpha, \quad \|g\|_2^2 \leq C\alpha \|f\|_1, \quad \sum_Q \|b_Q\|_1 \leq C\|f\|_1, \quad \|b_Q\|_1 \leq C\alpha|Q|. \quad (244)$$

Indeed, maximality gives $|Q|^{-1} \int_Q |f| \leq 2^n \alpha$, and hence $|a_Q| \leq 2^n \alpha$ and $\|b_Q\|_1 \leq 2 \int_Q |f|$. The cubes are disjoint. Off their union $|f| \leq \alpha$ almost everywhere, so these observations give all four estimates in (244), including

$$\|g\|_2^2 \leq \alpha \int_{(\cup \mathcal{Q})^c} |f| + C\alpha \sum_Q |a_Q| |Q| \leq C\alpha \|f\|_1.$$

Most importantly, phase adaptation gives the exact cancellation

$$h_Q(y) := e^{iP(c_Q, y)} b_Q(y), \quad \int_Q h_Q(y) dy = 0, \quad |h_Q(y)| = |b_Q(y)|. \quad (245)$$

No pointwise bound for b_Q or h_Q is asserted. The packet estimates below are deliberately formulated with only the density $\|h_Q\|_1/|Q|$ and are applied directly to the original blocks; thus no charge-neutralized stopping-tree decomposition is used.

10 The far-field weak endpoint

This section proves the weak critical-jump estimate for the normalized far-field trajectory. Assume that the highest nonpure homogeneous block P_m has been normalized by

$$\|P_m\|_{\text{coeff}} = 1. \quad (246)$$

Define the far family by

$$\mathcal{F}_\varepsilon^P f(x) = \int_{\substack{|x-y| > \varepsilon \\ |x-y| > 1}} e^{iP(x, y)} K_\Omega(x-y) f(y) dy,$$

extended constantly for $0 < \varepsilon \leq 1$. The strong $p = 2$ theorem gives

$$\sup_{\lambda > 0} \|\lambda N_\lambda(T_{\Omega, \varepsilon}^P h : \varepsilon > 0)^{1/2}\|_2 \leq C_{n,d} \mathfrak{L}(\Omega) \|h\|_2. \quad (247)$$

Proposition 10.1 (Far weak endpoint). *Let P be a real polynomial of degree at most d with highest nonpure homogeneous block P_m , and assume $\|P_m\|_{\text{coeff}} = 1$. Let $\Omega \in L \log L(\mathbb{S}^{n-1})$ have mean zero. Then, for every bounded compactly supported f ,*

$$\sup_{\lambda > 0} \|\lambda N_\lambda(\mathcal{F}_\varepsilon^P f : \varepsilon > 0)^{1/2}\|_{L^{1,\infty}} \leq C_{n,d} \mathfrak{L}(\Omega) \|f\|_1. \quad (248)$$

After the completion in Section 12, the same estimate holds for every $f \in L^1(\mathbb{R}^n)$.

Proof. We first take bounded compactly supported f and impose finite cube, shell, and rational-parameter cutoffs, exactly as in Proposition 8.1. Every estimate below is uniform in those cutoffs.

If $\mathfrak{L}(\Omega) = 0$, the assertion is immediate. By angular homogeneity normalize $\mathfrak{L}(\Omega) = 1$, fix a Calderón–Zygmund height $\alpha > 0$, and use (243). Equations (247) and (244) control the good part outside a set of measure $C\alpha^{-1}\|f\|_1$. Enlarge the bad cubes by a fixed factor chosen as follows. Let $s_0 \geq \max\{5, t_0\}$, where t_0 is the threshold in Proposition 7.10, and use dilation factor $C_n 2^{s_0}$. This produces another exceptional set E_0 of measure $C\alpha^{-1}\|f\|_1$. Outside E_0 , a cube meeting a shell $R = 2^j \geq 1$ has

$$\ell(Q) = R2^{-s}, \quad s \geq s_0. \quad (249)$$

Put $b = \sum_Q b_Q$ and $\Sigma = \sum_Q \|b_Q\|_1$. For every fixed s , the cubes in (249), as j varies, form a subfamily of the original disjoint Calderón–Zygmund cubes and

$$\sum_j \sum_{\ell(Q)=2^{j-s}} \|h_Q\|_1 \leq \Sigma, \quad h_Q = e^{iP(c_Q, \cdot)} b_Q. \quad (250)$$

Split $\Omega = \omega_s + \Omega_s^\sharp$ by (27). The angular tails have summable total $V^1 L^1$ norm, while

$$\|\omega_s\|_\infty \leq C2^{\delta a s} \|\Omega\|_1, \quad \|\omega_s\|_1 \leq 2\|\Omega\|_1. \quad (251)$$

Short variation.

We first control the short variation inside every dyadic annulus. Apply Proposition 7.10 directly to the h_Q at each (j, s) , with $\Lambda = C\alpha$ and $t = s$. Square-summing in j and using (250) gives

$$\left\| \left(\sum_j V^2(G_{j,s})^2 \right)^{1/2} \right\|_2^2 \leq C \|\omega_s\|_\infty^2 2^{-c_P s} \alpha \Sigma, \quad (252)$$

and the packet errors satisfy

$$\sum_j \|V^1(E_{j,s})\|_1 \leq C \|\omega_s\|_1 2^{-c_E s} \Sigma. \quad (253)$$

Long variation: low shells.

It remains to control the dyadic long variation. Fix a small $\kappa > 0$ and split the shells by the single global top block:

$$\text{low: } 1 \leq R^m \leq 2^{\kappa s}, \quad \text{high: } R^m > 2^{\kappa s}. \quad (254)$$

There are at most $C(1+s)$ low shells. If $F_{j,s}$ is their full annular output, then

$$V^2 \left(\sum_{j \geq k, j \text{ low}} F_{j,s} : k \right) \leq 2 \sum_{j \text{ low}} |F_{j,s}|.$$

The endpoint difference of a one-shell trajectory is dominated by its full variation. More explicitly write $F_{j,s} = F_{j,s}^G + F_{j,s}^E$ using the endpoint differences of the decomposition in Proposition 7.10; then

$$|F_{j,s}^G| \leq V^2(G_{j,s}), \quad |F_{j,s}^E| \leq V^1(E_{j,s}).$$

Cauchy–Schwarz in the $O(1+s)$ low shells and (173)–(174) therefore give an L^2 bound for $2 \sum_{j \text{ low}} |F_{j,s}^G|$ which, after absorbing $1+s$ and taking $c = c_P/2$, is

$$C \|\omega_s\|_\infty^2 2^{-c s} \alpha \Sigma \quad (255)$$

and a summable $V^1 L^1$ error. This is the only place where translated lower blocks are Fourier-expanded across low shells.

Long variation: high shells.

On the high shells no phase block is expanded. Return to the original form $e^{iP(x,y)}b_Q(y)$; every packet and shell now has the same global phase P , and (254) is precisely the threshold (223). Their short variation has already been included in the square sum (252); the collective estimate is needed only for the dyadic long variation.

For the high dyadic long variation use exactly the output localization and fixed rank from the proof of Proposition 8.1. More explicitly, assign the natural annular output of every Q at shell 2^j to a cube $R(Q)$ in one of the finitely many nested adjacent systems (236), with $\ell(R(Q)) \simeq 2^j$. Split into the finitely many classes determined by the adjacent system and the generation offset, and treat the classes separately. Thus within the current class all $R(Q)$ lie in one nested system and $\ell(R(Q)) = 2^{j+\iota_{\text{gen}}}$ for one fixed integer ι_{gen} . Define the current bounded-angular annular operator by

$$T_j^{P,\omega_s} g(x) = \int_{2^j < |x-y| \leq 2^{j+1}} e^{iP(x,y)} K_{\omega_s}(x-y) g(y) dy,$$

and put

$$F_R = T_j^{P,\omega_s} \sum_{\substack{Q: R(Q)=R \\ \ell(Q)=2^{j-s}}} b_Q.$$

Then $\text{supp } F_R \subset R$ and

$$\sum_R |R| \leq C 2^{ns} \sum_Q |Q|. \quad (256)$$

Let $D_s = \lceil 2^{(n+1)s} \rceil$ and

$$X_s = \left\{ \sum_R \mathbf{1}_R > D_s \right\}.$$

By (256),

$$|X_s| \leq C 2^{-s} \alpha^{-1} \|f\|_1. \quad (257)$$

Discard the cubes $R \subset X_s$ and rank the retained finite family by strict inclusion, as in (240). At every $x \notin X_s$, a high-shell tail is a suffix of the rank sequence; gaps caused by low shells only repeat a suffix value. A fixed rank sign induces a fixed coefficient in $\{-1, 0, 1\}$ on each original packet.

We verify explicitly the hypotheses of Proposition 7.16. For the fixed s , the retained set of high radii is finite because the shell cutoff is finite. Across all of those radii, the packet cubes are still a subfamily of the original globally disjoint Calderón–Zygmund cubes. Take $d\nu_Q = b_Q(y) dy$. Then

$$\text{supp } \nu_Q \subset Q, \quad |h_Q| = |b_Q|, \quad |\nu_Q|(Q) = \|h_Q\|_1 \leq C\alpha|Q|.$$

The normalized sharp-annulus profiles have uniformly bounded $L^\infty + \text{BV}$ norms. Finally, the Rademacher–Menshov signs are fixed packet by packet before the output point x is evaluated. Thus Lemma 5.2 and Proposition 7.16 give

$$\|\mathbf{1}_{X_s^c} V^2(\text{high dyadic tails})\|_2^2 \leq C s^2 \|\omega_s\|_\infty^2 2^{-c_\kappa s} \alpha \Sigma. \quad (258)$$

This is where the estimate must allow packetwise, rather than merely shellwise, signs.

Choose $0 < \delta_a < \frac{1}{4} \min(c_P, c_E, c_\kappa)$. Equations (252)–(258), polynomial factors in s , and (251) imply the following precise majorant bounds. Put

$$\begin{aligned} \mathcal{S}_s &= \left(\sum_j V^2(G_{j,s})^2 \right)^{1/2}, \\ \mathcal{L}_s &= 2 \sum_{j \text{ low}} |F_{j,s}^G|, \\ \mathcal{H}_s &= \mathbf{1}_{X_s^c} V^2(\text{high dyadic tails}), \end{aligned}$$

and let $\mathcal{W}$ be the total 1-variation majorant of all angular tail paths plus $C \sum_{s,j} V^1(E_{j,s})$; the constant also covers the low-shell endpoint errors $2 \sum_{j \text{ low}} |F_{j,s}^E|$. Set

$$\mathcal{G} = \sum_s (\mathcal{S}_s + \mathcal{L}_s + \mathcal{H}_s), \quad X = \bigcup_s X_s.$$

Minkowski's inequality, applied to the square roots of the fixed- s estimates, now gives

$$\|\mathcal{G}\|_2^2 \leq C\alpha\Sigma, \quad \|\mathcal{W}\|_1 \leq C\Sigma, \quad |X| \leq C\alpha^{-1}\|f\|_1. \quad (259)$$

The geometric decay makes all three sums finite. For each bounded angular piece, the deterministic full-variation long–short inequality (77) bounds its 2-variation by the corresponding short square function and dyadic 2-variation. The latter is bounded by $\mathcal{L}_s + \mathcal{H}_s$, while the former is $\mathcal{S}_s$. Summing these bounds over s with (11), using $V^2 \leq V^1$ for the error path majorized by $\mathcal{W}$, and finally applying (10), gives on $E_0^c \cap X^c$

$$\lambda N_\lambda(\mathcal{F}_\varepsilon^P b : \varepsilon > 0)^{1/2} \leq C(\mathcal{G} + \mathcal{W}). \quad (260)$$

Chebyshev applied to (260), together with the good-function estimate, the measures of E_0 and X , and (259), gives

$$|\{x : \lambda N_\lambda(\mathcal{F}_\varepsilon^P f : \varepsilon > 0)^{1/2} > C\alpha\}| \leq C\alpha^{-1}\|f\|_1.$$

The proof in dimension one is identical after using the one-dimensional parts of Corollary 7.14 and Proposition 7.16; mean zero on $\mathbb{S}^0$ makes the angular function a Hilbert-kernel multiple, so no angular-chart issue arises.

Removal of cutoffs.

Finally remove the cutoffs in the same order as in the proof of Proposition 8.1. With the shell interval and the finite rational parameter set fixed, exhaust the phase-adapted bad-cube family. Since $\sum_Q \|b_Q\|_1 < \infty$, the corresponding inputs converge in L^1 ; Young's inequality for each finite-annulus coordinate, followed by a subsequence, gives pointwise convergence simultaneously on the finite parameter set. Lemma 2.6 and Fatou preserve the uniform distribution estimate.

Next exhaust the dyadic shell interval. For a fixed output point x and truncation $q > 0$, every shell with $2^{j+1} \leq q$ lies below the truncation, while for all sufficiently large j the annulus centered at x misses the compact support of the input. Thus each coordinate is eventually a finite shell sum, and the shell-cutoff limit is pointwise. Apply Lemma 2.6 and Fatou once more. Finally exhaust $\mathbb{Q}_+$. For bounded compactly supported data, Fubini on finite annuli gives local continuity in the truncation parameter, so rational and full-parameter jump counts agree. The ranks, signs, and exceptional sets are auxiliary devices used only to prove estimates uniform in the cutoffs; none of them is passed to the limit. Only the original finite-cutoff trajectories are passed to the limit. In particular, no convergence of the ranks, signs, or exceptional sets is asserted or needed. This proves (248) for core data; Section 12 gives the general L^1 trajectory. Angular homogeneity restores $\mathfrak{L}(\Omega)$. $\square$

11 Local weak estimates and homogeneous-degree induction

We close the weak endpoint with a coarser triangular order than in the strong argument: each step removes the entire highest nonpure homogeneous class rather than a single flag coordinate.

Proposition 11.1 (Local homogeneous-degree reduction). *Let P be a real polynomial of degree at most d whose highest nonpure homogeneous block P_m satisfies $\|P_m\|_{\text{coeff}} = 1$. Let $\Omega \in L^1(\mathbb{S}^{n-1})$ and let f be bounded and compactly supported. On each unit output cube, the local path (93) is, modulo pure input and output phases, the sum of*

- (i) *a localized path of the form (95) whose phase has nonpure total degree at most $m - 1$;*

(ii) an error E satisfying

$$V^1(E)(x) \leq C_{n,d} \int_{|x-y|<1} \frac{|\Omega((x-y)/|x-y|)|}{|x-y|^{n-1}} |f(y)| dy. \quad (261)$$

The majorant has L^1 norm at most $C_{n,d} \|\Omega\|_1 \|f\|_1$.

Proof. Let the output cube be centered at a and write $x = a + X, y = a + Y$. Simultaneous translation preserves the degree- m class and creates only lower degrees. Remove all pure terms and let H_m be the mixed representative of the normalized top class. Replace it by the anchored representative

$$\widehat{H}_m(X, Y) = H_m(X, Y) - H_m(Y, Y),$$

which differs by a pure input phase and satisfies $\widehat{H}_m(Y, Y) = 0$. On the fixed bounded set relevant to $|X - Y| < 1$,

$$|\widehat{H}_m(X, Y)| \leq |X - Y| \int_0^1 |\partial_x H_m(Y + t(X - Y), Y)| dt \leq C_{n,d} |X - Y|.$$

The last constant is uniform over the compact unit sphere of $\mathcal{V}_m$. Replacing $e^{i\widehat{H}_m}$ by one therefore creates exactly the total-variation majorant (261). What remains has nonpure degree at most $m - 1$. Young's inequality proves the L^1 assertion. $\square$

Proposition 11.2 (Weak polynomial critical jumps on core data). *For every real polynomial P of degree at most d , every mean-zero $\Omega \in L \log L(\mathbb{S}^{n-1})$, and every bounded compactly supported f ,*

$$\sup_{\lambda>0} \|\lambda N_\lambda(T_{\Omega,\varepsilon}^P f : \varepsilon > 0)^{1/2}\|_{L^{1,\infty}} \leq C_{n,d} \mathfrak{Q}(\Omega) \|f\|_1,$$

with a constant independent of the coefficients of P .

Proof. Induct on the largest total degree m carrying a nonpure homogeneous class. If there is no such class, the phase is pure and Proposition 8.1 applies after input and output conjugation. Assume the assertion for nonpure degree at most $m - 1$ and let

$$C_m = \|P_m\|_{\text{coeff}} > 0, \quad \rho = C_m^{-1/m}.$$

Under $x = \rho X, y = \rho Y$, the entire degree- m class, rather than one scalar coordinate, has norm one. Proposition 10.1 controls the normalized far path.

For the local path use Proposition 11.1. The error is strong $(1, 1)$ in total variation by (261). For the residual phase, subtracting the fixed truncation at one does not change increments below 1; adjoining zero for $\varepsilon \geq 1$ creates no additional jump beyond the precise domination (96). Apply the induction hypothesis to f restricted to the fixed enlargement of each output cube. The output cubes are disjoint and the input enlargements have bounded overlap, so their weak distribution bounds sum to $C\alpha^{-1} \mathfrak{Q}(\Omega) \|f\|_1$.

Combine local and far paths with (9). Finally the exact conjugacy

$$T_{\Omega,\varepsilon}^P f(\rho X) = T_{\Omega,\varepsilon/\rho}^{P_\rho} f_\rho(X), \quad f_\rho(X) = f(\rho X),$$

makes both input and weak output L^1 norms acquire the same factor ρ^{-n} . Thus the bound is independent of C_m . The finite degree induction proves the proposition for every polynomial phase. $\square$

12 Completion, consequences, and radial BV transfer

12.1 Completion of the critical-jump trajectory

For core data, the weak endpoint propositions already remove the auxiliary cube, shell, and rational-parameter cutoffs. We pass to general L^1 data and construct the complete trajectory over all positive truncation parameters.

For bounded compactly supported f , finite annular differences are absolutely convergent. On each compact interval $0 < a < b < \infty$,

$$\int_{a < |z| < b} |K_\Omega(z)| |f(x-z)| dz < \infty$$

for almost every x by Fubini. Hence the truncation path is continuous in ε on compact subintervals of $(0, \infty)$ for almost every x . Jumps over positive rationals therefore equal jumps over all positive parameters.

For such core data the path is zero for sufficiently large truncation at each fixed x . Thus the jump estimate implies the maximal estimate

$$\| \sup_{q \in \mathbb{Q}_+} |T_{\Omega,q}^P h| \|_{L^{1,\infty}} \leq C_{n,d} \mathfrak{Q}(\Omega) \|h\|_1. \quad (262)$$

Indeed, if $|T_{\Omega,q}^P h| > \tau$, compare it with a sufficiently large rational truncation where the value is zero and use one strict $\tau/2$ -jump.

Let f_k be bounded compactly supported functions with $f_k \rightarrow f$ in L^1 and $\|f_{k+1} - f_k\|_1 \leq 2^{-3k}$. By (262),

$$\sum_k \left| \left\{ \sup_{q \in \mathbb{Q}_+} |T_{\Omega,q}^P (f_{k+1} - f_k)| > 2^{-k} \right\} \right| < \infty.$$

Borel–Cantelli gives almost-everywhere uniform convergence on $\mathbb{Q}_+$. This defines $T_{\Omega,q}^P f$ simultaneously for all rational $q > 0$. More explicitly, outside the Borel–Cantelli exceptional set there is $k_0(x)$ such that, for every $k \geq k_0(x)$,

$$\sup_{q \in \mathbb{Q}_+} |T_{\Omega,q}^P (f_{k+1} - f_k)(x)| \leq 2^{-k}.$$

Consequently, for every $m \geq k_0(x)$,

$$\sup_{q \in \mathbb{Q}_+} |T_{\Omega,q}^P (f - f_m)(x)| \leq \sum_{k \geq m} 2^{-k}. \quad (263)$$

For the core function f_m , the trajectory is zero at x for all sufficiently large q . Taking the upper limit as $q \rightarrow \infty$ in (263) and then letting $m \rightarrow \infty$ proves the rational zero-at-infinity anchor. After passing to a further subsequence, Young’s inequality for the finite-annulus kernels and a diagonal argument ensure, on the same full-measure set, convergence of the finite-annulus identities for every rational pair of endpoints. Define the value at arbitrary ε from the base point 1 by

$$T_{\Omega,\varepsilon}^P f = T_{\Omega,1}^P f + \begin{cases} \int_{\varepsilon < |z| \leq 1} e^{iP(x,x-z)} K_\Omega(z) f(x-z) dz, & \varepsilon < 1, \\ - \int_{1 < |z| \leq \varepsilon} e^{iP(x,x-z)} K_\Omega(z) f(x-z) dz, & \varepsilon > 1. \end{cases}$$

The added kernels are L^1 on every finite annulus, so the definition is valid almost everywhere and locally continuous in ε . It agrees with the rational limits.

For every finite rational parameter set, Lemma 2.6 and Fatou’s lemma transfer the distribution estimate to f . Increasing finite rational sets, followed by local continuity, gives all positive truncation parameters. The rational anchor just proved and the finite-annulus definition show that the path tends to zero at infinity over all positive radii. Two paths with the same finite-shell increments differ by a function independent of ε ; the zero anchor therefore makes this completion unique. This completes the unweighted weak critical-jump construction and, together with the strong theorem already proved, establishes Theorem 1.1.

12.2 Abstract consequences

The strong variation estimate is (100). At the endpoint, apply (18) first on an increasing sequence of finite ordered subsets of $\mathbb{Q}_+$. Its bound is uniform in the finite set. Monotonicity of V^r under enlargement of the parameter set and Fatou's lemma for distribution functions pass the estimate to $\mathbb{Q}_+$. Local continuity and Lemma 2.6 then recover all positive parameters. The unweighted weak jump theorem gives

$$\|V^r(T_{\Omega,\varepsilon}^P f : \varepsilon > 0)\|_{L^{1,\infty}} \leq C_{n,d} \frac{r}{r-2} \mathfrak{L}(\Omega) \|f\|_1. \quad (264)$$

The constants remain coefficient-uniform because both abstract implications are uniform over the parameter set. The zero anchor gives the maximal estimate, and finite r -variation gives the limit at zero. This proves Corollary 1.2.

12.3 Radial BV transfer

Write

$$F_\varepsilon = T_{\Omega,\varepsilon}^P f, \quad G_\varepsilon = T_{\Omega,\varepsilon}^{P,b} f.$$

For test data, Stieltjes integration by parts on a finite interval gives

$$G_\varepsilon - G_R = b(\varepsilon)F_\varepsilon - b(R)F_R + \int_{(\varepsilon,R]} F_s db(s). \quad (265)$$

Here $db((s,t]) = b(t) - b(s)$ is the Stieltjes measure associated with the fixed right-continuous representative of b . Altering b at a single radius does not change the original Lebesgue shell integral, whereas the Stieltjes formula depends on the chosen representative. We next justify the passage to general data on one common full-measure set.

Use the representatives of F_ε constructed in the strong completion and in the preceding weak endpoint completion. In both cases the base coordinate F_1 is the almost-everywhere limit of core-data coordinates, and every other coordinate is fixed by its signed finite-annulus increment from 1. Choose the corresponding core approximation f_k , sufficiently rapidly that the maximal norms of its successive differences are summable. The maximal estimates used in those constructions then allow a subsequence for which $F_q^{(k)}$ converges uniformly in $q \in \mathbb{Q}_+$ outside one null set. Since every core-data difference is locally continuous in the truncation parameter, the same subsequence is uniformly Cauchy on each compact parameter interval: for a continuous difference, the supremum on such an interval is its supremum on the rational parameters there.

For every rational pair $0 < a < c < \infty$, Young's inequality for

$$\|K_\Omega \mathbf{1}_{\{a < |\cdot| \leq c\}}\|_1 \leq \|\Omega\|_1 \log(c/a) \quad (266)$$

passes the finite-annulus identity to the limit. A diagonal subsequence and countability leave one full-measure set on which all rational identities and the convergence of the anchored base coordinate hold simultaneously. Enlarge its null complement by the Fubini null sets on $\{2^{-j} < |z| < 2^j\}$, $j \geq 1$, and by the null sets on which the zero anchor or finite 3-variation fails. On the resulting common full-measure set,

$$F_R \longrightarrow 0 \quad (R \rightarrow \infty), \quad M_F := \sup_{s>0} |F_s| < \infty,$$

and F is locally absolutely continuous in the truncation parameter. Indeed, with

$$\mathcal{A}_x(r) = \int_{\mathbb{S}^{n-1}} e^{iP(x,x-r\theta)} \Omega(\theta) f(x-r\theta) d\sigma(\theta),$$

Fubini locally in x (using Hölder's inequality on bounded x -sets when $f \in L^p$) gives, for every compact $0 < a < c < \infty$,

$$\int_a^c |\mathcal{A}_x(r)| \frac{dr}{r} < \infty, \quad F_a(x) - F_c(x) = \int_a^c \mathcal{A}_x(r) \frac{dr}{r}. \quad (267)$$

The rational identities and absolute continuity give (267) for all real $0 < a < c < \infty$.

On this same set define

$$G_\varepsilon = b(\varepsilon)F_\varepsilon + \int_{(\varepsilon, \infty)} F_s db(s). \quad (268)$$

The Stieltjes integral is absolutely convergent because $|db|(0, \infty) < \infty$ and $M_F < \infty$. For $0 < \varepsilon < R < \infty$, integration by parts for the locally absolutely continuous path F and the fixed right-continuous representative of b uses the product formula

$$b(R)F_R - b(\varepsilon)F_\varepsilon = \int_{(\varepsilon, R]} F_s db(s) + \int_\varepsilon^R b(s) dF_s. \quad (269)$$

There is no ambiguity between $b(s)$ and $b(s-)$ in the second integral, because dF has no atoms. By (267),

$$dF_s = -\mathcal{A}_x(s) \frac{ds}{s}.$$

Consequently, (269) gives

$$\begin{aligned} G_\varepsilon - G_R &= b(\varepsilon)F_\varepsilon - b(R)F_R + \int_{(\varepsilon, R]} F_s db(s) \\ &= - \int_{(\varepsilon, R]} b(s) dF_s \\ &= \int_{\varepsilon < |z| \leq R} e^{iP(x, x-z)} K_\Omega(z) b(|z|) f(x-z) dz. \end{aligned} \quad (270)$$

Indeed, $-\int_\varepsilon^R b(s) dF_s = \int_\varepsilon^R b(s) \mathcal{A}_x(s) ds/s$, and polar coordinates give exactly the last line, including its sign and half-open endpoint convention. The last integral is absolutely meaningful on the same set; its kernel has L^1 norm at most $\|b\|_\infty \|\Omega\|_1 \log(R/\varepsilon)$. Hence (268) has exactly the required finite-annulus increments for general data. Right continuity fixes the Stieltjes representative, while altering b at one radius does not change the last integral. Finally,

$$|G_\varepsilon| \leq \|b\|_\infty |F_\varepsilon| + M_F |db|((\varepsilon, \infty)) \longrightarrow 0 \quad (\varepsilon \rightarrow \infty), \quad (271)$$

so the zero anchor uniquely determines the extension.

Put $B = \|b\|_\infty$, $V = \text{Var}(b)$, and $M_F = \sup_{s>0} |F_s|$. The zero anchor gives $M_F \leq V^r(F)$ for every $r \geq 1$. Product variation and (268) give, pointwise,

$$\begin{aligned} V^r(bF) &\leq BV^r(F) + VM_F, \\ V^r\left(\int_{(\varepsilon, \infty)} F_s db(s) : \varepsilon > 0\right) &\leq VM_F, \\ V^r(G) &\leq (B + 2V)V^r(F). \end{aligned} \quad (272)$$

Indeed, the second line follows by summing the $|db|$ -masses of the disjoint parameter intervals in a variation partition.

We record separately the critical-jump transfer. If $M_F = 0$, then $F_s = 0$ for every $s > 0$, and (268) gives $G_\varepsilon = 0$; all assertions are then immediate. Hence assume $M_F > 0$. Suppose first that $B > 0$. For an ordered family of λ -jumps of bF , write

$$b(t)F_t - b(s)F_s = b(t)(F_t - F_s) + (b(t) - b(s))F_s.$$

Assign the pair to the first term if its modulus is $> \lambda/2$, and otherwise to the second. The first subfamily contributes at most $N_{\lambda/(2B)}(F)$. The parameter intervals in the second subfamily are disjoint, and hence

$$\frac{\lambda}{2M_F} N_{\text{second}} < \sum_{\text{second}} |b(t) - b(s)| \leq V.$$

This proves

$$N_\lambda(bF) \leq N_{\lambda/(2B)}(F) + \frac{2M_F V}{\lambda}. \quad (273)$$

On the set where the left side is nonzero, $\lambda \leq 2BM_F$. With $\mu = \lambda/(2B)$, the second term after taking square roots is bounded by

$$\sqrt{2\lambda M_F V} \leq 2M_F \sqrt{BV} \leq (B + V)M_F.$$

It follows that

$$\lambda N_\lambda(bF)^{1/2} \leq 2B\mu N_\mu(F)^{1/2} + C(B + V)M_F. \quad (274)$$

For $B = 0$ the path is zero. Define the Stieltjes-tail path by

$$H_\varepsilon = \int_{(\varepsilon, \infty)} F_s db(s).$$

It satisfies $V^1(H) \leq VM_F$; since a nonzero λ -jump of H also forces $\lambda \leq 2VM_F$, while the disjoint jump intervals give $N_\lambda(H) \leq VM_F/\lambda$. Therefore

$$\lambda N_\lambda(H)^{1/2} \leq CVM_F. \quad (275)$$

Taking the supremum in λ after applying (9), (274), and (275) proves the critical-jump bounds for G . In strong L^p use (100) with, say, $r = 3$ to control M_F ; in weak L^1 use (264) with $r = 3$ and the quasi-triangle inequality in $L^{1,\infty}$. Formula (272) gives the asserted strong and weak r -variation bounds. This proves the estimates in Corollary 1.3.

The almost-everywhere zero anchor was proved in (271); for $f \in L^p$, both terms there also tend to zero in L^p . Since a scalar path of finite r -variation has one-sided limits at both endpoints, G_ε has an almost-everywhere limit as $\varepsilon \downarrow 0$. The zero anchor gives $\sup_{\varepsilon > 0} |G_\varepsilon| \leq V^r(G)$ and hence the maximal truncation assertion. This completes the proof of Corollary 1.3.

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